

Vectors in the plane
 Addition and scalar multiplication
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1 — VECTORS

1.1 Vectors in the plane

Definition 1.1.1 A scalar is a quantity that has magnitude but no direction.
 A vector is usually described as a quantity that has both magnitude and direction.

Geometrically, a vector is represented by a directed line segment that is an arrow and is written either as a boldface symbol $\mathbf{v}$ or $\vec{v}$ or $\overrightarrow{AB}$ for instance weight, velocity, frictional force are vector quantity. The arrow points in the direction of the vector, and the length of the arrow gives the magnitude of the vector.

Notations and Terminologies

A vector whose initial point is A and whose terminal point is B is given by $\overrightarrow{AB}$ and the magnitude (or length) of vector $\overrightarrow{AB}$ is denoted by $\|\overrightarrow{AB}\|$.

Figure 1.1a gives an aerial view of a tugboat trying to free a cruise liner that has run aground in shallow waters. The magnitude and direction of the force exerted by the tugboat are represented by the vector shown in the figure.

In Figure 1.1b the vectors (arrows) give the magnitude and direction of blood cells flowing through

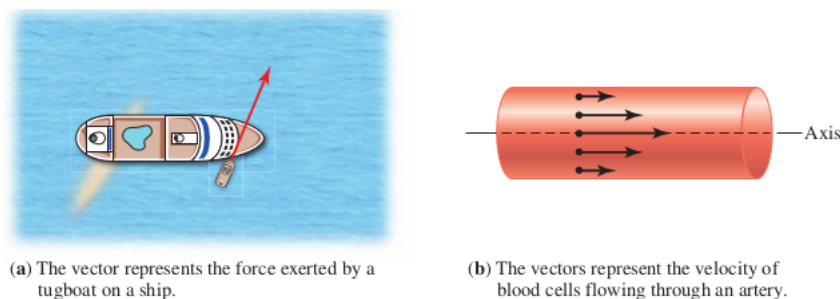


Figure 1.1:

an artery. Observe that the lengths of the vectors vary; this reflects the fact that the blood cells near the central axis have a greater velocity than those near the walls of the artery.

Two vectors, $\mathbf{v}$ and $\mathbf{w}$, that have the same magnitude and direction are said to be equal, written

$\mathbf{v} = \mathbf{w}$. Thus, the vectors $\mathbf{v} = \overrightarrow{AB}$ and $\mathbf{w} = \overrightarrow{CD}$. Graphically, this means that the arrows representing $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel, have the same length and are pointing in the same direction.

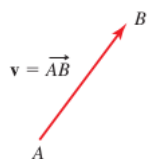


Figure 1.2: $\mathbf{v}$ is the directed line segment from A to B.

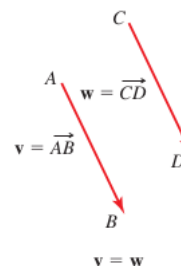


Figure 1.3: $\mathbf{v}$ and $\mathbf{w}$ have the same length and direction.

1.2 Addition and scalar multiplication

Consider the vector $\overrightarrow{AB}$ as representing the displacement of a particle from the point A to the point B, notice that the end result of displacing the particle from A to B (corresponding to the vector $\overrightarrow{AB}$), followed by displacing the particle from B to C (corresponding to the vector $\overrightarrow{BC}$) is the same as displacing the particle directly from A to C, which corresponds to the vector $\overrightarrow{AC}$ (called the resultant vector). We call AC the sum of AB and BC and write

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}.$$

To add two vectors, we locate the initial point of one at the terminal point of the other and complete the parallelogram, as indicated in Figure 1.5. The vector lying along the diagonal, with initial point at A and terminal point at C is the sum

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}.$$

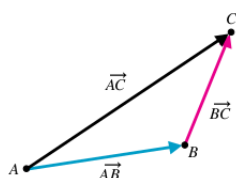


Figure 1.4: Resultant vector

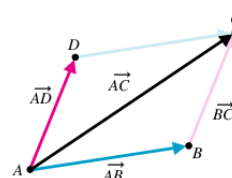


Figure 1.5: Sum of two vectors.

A vector can be multiplied by a scalar. If $c \neq 0$ is a scalar and $\mathbf{v}$ is a vector, then the scalar multiple of c and $\mathbf{v}$ is a vector $c\mathbf{v}$. The magnitude of $c\mathbf{v}$ is $|c||\mathbf{v}|$.

Two nonzero vectors are parallel if they are scalar multiples of one another. In figure 1.5 vector $\overrightarrow{AD}$ and $\overrightarrow{BC}$ are parallel.

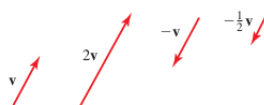


Figure 1.6: Scalar multiples of $\mathbf{v}$

The difference of two vectors $\mathbf{v}$ and $\mathbf{w}$, written $\mathbf{v} - \mathbf{w}$, is defined by

$$\mathbf{v} - \mathbf{w} = \mathbf{v} + (-\mathbf{w})$$

To describe this operation geometrically, consider once again the two vectors $\mathbf{v}$ and $\mathbf{w}$ of Figure 1.7, which are reproduced in Figure 1.8. If we translate $\mathbf{w}$, reverse it to obtain $-\mathbf{w}$, and then use the parallelogram law to add $\mathbf{v}$ to $-\mathbf{w}$, we obtain $\mathbf{v} - \mathbf{w}$, as shown in Figure 1.8



Figure 1.7: The vectors $\mathbf{v}$ and $\mathbf{w}$

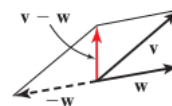


Figure 1.8: The vector $\mathbf{v} - \mathbf{w}$.

Vectors in the Coordinate Plane

The vector $\mathbf{v}$ with initial point at the origin and terminal point $P(v_1, v_2)$ is called the **position vector** of the point $P(v_1, v_2)$ and is denoted by $\langle v_1, v_2 \rangle$.

Definition 1.2.1 A vector in the plane is an ordered pair $\mathbf{v} = \langle v_1, v_2 \rangle$ of real numbers, v_1 and v_2 , called the **scalar components** of $\mathbf{v}$. The **zero vector** is $\mathbf{0} = \langle 0, 0 \rangle$.

Definition 1.2.2 Given the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$, the vector $\overrightarrow{P_1P_2}$ is represented by the position vector

$$\mathbf{v} = \overrightarrow{P_1P_2} = \langle x_2 - x_1, y_2 - y_1 \rangle \quad (1.1)$$

■ **Example 1.1** Find the vector $\mathbf{v}$ with initial point $A(-3, -2)$ and terminal point $B(2, 1)$. ■

Solution: Using Equation 1.1, we find the vector $\mathbf{v}$ to be

$$\mathbf{v} = \langle 2 - (-3), 1 - (-2) \rangle = \langle 5, 3 \rangle$$

■ **Example 1.2** Let $\mathbf{v}$ be a vector with initial point $A(0, 0)$ and terminal point $B(3, 2)$, and let $\mathbf{u}$ be a vector with initial point $C(1, 3)$ and terminal point $D(4, 5)$. Show that $\mathbf{v} = \mathbf{u}$. ■

Solution: To show that $\mathbf{v} = \mathbf{u}$, we need to show that both vectors have the same length and direction. Using the distance formula, we find

$$\text{length of } \overrightarrow{AB} = \sqrt{(3-0)^2 + (2-0)^2} = \sqrt{13}$$

and

$$\text{length of } \overrightarrow{CD} = \sqrt{(4-1)^2 + (5-3)^2} = \sqrt{13}$$

so $\mathbf{v}$ and $\mathbf{u}$ have the same length. Next, we find

$$\text{slope of } \overrightarrow{AB} = \frac{2-0}{3-0} = \frac{2}{3}$$

and

$$\text{slope of } \overrightarrow{CD} = \frac{5-3}{4-1} = \frac{2}{3}$$

so $\mathbf{v}$ and $\mathbf{u}$ have the same direction. This proves that $\mathbf{v} = \mathbf{u}$

Length of a Vector

Definition 1.2.3 The length or magnitude of $\mathbf{v} = \langle v_1, v_2 \rangle$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2} \quad (1.2)$$

Vector Addition in the Coordinate Plane

Parallelogram Law for Vector Addition

If $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$, then

$$\mathbf{u} + \mathbf{v} = \langle u_1, u_2 \rangle + \langle v_1, v_2 \rangle = \langle u_1 + v_1, u_2 + v_2 \rangle$$

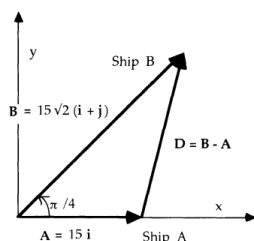
■ **Example 1.3** If $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 3 \rangle$, then

$$\mathbf{u} + \mathbf{v} = \langle 3 + (-1), -2 + 3 \rangle = \langle 2, 1 \rangle$$

■ **Example 1.4** Suppose ship A is cursing in the easterly direction at a speed of 15km/hr while ship B is speeding at 30km/hr in the north east direction. Assuming that both ships started from the same place at the same time, find the rate at which the two ships are separating.

solution: Taking the x-axis the easterly direction and y-axis the northerly direction see figure 1.9, we see that the velocity of the ship A is represented by the vector $A = 15i$ and the velocity of the ship B by the vector

$$B = 30 \left(\cos \frac{\pi}{4} i + \sin \frac{\pi}{4} j \right)$$



The displacement of the ship B relative to the ship A is represented by the vector $D = B - A$. Hence the rate at which the ships separating is equal to the magnitude of the displacement vector D . Thus,

$$\begin{aligned} D &= B - A = 15(\sqrt{2} - 1)i + 15\sqrt{2}j \\ \Rightarrow \|D\| &= 15\sqrt{(\sqrt{2} - 1)^2 + 2} = 22.1 \text{ km/hr} \end{aligned}$$

Figure 1.9: Rate of displacement between two ships

Scalar Multiplication

If $\mathbf{u} = \langle u_1, u_2 \rangle$ and c is a scalar, then,

$$c\mathbf{u} = \langle cu_1, cu_2 \rangle$$

Properties of Vectors

Theorem 1.2.1 — Rules for Vector Addition and Scalar Multiplication. Suppose that $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors and that c and d are scalars. Then

- | | |
|--|---|
| 1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ | 5. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ |
| 2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ | 6. $c(d\mathbf{u}) = (cd)\mathbf{u}$ |
| 3. $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ | 7. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ |
| 4. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ | 8. $1\mathbf{u} = \mathbf{u}$ |

Unit Vectors

A unit vector is a vector of length 1. Unit vectors are primarily used as indicators of direction. For example, if $\mathbf{w}$ is a nonzero vector, then the vector

$$\mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|}$$

is a unit vector having the same direction as $\mathbf{w}$. Furthermore, by writing $\mathbf{w}$ in the form

$$\mathbf{w} = \|\mathbf{w}\| \left(\frac{\mathbf{w}}{\|\mathbf{w}\|} \right) = \|\mathbf{w}\| \mathbf{u}$$

the two properties of magnitude and direction that define a vector are clearly displayed.

■ **Example 1.5** Find a unit vector in the same direction as $\mathbf{v} = \langle 4, 3 \rangle$ ■

Solution: $\|\mathbf{v}\| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$

The unit vector in the same direction as $\mathbf{v} = \langle 4, 3 \rangle$ is then

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\langle 4, 3 \rangle}{5} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$$

Standard Basis Vectors

There are two unit vectors in the coordinate plane that are singled out for a special role. They are the vectors $\mathbf{i}$ and $\mathbf{j}$ defined by

$$\mathbf{i} = \langle 1, 0 \rangle \quad \text{and} \quad \mathbf{j} = \langle 0, 1 \rangle$$

The vector $\mathbf{i}$ points in the positive x -direction, whereas the vector $\mathbf{j}$ points in the positive y -direction.

Let $\mathbf{u} = \langle u_1, u_2 \rangle$ be a vector in the coordinate plane. Then

$$\begin{aligned} \mathbf{u} &= \langle u_1, u_2 \rangle = \langle u_1, 0 \rangle + \langle 0, u_2 \rangle \\ &= u_1 \langle 1, 0 \rangle + u_2 \langle 0, 1 \rangle \\ &= u_1 \mathbf{i} + u_2 \mathbf{j} \end{aligned}$$

This shows that any vector in the plane can be expressed in terms of the vectors $\mathbf{i}$ and $\mathbf{j}$. For this reason the vectors $\mathbf{i}$ and $\mathbf{j}$ are referred to as standard basis vectors.

■ **Example 1.6** Let $\mathbf{u} = -4\mathbf{i} + 2\mathbf{j}$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j}$, and $\mathbf{w} = 2\mathbf{i} + 3\mathbf{j}$. Find the scalar m and n such that $\mathbf{w} = m\mathbf{u} + n\mathbf{v}$ ■

Solution: We have

$$\begin{aligned} m\mathbf{u} + n\mathbf{v} &= m(-4\mathbf{i} + 2\mathbf{j}) + n(2\mathbf{i} + \mathbf{j}) \\ &= (-4m + 2n)\mathbf{i} + (2m + n)\mathbf{j} = 2\mathbf{i} + 3\mathbf{j} \end{aligned}$$

Equating the corresponding components, we obtain

$$-4m + 2n = 2 \quad \text{and} \quad 2m + n = 3$$

Solving for m and n , we get $m = \frac{1}{2}$ and $n = 2$

■ **Example 1.7** At a certain point during a jump, there are two principal forces acting on a sky diver: gravity exerting a force of 180 pounds straight down and air resistance exerting a force of 180 pounds up and 30 pounds to the right. What is the net force acting on the sky diver? ■

Solution: We write the gravity force vector as $\mathbf{g} = \langle 0, -180 \rangle$ and the air resistance force vector as $\mathbf{r} = \langle 30, 180 \rangle$. The net force on the sky diver is the sum of the two forces, $\mathbf{g} + \mathbf{r} = \langle 30, 0 \rangle$. We illustrate the forces in Figure 1.10. Notice that at this point, the vertical forces are balanced, producing a “free-fall” vertically, so that the sky diver is neither accelerating nor decelerating vertically. The net force is purely horizontal, combating the horizontal motion of the sky diver after jumping from the plane.

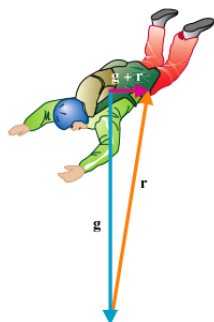


Figure 1.10: Forces on a sky diver.

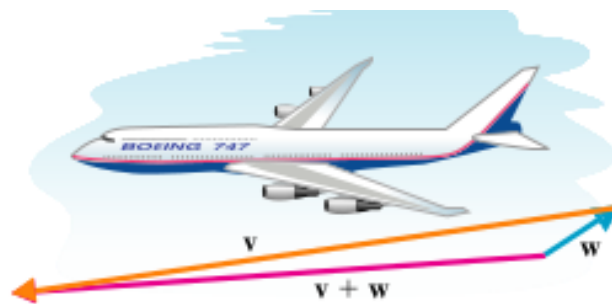


Figure 1.11: Forces on an airplane.

■ **Example 1.8** An airplane has an airspeed of 400 mph. Suppose that the wind velocity is given by the vector $\mathbf{w} = \langle 20, 30 \rangle$. In what direction should the airplane head in order to fly due west i.e., in the direction of the unit vector $-\mathbf{i} = \langle -1, 0 \rangle$? ■

Solution: We illustrate the velocity vectors for the airplane and the wind in Figure 1.11. We let the airplane's velocity vector be $\mathbf{v} = \langle x, y \rangle$. The effective velocity of the plane is then $\mathbf{v} + \mathbf{w}$, which we set equal to $\langle c, 0 \rangle$, for some negative constant c . Since

$$\mathbf{v} + \mathbf{w} = \langle x + 20, y + 30 \rangle = \langle c, 0 \rangle$$

we must have $x + 20 = c$ and $y + 30 = 0$, so that $y = -30$. Further, since the plane's airspeed is 400 mph we must have $400 = \|\mathbf{v}\| = \sqrt{x^2 + y^2} = \sqrt{x^2 + 900}$. Squaring this gives us $x^2 + 900 = 160,000$, so that $x = -159,100$. (We take the negative square root so that the plane heads westward.) Consequently, the plane should head in the direction of $\mathbf{v} = \langle -159,100, -30 \rangle$ which points left and down, or southwest, at an angle of $\tan^{-1}(30/159,100) \approx 4^\circ$ below due west.

■ **Example 1.9** A body weighing 100 kg is suspended from two ropes as shown in figure 1.12. Find the tension in the ropes. ■

Solution: Let the tension in the ropes be denoted by $F_1 = a_1\mathbf{i} + a_2\mathbf{j}$ and $F_2 = b_1\mathbf{i} + b_2\mathbf{j}$. The weight of the body is represented by the vector $\mathbf{W} = -100\mathbf{j}$. Since the system is in equilibrium the sum of all the forces must be equal to zero. Thus,

$$F_1 + F_2 + \mathbf{W} = (a_1 + b_1)\mathbf{i} + (a_2 + b_2 - 100)\mathbf{j} = \mathbf{0}. \text{ This implies}$$

$$a_1 + b_1 = 0, \quad a_2 + b_2 - 100 = 0$$

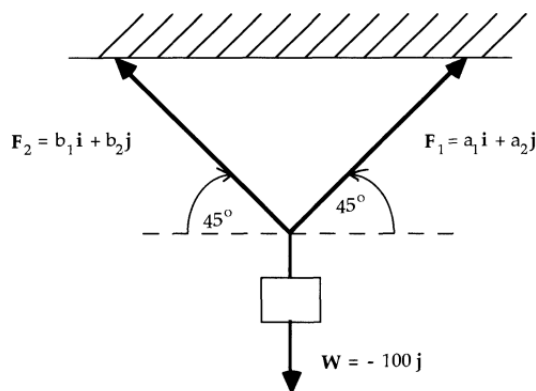


Figure 1.12: Tension in ropes from which a weight is suspended.

Hence, $a_1 = -b_1$ and $a_2 + b_2 = 100$. By symmetry, we know that $\|F_1\| = \|F_2\|$, which implies that $a_2 = b_2$. Therefore, $a_2 = b_2 = 50$. Since $a_1 = \|F_1\| \cos 45$ and $a_2 = \|F_2\| \sin 45$, it follows that $\|F_1\| = \|F_2\| = 50\sqrt{2}$ and $a_1 = -b_1 = 50$. Thus, $F_1 = 50(i + j)$ and $F_2 = 50(-i + j)$. Notice that $F_1 + F_2 = 100j$

1.3 Vectors in space

A vector in 3-space is an ordered triple of real numbers

$$\mathbf{a} = \langle a_1, a_2, a_3 \rangle,$$

where a_1, a_2 , and a_3 are the components of the vector. In particular, the position vector of a point $P(x_1, y_1, z_1)$ is the vector $\overrightarrow{OP_1}$. We represent the vector $\langle a_1, a_2, a_3 \rangle$ as an arrow from the origin $(0, 0, 0)$ to the point (a_1, a_2, a_3) in 3-space. In this way, the direction indicated by the arrow, as viewed from the origin, gives the direction of the vector.

Definition 1.3.1 If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ are vectors in 3-space and c is a scalar, then

1. $\mathbf{a} = \mathbf{b}$ if and only if $a_1 = b_1, a_2 = b_2, a_3 = b_3$
2. $\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$
3. $c\mathbf{a} = \langle ca_1, ca_2, ca_3 \rangle$
4. $\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ (Length or norm of a vector)

The vector with initial point $P_1(x_1, y_1, z_1)$ and terminal point $P_2(x_2, y_2, z_2)$ is

$$\overrightarrow{P_1P_2} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Thus, we can find the components of a vector by subtracting the respective coordinates of its initial point from the coordinates of its terminal point, as illustrated in Figure 1.13. The vectors $\overrightarrow{OP_1}$ and $\overrightarrow{OP_2}$ are the position vectors of the point $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$. As a natural extension of the definition of vector subtraction in 2-space into 3-space, we have

$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1} = \langle x_2, y_2, z_2 \rangle - \langle x_1, y_1, z_1 \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

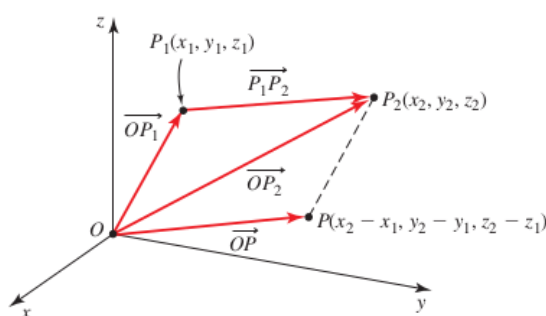


Figure 1.13:

By considering the parallelogram OPP_2P_1 in Figure 1.13, you can convince yourself that $\overrightarrow{P_1P_2}$ is represented by the position vector $\overrightarrow{OP}$ of the point $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$.

■ **Example 1.10** Let $P(2, -1, 2)$ and $Q(1, 4, 5)$ be two points in 3-space.

- (a) Find the vector $\overrightarrow{PQ}$.
- (b) Find $\|\overrightarrow{PQ}\|$.
- (c) Find a unit vector having the same direction as $\overrightarrow{PQ}$

■

Solution:

$$(a) \vec{PQ} = \langle 1 - 2, 4 - (-1), 5 - 2 \rangle = \langle -1, 5, 3 \rangle$$

$$(b) \|\vec{PQ}\| = \sqrt{(-1)^2 + 5^2 + 3^2} = \sqrt{35}$$

(c) Using the results of parts (a) and (b), we obtain the unit vector

$$\mathbf{u} = \frac{\vec{PQ}}{\|\vec{PQ}\|} = \frac{\langle -1, 5, 3 \rangle}{\sqrt{35}} = \left\langle -\frac{1}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{3}{\sqrt{35}} \right\rangle$$

Standard Basis Vectors in Space

In three-dimensional space, the 3-space vectors

$$\mathbf{i} = \langle 1, 0, 0 \rangle, \quad \mathbf{j} = \langle 0, 1, 0 \rangle \quad \text{and} \quad \mathbf{k} = \langle 0, 0, 1 \rangle$$

form a basis for the space, in the sense that any vector in the space can be expressed in terms of these vectors. If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ be a vector in space. Then

$$\begin{aligned} \mathbf{a} &= \langle a_1, a_2, a_3 \rangle = \langle a_1, 0, 0 \rangle + \langle 0, a_2, 0 \rangle + \langle 0, 0, a_3 \rangle \\ &= a_1 \langle 1, 0, 0 \rangle + a_2 \langle 0, 1, 0 \rangle + a_3 \langle 0, 0, 1 \rangle \\ &= a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k} \end{aligned}$$

1.4 The Scalar (Dot) Product

Definition 1.4.1 Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ be any two vector in space. Then the dot product of $\mathbf{a}$ and $\mathbf{b}$ is the number $\mathbf{a} \cdot \mathbf{b}$ defined by

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

■ **Example 1.11** Find the dot product of each pair of vectors:

1. $\mathbf{v} = \langle 2, -3 \rangle$ and $\mathbf{u} = \langle 3, 1 \rangle$
2. $\mathbf{v} = \langle 1, -2, 4 \rangle$ and $\mathbf{u} = \langle 1, 1, 0 \rangle$

Properties of the Dot Product

Let $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ be vectors in 2- or 3-space and let c be a scalar. Then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
3. $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$
4. $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$
5. $\mathbf{0} \cdot \mathbf{u} = \mathbf{0}$

The Angle Between Two Vectors

The **angle between two nonzero vectors** is the angle θ between their corresponding position vectors, where $0 \leq \theta \leq \pi$.



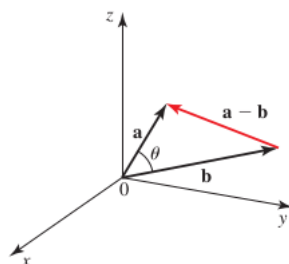
If two vectors are parallel, then $\theta = 0$ or $\theta = \pi$
The angle between the zero vector and another vector is not defined.

Theorem 1.4.1 Let θ be the angle between two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$. Then

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

Proof. Consider the triangle determined by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ as shown in Figure 1.14. Using the law of cosines, we have

$$\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos \theta$$

Figure 1.14: The angle between $\mathbf{a}$ and $\mathbf{b}$ is θ .

But

$$\begin{aligned}\|\mathbf{a} - \mathbf{b}\|^2 &= (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b} \\ &= \|\mathbf{a}\|^2 - 2\mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2\end{aligned}$$

so we have

$$\begin{aligned}\|\mathbf{a}\|^2 - 2\mathbf{a} \cdot \mathbf{b} + \|\mathbf{b}\|^2 &= \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta \\ \implies -2\mathbf{a} \cdot \mathbf{b} &= -2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta \\ \implies \cos\theta &= \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}\end{aligned}$$

■

■ **Example 1.12** If the vectors $\mathbf{a}$ and $\mathbf{b}$ have lengths 4 and 6, and the angle between them is $\frac{\pi}{3}$, find $\mathbf{a} \cdot \mathbf{b}$. ■

Solution: We have

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta = 4 \cdot 6 \cdot \cos\frac{\pi}{3} = 24 \cdot \frac{1}{2} = 12$$

■ **Example 1.13** Find the angle between the vectors $\mathbf{a} = \langle 2, -1, -2 \rangle$ and $\mathbf{b} = \langle 1, 2, 2 \rangle$ ■

Solution: We have

$$\|\mathbf{a}\| = \sqrt{2^2 + (-1)^2 + (-2)^2} = 3, \quad \|\mathbf{b}\| = \sqrt{1^2 + 2^2 + 2^2} = 3 \text{ and } \mathbf{a} \cdot \mathbf{b} = 2(1) + (-1)(2) + (-2)(2) = -4$$

$$\text{Hence, } \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} = \frac{-4}{9} \implies \theta = \cos^{-1}\left(\frac{-4}{9}\right) =$$

Theorem 1.4.2 Two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.

■ **Example 1.14** Determine whether the following pairs of vectors are orthogonal:

- (a) $\mathbf{a} = -i + 2j + 5k$, $\mathbf{b} = 3i + 4j - k$
 (b) $\mathbf{u} = i - j + 2k$, $\mathbf{v} = 2i - j + k$

■

Solution:

- (a) Since $\mathbf{a} \cdot \mathbf{b} = (-1)(3) + 2(4) + 5(-1) = 0$. So, $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.
 (b) Since $\mathbf{u} \cdot \mathbf{v} = 1(2) + (-1)(-1) + 2(1) = 5 \neq 0$. So, $\mathbf{u}$ and $\mathbf{v}$ are not orthogonal.

Theorem 1.4.3 — Cauchy-Schwartz Inequality. For any vectors $\mathbf{a}$ and $\mathbf{b}$,

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

Theorem 1.4.4 — The Triangle Inequality. For any vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$$

Direction Cosines

We can describe the direction of a nonzero vector $\mathbf{a}$ by giving the angles α , β , and γ that $\mathbf{a}$ makes with the positive x -, y -, and z -axes, respectively. (See Figure 1.15.) These angles are called the direction angles of $\mathbf{a}$. The cosines of these angles, $\cos \alpha$, $\cos \beta$, and $\cos \gamma$, are called the direction cosines of the vector $\mathbf{a}$.

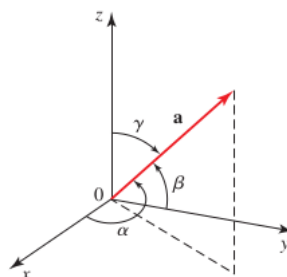


Figure 1.15: The direction angles of a vector

Let $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$ be a nonzero vector in 3-space. Then

$$\mathbf{a} \cdot \mathbf{i} = (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \cdot \mathbf{i} = a_1$$

So,

$$\cos \alpha = \frac{\mathbf{a} \cdot \mathbf{i}}{\|\mathbf{a}\| \|\mathbf{i}\|} = \frac{a_1}{\|\mathbf{a}\|}$$

Similarly,

$$\cos \beta = \frac{\mathbf{a} \cdot \mathbf{j}}{\|\mathbf{a}\| \|\mathbf{j}\|} = \frac{a_2}{\|\mathbf{a}\|}, \text{ and } \cos \gamma = \frac{\mathbf{a} \cdot \mathbf{k}}{\|\mathbf{a}\| \|\mathbf{k}\|} = \frac{a_3}{\|\mathbf{a}\|}$$

By squaring and adding the three direction cosines, we obtain

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{a_1^2}{\|\mathbf{a}\|^2} + \frac{a_2^2}{\|\mathbf{a}\|^2} + \frac{a_3^2}{\|\mathbf{a}\|^2} = \frac{\|\mathbf{a}\|^2}{\|\mathbf{a}\|^2} = 1$$

R

1. If $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}$ is nonzero, then the unit vector having the same direction as $\mathbf{a}$ is

$$\begin{aligned} \mathbf{u} &= \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{a_1}{\|\mathbf{a}\|} \mathbf{i} + \frac{a_2}{\|\mathbf{a}\|} \mathbf{j} + \frac{a_3}{\|\mathbf{a}\|} \mathbf{k} \\ &= (\cos \alpha) \mathbf{i} + (\cos \beta) \mathbf{j} + (\cos \gamma) \mathbf{k} \end{aligned} \quad (1.3)$$

This shows that the direction cosines of $\mathbf{a}$ are the components of the unit vector in the direction of $\mathbf{a}$.

2. From Equation (1.3) we see that

$$\mathbf{a} = \|\mathbf{a}\| [(\cos \alpha) \mathbf{i} + (\cos \beta) \mathbf{j} + (\cos \gamma) \mathbf{k}]$$

Example 1.15 Find the direction angles of the vector $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$

Solution: We have $\|\mathbf{a}\| = \sqrt{2^2 + 1^2 + 2^2} = 3$. So,

$$\cos \alpha = \frac{2}{3}, \cos \beta = \frac{1}{3}, \text{ and } \cos \gamma = \frac{2}{3}$$

Therefore,

$$\alpha = \cos^{-1} \left(\frac{2}{3} \right) \approx 48.19^\circ, \beta = \cos^{-1} \left(\frac{1}{3} \right) \approx 70.53^\circ, \text{ and } \gamma = \cos^{-1} \left(\frac{2}{3} \right) \approx 48.19^\circ$$

1.5 Vector Projections and Components

Figure 1.16a depicts a child pulling a sled with a constant force represented by the vector $\mathbf{F}$. The force $\mathbf{F}$ can be expressed as the sum of two forces: a horizontal component $\mathbf{F}_1$ and a vertical component $\mathbf{F}_2$, as shown in Figure 1.16b.



Figure 1.16:

Observe that $\mathbf{F}_1$ acts in the direction of motion, whereas $\mathbf{F}_2$ acts in a direction perpendicular to the direction of motion.

More generally, we are interested in the component of one vector $\mathbf{b}$ in the direction of another nonzero vector $\mathbf{a}$. The vector that is obtained by projecting $\mathbf{b}$ onto the line containing the vector $\mathbf{a}$ is called the vector projection of $\mathbf{b}$ onto $\mathbf{a}$ (also called the vector component of $\mathbf{b}$ along $\mathbf{a}$) and denoted by $\text{proj}_{\mathbf{a}}^{\mathbf{b}}$

(See Figure 1.17.) The scalar projection of $\mathbf{b}$ onto $\mathbf{a}$ (also called the scalar component of $\mathbf{b}$ along $\mathbf{a}$) is

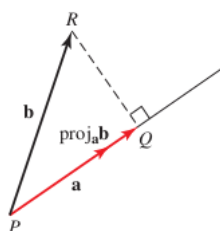


Figure 1.17: $\text{proj}_{\mathbf{a}}^{\mathbf{b}}$: ($\vec{PQ}$) is the vector projection of $\mathbf{b}$ onto $\mathbf{a}$.

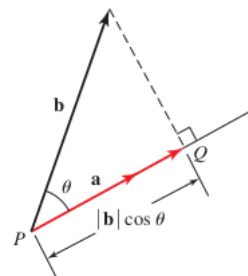


Figure 1.18: $\text{comp}_{\mathbf{a}}^{\mathbf{b}} = \|\mathbf{b}\| \cos \theta$.

the length of $\text{proj}_{\mathbf{a}}^{\mathbf{b}}$ if the projection has the same direction as $\mathbf{a}$ and the negative of the length of $\text{proj}_{\mathbf{a}}^{\mathbf{b}}$ if the projection has the opposite direction. We denote this scalar projection by

$$\text{comp}_{\mathbf{a}}^{\mathbf{b}}$$

Since

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

we can write

$$\|\mathbf{b}\| \cos \theta = \frac{|\mathbf{b}|(\mathbf{a} \cdot \mathbf{b})}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}$$

Therefore, the component of $\mathbf{b}$ along $\mathbf{a}$ is

$$\text{comp}_{\mathbf{a}}^{\mathbf{b}} = \frac{\mathbf{b} \cdot \mathbf{a}}{\|\mathbf{a}\|}$$

The vector projection of $\mathbf{b}$ along $\mathbf{a}$ is the component of $\mathbf{b}$ along $\mathbf{a}$ times the direction of $\mathbf{a}$.

$$\text{proj}_{\mathbf{a}}^{\mathbf{b}} = \left(\frac{\mathbf{b} \cdot \mathbf{a}}{\|\mathbf{a}\|} \right) \frac{\mathbf{a}}{\|\mathbf{a}\|} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a}$$

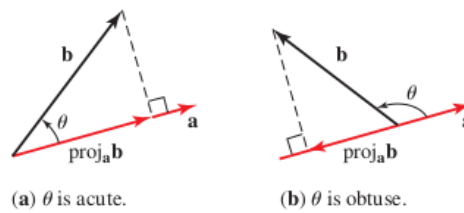


Figure 1.19: $\text{proj}_{\mathbf{a}}^{\mathbf{b}}$ points in the same direction as $\mathbf{a}$ if θ is acute and points in the opposite direction as $\mathbf{a}$ if θ is obtuse.

■ **Example 1.16** Let $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$. Find
 (a) the component of $\mathbf{b}$ along $\mathbf{a}$ (b) $\text{proj}_{\mathbf{a}}^{\mathbf{b}}$ (c) $\text{proj}_{\mathbf{b}}^{\mathbf{a}}$ ■

Solution: (a) The scalar component of $\mathbf{b}$ along $\mathbf{a}$ is

$$\text{comp}_{\mathbf{a}}^{\mathbf{b}} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} = \frac{2(3) + 3(-2) + (-4)(1)}{\sqrt{2^2 + 3^2 + (-4)^2}} = -\frac{4}{\sqrt{29}}$$

(b) The projection of vector $\mathbf{b}$ along $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}}^{\mathbf{b}} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} = -\frac{4}{29} (2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}) = -\frac{8}{29}\mathbf{i} - \frac{12}{29}\mathbf{j} + \frac{16}{29}\mathbf{k}$$

(c) The projection of vector $\mathbf{a}$ along $\mathbf{b}$ is

$$\text{proj}_{\mathbf{b}}^{\mathbf{a}} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \right) \mathbf{b} = -\frac{4}{3^2 + (-2)^2 + 1^2} (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = -\frac{6}{7}\mathbf{i} + \frac{4}{7}\mathbf{j} - \frac{2}{7}\mathbf{k}$$

Using vector projections, we can express any vector $\mathbf{b}$ as the sum of a vector parallel to a vector $\mathbf{a}$ and a vector perpendicular to $\mathbf{a}$. In fact, from Figure 1.20 we see that.

$$\begin{aligned} \mathbf{b} &= \text{proj}_{\mathbf{a}}^{\mathbf{b}} + (\mathbf{b} - \text{proj}_{\mathbf{a}}^{\mathbf{b}}) \\ &= \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} + \left[\mathbf{b} - \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} \right] \end{aligned}$$

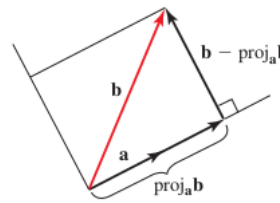


Figure 1.20: The direction angles of a vector

■ **Example 1.17** Write $\mathbf{b} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ as the sum of a vector parallel to $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and a vector perpendicular to $\mathbf{a}$. ■

Solution: We have

$$\mathbf{a} \cdot \mathbf{b} = 2(3) + (-1)(-1) + 1(2) = 9 \text{ and } \|\mathbf{a}\|^2 = \mathbf{a} \cdot \mathbf{a} = 2^2 + (-1)^2 + 1^2 = 6$$

Hence,

$$\begin{aligned} \mathbf{b} &= \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} + \left[\mathbf{b} - \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} \right] \\ &= \frac{9}{6} (2\mathbf{i} - \mathbf{j} + \mathbf{k}) + \left[(3\mathbf{i} - \mathbf{j} + 2\mathbf{k}) - \frac{9}{6} (2\mathbf{i} - \mathbf{j} + \mathbf{k}) \right] \\ &= \left(3\mathbf{i} - \frac{3}{2}\mathbf{j} + \frac{3}{2}\mathbf{k} \right) + \left(\frac{1}{2}\mathbf{j} + \frac{1}{2}\mathbf{k} \right) \end{aligned}$$

Therefore, the vector $\left(3\mathbf{i} - \frac{3}{2}\mathbf{j} + \frac{3}{2}\mathbf{k} \right)$ is parallel to $\mathbf{a}$ and the vector $\left(\frac{1}{2}\mathbf{j} + \frac{1}{2}\mathbf{k} \right)$ is perpendicular to $\mathbf{a}$.

Work

One application of vector projections lies in the computation of the work done by a force.

If a constant force $\mathbf{F}$ moves an object from point P to point Q , we refer to the vector $\mathbf{d} = \overrightarrow{PQ}$ as the displacement vector. The work done is the product of the component of $\mathbf{F}$ along $\mathbf{d}$ and the distance:

$$\begin{aligned} W &= \text{comp}_{\mathbf{d}}^{\mathbf{F}} \|\mathbf{d}\| \\ &= \frac{\mathbf{F} \cdot \mathbf{d}}{\|\mathbf{d}\|} \|\mathbf{d}\| = \mathbf{F} \cdot \mathbf{d} \end{aligned}$$

■ **Example 1.18** A force $\mathbf{F} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ moves a particle along the line segment from the point $P(1, 2, 1)$ to the point $Q(3, 6, 5)$. Find the work done by the force if $\|\mathbf{F}\|$ is measured in newtons and $\|\mathbf{d}\|$ is measured in meters. ■

solution: The displacement vector is $\|\mathbf{d}\| = \overrightarrow{PQ} = 2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$. Therefore, the work done by $\mathbf{F}$ is
 $W = \mathbf{F} \cdot \mathbf{d} = 32$ joules.

■ **Example 1.19** A wagon is pulled a distance of 100 m along a horizontal path by a constant force of 70 N. The handle of the wagon is held at an angle of 30° above the horizontal. Find the work done by the force. ■

Solution: $W = \mathbf{F} \cdot \mathbf{d} = \|\mathbf{F}\| \|\mathbf{d}\| \cos \theta = 70(100)\left(\frac{\sqrt{3}}{2}\right) = 3500\sqrt{3} = 6062.18 \text{ J}$

Exercise 1.1 1. Determine whether the following vectors in $\mathbb{R}^3$ are parallel or not

(a) $\vec{A} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{B} = -3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

(b) $\vec{A} = 2\mathbf{i} + 4\mathbf{j}$ and $\vec{B} = -\mathbf{i} - 2\mathbf{j} + \frac{1}{2}\mathbf{k}$

2. Find the value(s) of x such that the vector $\vec{A} = (1, 4, 3)$ and $\vec{B} = (x, -1, 2)$ are orthogonal.

3. Find the angle between $\vec{A} = \langle -1, -1, 1 \rangle$ and $\vec{B} = \langle \sqrt{6}, 1, 1 \rangle$

4. Find any unit vectors that are perpendicular to the vector $\vec{A} = \langle 3, 4 \rangle$

5. If the angle between the vectors $\vec{A}$ and $\vec{B}$ is $\theta = \frac{\pi}{6}$ with each other and $\|\vec{A}\| = \sqrt{3}$ and $\|\vec{B}\| = 1$, then calculate the cosine the angle between the vectors $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$.

6. Find the projection of $\vec{A} = \langle -1, 3, 1 \rangle$ on to $\vec{B} = \langle 2, 4, 3 \rangle$

7. Suppose $\vec{a}$ and $\vec{b}$ are orthogonal, show that $\text{proj}_{\vec{b}}^{\vec{a}} = 0$.

8. Suppose $\vec{a}$ and $\vec{b}$ are parallel, show that $\text{proj}_{\vec{a}}^{\vec{b}} = \vec{b}$.

9. Express the vector $\vec{b} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ as the sum of a vector parallel to $\vec{a} = 3\mathbf{i} - \mathbf{j}$ and a vector orthogonal to $\vec{a}$

10. Find the work done by a force $\mathbf{F} = 5\mathbf{k}$ in moving an object along the line from the origin to the point $P(1, 1, 1)$ distance in meters. ■

1.6 The Cross Product

Definition 1.6.1 Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ be any two vectors in space. Then the **cross product** of $\mathbf{a}$ and $\mathbf{b}$ is the vector

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$$

■ **Example 1.20** Let $\mathbf{a} = \langle 1, 3, 2 \rangle$, and $\mathbf{b} = \langle 2, -1, -1 \rangle$. Find $\mathbf{a} \times \mathbf{b}$ ■

■ **Example 1.21** Show that $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ ■

Theorem 1.6.1 The vector $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

Proof. Exercise ■

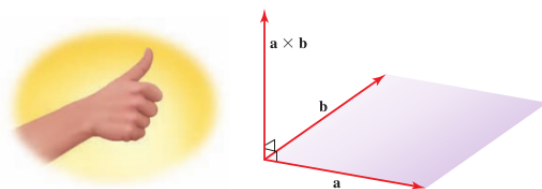


Figure 1.21: The vector $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$ with direction determined by the right-hand rule.

Theorem 1.6.2 Let $\mathbf{a}$ and $\mathbf{b}$ be vectors in space. Then

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ and $0 \leq \theta \leq \pi$

■ **Example 1.22** Let $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 2\mathbf{j} + \mathbf{k}$. Find a unit vector $\mathbf{n}$ that is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$. ■

Solution: A vector that is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 0 \\ 0 & 2 & 1 \end{vmatrix} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$$

The length of $\mathbf{a} \times \mathbf{b}$ is

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{3^2 + (-2)^2 + 4^2} = \sqrt{29}$$

Therefore, a unit vector that is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{n} = \frac{\mathbf{a} \times \mathbf{b}}{\|\mathbf{a} \times \mathbf{b}\|} = \frac{3}{\sqrt{29}}\mathbf{i} - \frac{2}{\sqrt{29}}\mathbf{j} + \frac{4}{\sqrt{29}}\mathbf{k}$$

Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$

1.6.1 Finding the Area of a parallelogram

Consider the parallelogram determined by the vectors $\mathbf{a}$ and $\mathbf{b}$ shown in Figure 1.22a. The altitude of the parallelogram is $\|\mathbf{b}\| \sin \theta$ and the length of its base is $\|\mathbf{a}\|$, so its area is

$$A = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta = \|\mathbf{a} \times \mathbf{b}\|$$

Thus, the length of the cross product $\mathbf{a} \times \mathbf{b}$ and the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$ have the same numerical value. (See Figure 1.22b.)

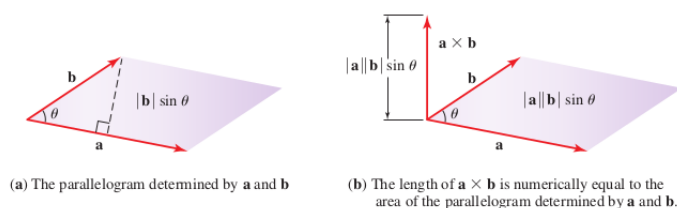


Figure 1.22:

R The area of the triangle determined by $\mathbf{a}$ and $\mathbf{b}$ is $\frac{1}{2} \|\mathbf{a} \times \mathbf{b}\|$

■ **Example 1.23** Find the area of the triangle with vertices $P(3, -3, 0)$, $Q(1, 2, 2)$, and $R(1, -2, 5)$.

Solution: The area of $\triangle PQR$ is half the area of the parallelogram determined by the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. Now $\overrightarrow{PQ} = \langle -2, 5, 2 \rangle$ and $\overrightarrow{PR} = \langle -2, 1, 5 \rangle$ so

$$\overrightarrow{PQ} \times \overrightarrow{PR} = 23\mathbf{i} + 6\mathbf{j} + 8\mathbf{k}$$

Therefore, the area of the parallelogram is

$$\|\overrightarrow{PQ} \times \overrightarrow{PR}\| = \sqrt{23^2 + 6^2 + 8^2} = \sqrt{629} \approx 25.1$$

so the area of the required triangle is $\frac{1}{2} \sqrt{629}$ or approximately 12.5.

Exercise 1.2 Let $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$. Find the area of the triangle formed by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{a} - \mathbf{b}$. Ans $2\sqrt{17}$ square units. ■

Properties of the Cross Product

If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors and c is a scalar, then

- | | |
|--|---|
| 1. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ | 5. $\mathbf{a} \times \mathbf{0} = \mathbf{0} \times \mathbf{a} = \mathbf{0}$ |
| 2. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$ | 6. $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ |
| 3. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$ | 7. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ |
| 4. $c(\mathbf{a} \times \mathbf{b}) = (c\mathbf{a}) \times \mathbf{b} = \mathbf{a} \times (c\mathbf{b})$ | 8. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ |

The Scalar Triple Product

If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in three-dimensional space. The dot product of $\mathbf{a}$ and $\mathbf{b} \times \mathbf{c}$, $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$, is called the scalar triple product. If we write $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, then by direct computation,

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

The geometric significance of the scalar triple product can be seen by examining the parallelepiped determined by the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. (See Figure 1.23.) The base of the parallelepiped is a parallelogram with adjacent sides determined by $\mathbf{b}$, and $\mathbf{c}$ with area $\|\mathbf{b} \times \mathbf{c}\|$. If θ is the angle between $\mathbf{a}$ and $\mathbf{b} \times \mathbf{c}$, then the height of the parallelepiped is given by $h = \|\mathbf{a}\| \cos \theta$. Therefore, the volume

of the parallelepiped is

$$\begin{aligned} V &= \|\mathbf{b} \times \mathbf{c}\| \|\mathbf{a}\| \cos \theta && \text{(area of base . height)} \\ &= |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| \end{aligned}$$

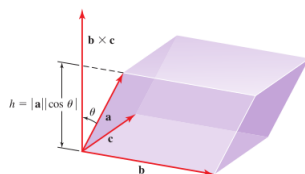


Figure 1.23: The volume V of the parallelepiped is equal to $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$

■ **Example 1.24** Find the volume of the parallelepiped with three adjacent edges formed by the vectors $\mathbf{a} = i + 2j + 3k$, $\mathbf{b} = 4i + 5j + 6k$ and $\mathbf{c} = 7i + 8j$ ■

Solution: The volume of the parallelepiped is

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{vmatrix} = |1(0 - 48) - 2(0 - 42) + 3(32 - 35)| = 27$$

Test for Coplanar Vectors

The vector $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are coplanar if and only if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$

■ **Example 1.25** Show that the vectors $\mathbf{a} = \langle 1, 4, -7 \rangle$, $\mathbf{b} = \langle 2, -1, 4 \rangle$, and $\mathbf{c} = \langle 0, -9, 18 \rangle$ are coplanar. ■

Solution:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix} = |1(-1(18) - (-9)4) - 4(2(18) - 0) + -7(2(-9) - 0)| = 0$$

Torque

The idea of a cross product occurs often in physics. In particular, we consider a force $\mathbf{F}$ acting on a rigid body at a point given by a position vector $\mathbf{r}$. (For instance, if we tighten a bolt by applying a force to a wrench as in Figure 1.24, we produce a turning effect.) The torque (relative to the origin) is defined to be the cross product of the position and force vectors

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$$

and measures the tendency of the body to rotate about the origin. The direction of the torque vector indicates the axis of rotation. According to Theorem 1.6.2, the magnitude of the torque vector is

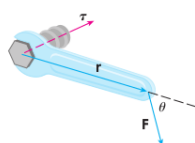


Figure 1.24:

$$\|\tau\| = \|\mathbf{r} \times \mathbf{F}\| = \|\mathbf{r}\| \|\mathbf{F}\| \sin \theta$$

where θ is the angle between the position and force vectors. Observe that the only component of $\mathbf{F}$ that can cause a rotation is the one perpendicular to $\mathbf{r}$, that is, $\|\mathbf{F}\| \sin \theta$. The magnitude of the torque is equal to the area of the parallelogram determined by $\mathbf{r}$ and $\mathbf{F}$.

■ **Example 1.26** A bolt is tightened by applying a 40-N force to a 0.25-m wrench as shown in Figure 1.25. Find the magnitude of the torque about the center of the bolt. ■

Solution:

The magnitude of the torque vector is

$$\begin{aligned} \tau &= \mathbf{r} \times \mathbf{F} \\ &= (0.25)(40) \sin 75^\circ = 10 \sin 75^\circ \end{aligned}$$

If the bolt is right-threaded, then the torque vector itself is

$$\tau = \|\tau\| \mathbf{n}$$

where $\mathbf{n}$ is a unit vector directed down into the

page.

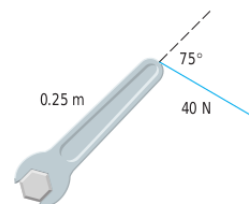


Figure 1.25:

Exercise 1.3

1. Find the area of a triangle whose vertices are $A = (1, -1, 0)$, $B = (2, 1, -1)$, and $C = (-1, 1, 2)$
2. Find the area of a parallelogram having diagonals $\vec{d} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\vec{b} = \mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$.
3. Find the volume of the parallelepiped with edges $\vec{u} = \mathbf{i} + \mathbf{k}$, $\vec{v} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$ and $\mathbf{j} + \mathbf{k}$.
4. Find a unit vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$ and $R(-1, 1, 2)$.
5. Show that the vectors $\mathbf{a} = \langle 1, 5, -2 \rangle$, $\mathbf{b} = \langle 3, -1, 0 \rangle$, and $\mathbf{c} = \langle 5, 9, -4 \rangle$ are coplanar. ■

1.7 Lines and Planes in Space

Definition 1.7.1 A vector $\mathbf{v} = \langle a, b, c \rangle$ is said to be parallel to a line L if $\mathbf{v}$ is parallel to $\overrightarrow{P_0P}$ for any two points p_0 and p_1 on L .

A line L in $\mathbb{R}^3$ is determined by a given point $p_0(x_0, y_0, z_0)$ on L and a parallel vector $\mathbf{v} = \langle a, b, c \rangle$ (directional vector) to L .

1.7.1 Equation of a line in space

Suppose that the line L passes through the point $P_0(x_0, y_0, z_0)$ and has the same direction as the vector $\mathbf{v} = \langle a, b, c \rangle$.

Let $P(x, y, z)$ be any point on L . Then the vector $\overrightarrow{P_0P}$ is parallel to $\mathbf{v}$. But two vectors are parallel if and only if one is a scalar multiple of the other. Therefore, there exists some number t , called a parameter, such that

$$\overrightarrow{P_0P} = t\mathbf{v}$$

or since, $\overrightarrow{P_0P} = \langle x - x_0, y - y_0, z - z_0 \rangle = t \langle a, b, c \rangle = \langle ta, tb, tc \rangle$ we have

$$\langle x - x_0, y - y_0, z - z_0 \rangle$$

Equating the corresponding components of the two vectors then yields

$$x - x_0 = ta, y - y_0 = tb, z - z_0 = tc$$

Solving these equations for x, y , and z , respectively, gives the following standard parametric equations of the line L .

Definition 1.7.2 — Parametric Equations of a Line. The parametric equations of the line passing through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $\mathbf{v} = \langle a, b, c \rangle$ are

$$\langle x = x_0 + ta, y = y_0 + tb, z = z_0 + tc \rangle$$

■ **Example 1.27** Find parametric equations for the line passing through the point $P_0(2, 1, 3)$ and parallel to the vector $\mathbf{v} = \langle 1, 2, 2 \rangle$. ■

Solution: $x = -2 + t$, $y = 1 + 2t$, and $z = 3 - 2t$

Definition 1.7.3 — Symmetric Equations of a Line. The symmetric equations of the line L passing through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

R Suppose $a = 0$ and both b and c are not equal to zero, then the parametric equations of the line take the form

$$x = x_0, y = y_0 + tb, z = z_0 + tc$$

and the line lies in the plane $x = x_0$ (parallel to the yz -plane). Solving the second and third equations for t leads to

$$x = x_0, \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

which are the symmetric equations of the line.

■ **Example 1.28**

- Find parametric equations and symmetric equations for the line L passing through the points $P(-3, 3, -2)$ and $Q(2, -1, 4)$.
- At what point does L intersect the xy -plane?

Solution:

- The direction of L is the same as that of the vector $\overrightarrow{PQ} = \langle 5, -4, 6 \rangle$. Since L passes through $P(-3, 3, -2)$, the parametric equations of the line is

$$x = -3 + 5t, y = 3 - 4t, \text{ and } z = -2 + 6t$$

The symmetric equations for L is:

$$\frac{x + 3}{5} = \frac{y - 3}{-4} = \frac{z + 2}{6}$$

- At the point where the line intersects the xy -plane, we have $z = 0$. So setting $z = 0$ in the third parametric equation, we obtain $t = \frac{1}{3}$. Substituting this value of t into the other parametric equations gives the required point as $\left(-\frac{4}{3}, \frac{5}{3}, 0\right)$.

■ **Example 1.29** Find the parametric equations of the line that pass through the point $(-1, 3, -2)$ and is perpendicular to the vectors $\vec{A} = 3\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ and $\vec{B} = \mathbf{i} + 2\mathbf{j}$ ■

Solution: Since the line is perpendicular to the vectors $\vec{A}$ and $\vec{B}$, it is parallel to the vector product of $\vec{A}$ and $\vec{B}$. Since

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 1 \\ 1 & 2 & 0 \end{vmatrix} = -2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$$

it follows that the parametric equations of the line are

$$x = -1 - 2t, y = 3 + t, z = -2 + 2t$$

Definition 1.7.4 Let l_1 and l_2 be two lines in $\mathbb{R}^3$, with parallel vectors $\mathbf{a}$ and $\mathbf{b}$, respectively, and let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$.

- (a) The lines l_1 and l_2 are **parallel** whenever $\mathbf{a}$ and $\mathbf{b}$ are parallel.
- (b) If l_1 and l_2 intersect, then
 - i. the angle between l_1 and l_2 is θ and
 - ii. the lines l_1 and l_2 are orthogonal whenever $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

■ **Example 1.30** Let L_1 be the line with parametric equations

$$x = 1 + 2t, y = 2 - 3t, \text{ and } z = 2 + t$$

and L_2 be the line with parametric equations

$$x = 3 - 4t, y = 1 + 4t, \text{ and } z = -3 + 4t$$

- (a) Show that the lines L_1 and L_2 are not parallel to each other.
- (b) Do the lines L_1 and L_2 intersect? If so, find their point of intersection.

Solution:

- (a) L_1 has the same direction as the vector $\vec{v}_1 = \langle 2, -3, 1 \rangle$. Similarly, L_2 has direction given by the vector $\vec{v}_2 = \langle -4, 4, 4 \rangle = 4 \langle -1, 1, 1 \rangle$. Since $\vec{v}_1$ is not a scalar multiple of $\vec{v}_2$, the vectors are not parallel, so L_1 and L_2 are not parallel as well.
- (b) Suppose that L_1 and L_2 intersect at the point $P_0(x_0, y_0, z_0)$. Then there must exist parameter values t_1 and t_2 such that

$$x_0 = 1 + 2t_1, y_0 = 2 - 3t_1, \text{ and } z_0 = 2 + t_1$$

and

$$x_0 = 3 - 4t_2, y_0 = 1 + 4t_2, \text{ and } z_0 = -3 + 4t_2$$

This leads to the system of three linear equations

$$1 + 2t_1 = 3 - 4t_2$$

$$2 - 3t_1 = 1 + 4t_2$$

$$2 + t_1 = -3 + 4t_2$$

Adding the first two equations gives $t_1 = -1$. Substituting this value of t_1 into either the first or the second equation then gives $t_2 = 1$. Finally, substituting these values of t_1 and t_2 into the third equation gives $2 - 1 = -3 + 4(1) = 1$, which shows that the third equation is also satisfied by these values. We conclude that L_1 and L_2 do indeed intersect at a point. To find the point of intersection, substitute $t_1 = -1$ into the parametric equations defining L_1 , or, equivalently substitute $t_2 = 1$ into the parametric equations defining L_2 . In both cases we find that $x_0 = -1, y_0 = 5$, and $z_0 = 1$, so the point of intersection is $(-1, 5, 1)$.

Definition 1.7.5 Two lines in space are said to be **skew** if they do not intersect and are not parallel.

■ **Example 1.31** As two planes fly by each other, their flight paths are given by the straight lines

$$L_1 : x = 1 - t, y = -2 - 3t, \text{ and } z = 4 + t$$

and

$$x = 2 - 2t, y = -4 + 3t, \text{ and } z = 1 + 4t$$

Show that the lines are skew and, therefore, that there is no danger of the planes colliding. ■

Solution: The directions of L_1 and L_2 are given by the directions of the vectors $\mathbf{v}_1 = \langle -1, -3, 1 \rangle$ and $\mathbf{v}_2 = \langle -2, 3, 4 \rangle$, respectively. Since one vector is not a scalar multiple of the other, the lines L_1 and L_2 are not parallel. Next, suppose that the two lines do intersect at some point $P_0(x_0, y_0, z_0)$. Then

$$x_0 = 1 - t_1, y_0 = -2 - 3t_1, \text{ and } z_0 = 4 + t_1$$

and

$$x_0 = 2 - 2t_2, y_0 = -4 + 3t_2, \text{ and } z_0 = 1 + 4t_2$$

for some t_1 and t_2 . Equating the values of x_0, y_0 , and z_0 then gives

$$\begin{aligned} 1 - t_1 &= 2 - 2t_2 \\ -2 - 3t_1 &= -4 + 3t_2 \\ 4 + t_1 &= 1 + 4t_2 \end{aligned}$$

Solving the first two equations for t_1 and t_2 yields $t_1 = \frac{1}{9}$ and $t_2 = \frac{5}{9}$ substituting these values of t_1 and t_2 into the third equation gives $\frac{37}{9} = \frac{29}{9}$ a contradiction. This shows that there are no values of t_1 and t_2 that satisfy the three equations simultaneously. Thus, L_1 and L_2 do not intersect. We have shown that L_1 and L_2 are skew lines, so there is no possibility of the planes colliding.

Distance from a point to a Line

Let d represent the distance from the point Q to the line through the points P and R . We have

$$D = \|\vec{PQ}\| \sin \theta$$

where θ is the angle between $\vec{PQ}$ and $\vec{PR}$. Thus,

$$\|\vec{PQ} \times \vec{PR}\| = \|\vec{PQ}\| \|\vec{PR}\| \sin \theta = \|\vec{PR}\| (D)$$

Solving this for D , we get

$$D = \frac{\|\vec{PQ} \times \vec{PR}\|}{\|\vec{PR}\|}$$

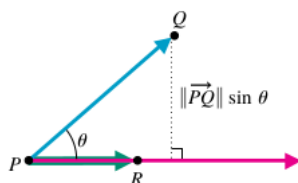


Figure 1.26:

■ **Example 1.32** Find the distance from the point $P(1, -2, 3)$ to the line with parametric equation $x = 3 + t, y = -1 - 2t, z = 4 + t$. ■

Solution: Choose any point from the line, say, $Q(3, -1, 4)$. Then $\vec{PQ} = \langle 2, 1, 1 \rangle$ and the direction vector is $\mathbf{v} = \langle 1, -2, 1 \rangle$, hence

$$\begin{aligned} \vec{PQ} \times \mathbf{v} &= \begin{vmatrix} i & j & k \\ 2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix} = 3i - j - 5k. \text{ Thus, the distance is} \\ D &= \frac{\|\vec{PQ} \times \mathbf{v}\|}{\|\mathbf{v}\|} = \frac{\sqrt{3^2 + (-1)^2 + (-5)^2}}{\sqrt{1^2 + (-2)^2 + 1^2}} = \frac{\sqrt{35}}{\sqrt{6}} \end{aligned}$$

■ **Example 1.33** Find the distance D between the two parallel lines L_1 and L_2 where $L_1: \frac{x-1}{2} = \frac{y+1}{-2} = \frac{z-2}{1}$, $L_2: x = 2 + t, y = -t, z = 3 + \frac{1}{2}t$ ■

Solution: $P = (1, -1, 2)$ and $Q = (2, 0, 3) \Rightarrow \vec{PQ} = \langle 2, 0, 3 \rangle$ and $\mathbf{v} = \langle 2, -2, 1 \rangle$. Thus

$$D = \frac{\|\vec{PQ} \times \mathbf{v}\|}{\|\mathbf{v}\|} = \frac{\sqrt{26}}{3}$$

1.7.2 Equations of Planes

A plane in space is uniquely determined by specifying a point $P_0(x_0, y_0, z_0)$ lying in the plane and a vector $\mathbf{n} = \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle$ that is normal (perpendicular) to it.

To find an equation of the plane, let $P(x, y, z)$ be any point in the plane. Then the vector $\overrightarrow{P_0P}$ must be orthogonal to $\mathbf{n}$. But two vectors are orthogonal if and only if their dot product is equal to zero. Therefore, we must have

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0$$

Since $\overrightarrow{P_0P} = \langle x - x_0, y - y_0, z - z_0 \rangle$, we have

$$\begin{aligned} \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle &= 0 \\ \implies a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \end{aligned}$$

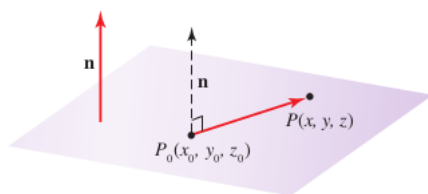


Figure 1.27:

Definition 1.7.6 The standard form of the equation of a plane containing the point $P_0(x_0, y_0, z_0)$ and having the normal vector $\mathbf{n} = \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

■ **Example 1.34** Find an equation of the plane containing the point $P_0(1, 2, 3)$ and having a normal vector, $\langle 4, 5, 6 \rangle$ and sketch the plane. ■

Solution: The equation of the plane is

$$\begin{aligned} a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \implies 4(x - 1) + 5(y - 2) + 6(z - 3) = 0 \\ \implies 4x + 5y + 6z &= 32 \end{aligned}$$

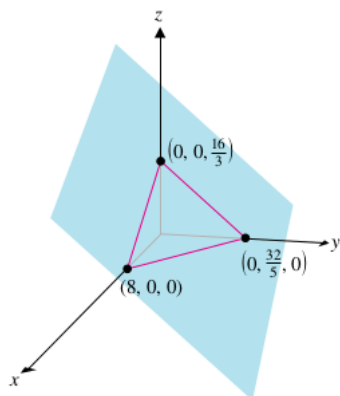


Figure 1.28: The plane through $(8, 0, 0)$, $(0, \frac{32}{5}, 0)$, $(0, 0, \frac{16}{3})$

■ **Example 1.35** Find an equation of the plane that passes through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$. ■

Solution: The vectors $\vec{PQ} = \langle 2, -4, 4 \rangle$ and $\vec{PR} = \langle 4, -1, -2 \rangle$ lie in the plane, so the vector $\|\vec{PQ} \times \vec{PR}\|$ is normal to the plane. Denoting this vector by $\mathbf{n}$, we have

$$\mathbf{n} = \|\vec{PQ} \times \vec{PR}\| = \begin{vmatrix} i & j & k \\ 2 & -4 & 4 \\ 4 & -1 & -2 \end{vmatrix} = 12i + 20j + 14k$$

Hence, using the point $P(1, 3, 2)$ in the plane (any of the other two points will also do) and the normal vector $\mathbf{n}$ just found, with $a = 12, b = 20, c = 14, x_0 = 1, y_0 = 3$, and $z_0 = 2$, we obtain the equation of the desired plane,

$$\begin{aligned} a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \implies 12(x - 1) + 20(y - 3) + 14(z - 2) = 0 \\ \implies 12x + 20y + 14z &= 100 \implies 6x + 10y + 7z = 50 \end{aligned}$$

Parallel and Orthogonal Planes

Two planes with normal vectors $\mathbf{m}$ and $\mathbf{n}$ are parallel to each other if $\mathbf{m}$ and $\mathbf{n}$ are parallel; the planes are orthogonal if $\mathbf{m}$ and $\mathbf{n}$ are orthogonal.

■ **Example 1.36** Find an equation of the plane containing $P(2, -1, 3)$ and parallel to the plane defined by $2x + 3y + 4z = 6$. ■

Solution: The normal vector of the given plane is $\mathbf{n} = \langle 2, -3, 4 \rangle$. Since the required plane is parallel to the given plane, it also has $\mathbf{n}$ as a normal vector. Hence, the equation of the plane is

$$2(x - 2) + (-3)(y - (-1)) + 4(z - 3) = 0 \implies 2x - 3y + 4z = 19$$

The Angle Between Two Planes

Two distinct planes in space are either parallel to each other or intersect in a straight line. If they do intersect, then the angle between the two planes is defined to be the acute angle between their normal vectors

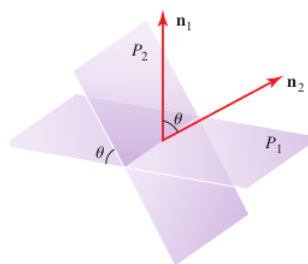


Figure 1.29: The angle between two planes is the angle between their normal vectors.

■ **Example 1.37** Find the angle between the two planes defined by $3x - y + 2z = 1$ and $2x + 3y - z = 4$. ■

Solution: The normal vectors of these planes are

$$\mathbf{n}_1 = \langle 3, -1, 2 \rangle \text{ and } \mathbf{n}_2 = \langle 2, 3, -1 \rangle$$

Therefore, the angle θ between the planes is given by

$$\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} = \frac{1}{14} \implies \theta = \cos^{-1} \left(\frac{1}{14} \right) \approx 86^\circ$$

■ **Example 1.38** Find parametric equations for the line of intersection of the planes defined by $3x - y + 2z = 1$ and $2x + 3y - z = 4$. ■

Solution:

The Distance Between a Point and a Plane

Let π be a plane with normal $\mathbf{n}$ and let P_1 be a point not on the plane. Then the distance D between P_1 and the plane π is given by

$$D = \frac{|\mathbf{n} \cdot \overrightarrow{P_1 P_0}|}{\|\mathbf{n}\|}$$

where P_0 is any point on the plane.

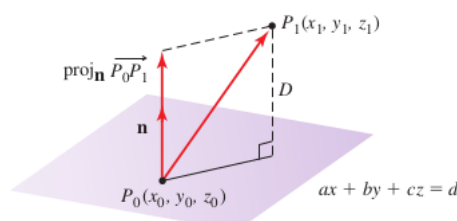


Figure 1.30:

■ **Example 1.39** Find the distance between the point $(-2, 1, 3)$ and the plane $2x - 3y + z = 1$. ■

Solution: $D = \frac{5}{\sqrt{14}}$

Exercise 1.4

- Find parametric and symmetric equations of the line passing through the point $(1, 2, -1)$ and parallel to the line with parametric equations $x = -1 + t$, $y = 2 + 2t$, and $z = -2 - 3t$. At what points does the line intersect the coordinate planes?
- Find symmetric equation of a line L containing the point $P_0(0, 1, -1)$ and perpendicular to the line L with parametric equation $x = 2 + 3t$, $y = -2 - 2t$, $z = 4t$
- Find parametric equations of the line that is parallel to the line with equation $\frac{x-1}{4} = \frac{y+4}{5} = \frac{z+1}{2}$ and contains the point of intersection of the lines $L_1: x = 4 + t, y = 5 + t, z = -1 + 2t$ and $L_2: x = 6 + 2t, y = 11 + 4t, z = -t$
- Find the equation of a plane containing the point $P_0(1, -1, 2)$ and having normal $\mathbf{n} = \frac{1}{2}\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.
- Find the equation of a plane containing the point $P(1, -1, 1)$, $Q(2, 3, 0)$ and $R(-1, 2, -2)$.
- Find the distance between the point $P_1(-4, 2, 7)$ and the plane $2x - 3y + 4z = 1$

Definition of Matrix

Types of Matrices

Matrix Addition and Scalar Multiplication

Transpose Matrix

Matrix Multiplication

Elementary Row Operations and Echelon Form

Rank of a matrix

Determinant of a Matrix and its Properties

Inverse of a Matrix

Gauss-Jordan elimination method

Cofactor matrix and adjoint

Systems of Linear Equations

Solving System of Equation using

Cramer's Rule

Solving System of Equation using Gaus-

sian Elimination Method

2 — MATRICES AND DETERMINANTS

2.1 Definition of Matrix

Definition 2.1.1 A matrix is defined as a rectangular array of numbers, enclosed by brackets.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

R The numbers in the array are called the elements of the matrix.

Some examples of matrices are:

$$\begin{bmatrix} -1 & 3 & 8 \\ -3 & 4 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ 7 & 11 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} -1 & 3 & 8 \\ 2 & 7 & 11 \\ -3 & 4 & 1 \end{pmatrix}, (-1.3 \quad 0.8), A = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}.$$

Note:

- If a matrix A has m rows(horizontals) and n columns(verticals), then the size (or dimension) of the matrix A is $m \times n$ (read as 'm by n').
- a_{ij} is the element that appears in the i^{th} row and in the j^{th} column.
- $A = (a_{ij})_{m \times n}$ is an $m \times n$ matrix.

■ **Example 2.1** Consider $A = \begin{pmatrix} 0 & 3 & 8 \\ 2 & 7 & 11 \end{pmatrix}$.

The size of matrix A is 2×3 because it has two rows and three columns.

The elements

$a_{11} = 0$, First row, first column, $a_{23} = 11$ Second row, third column ■

Definition 2.1.2 Two matrices are equal if they have the same size and their corresponding elements are equal.

■ **Example 2.2** Let $A = \begin{bmatrix} 3 & -5 & x \\ 2 & y+1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -5 & 2x-1 \\ 2 & 5 & 3 \end{bmatrix}$. Find the values of x and y such that $A = B$ ■

Solution: For the two matrices to be equal, we must have corresponding entries equal, so

$$x = 2x - 1 \implies x = 1 \quad a_{13} = b_{13}$$

$$y + 1 = 5 \implies y = 4 \quad a_{22} = b_{22}$$

■ **Example 2.3** Consider the two matrices given below

$$A = \begin{pmatrix} -1 & 3 & 8 \\ 2 & 7 & 11 \end{pmatrix} \text{ and } B = \begin{pmatrix} -1 & 4 & 8 \\ 2 & 7 & 11 \end{pmatrix}.$$

Solution: The size of the two matrix is 2×3 . Since $a_{12} = 3 \neq 4 = b_{12}$, we can say that $A \neq B$

2.1.1 Types of Matrices

1. **Row Matrix:** A matrix which has exactly one row ($1 \times n$) called a row matrix (row vector)
Example: $A = \begin{pmatrix} 2 & 5 & 6 \end{pmatrix}$ is a 1×3 row matrix.
2. **Column Matrix:** A matrix which has exactly one column ($m \times 1$) matrix is called a column matrix (or column vector).

Example: $A = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ is a 3×1 column matrix.

3. **Square Matrix:** A matrix in which the number of rows is equal to the number of columns is called a square matrix. ($n \times n$)

$$B = \begin{pmatrix} -1 & 4 & 8 \\ 2 & 7 & 11 \\ 3 & -6 & 0 \end{pmatrix}.$$

is a 3×3 square matrix.

4. **Null or zero matrix:** A matrix each of whose elements is zero is called a zero matrix.

Example: $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a 2×3 zero matrix.

5. **Diagonal matrix:** is a square matrix whose every element other than diagonal elements is zero.
The matrix

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 12 \end{pmatrix},$$

are example of diagonal matrix.

Ⓡ The elements a_{ii} are called diagonal elements of a square matrix (a_{ij}).

6. **Scalar matrix:** A scalar matrix is a diagonal matrix whose diagonal elements are equal

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

7. **Identity matrix:** is a diagonal matrix whose diagonal elements are equal to 1 (units)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

8. **Triangular matrix:**

A square matrix whose elements $a_{ij} = 0$ when ever $i < j$ is called a lower triangular matrix.

And square matrix whose elements $a_{ij} = 0$ when ever $i > j$ is called an upper triangular matrix.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 11 & 0 & 0 \\ 3 & 2 & 4 & 0 \\ 0 & -3 & 6 & 12 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 1.5 & 1 \end{pmatrix}$$

are lower triangular matrix. And

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 0 & 2 & 8 & 5 \\ 0 & 0 & 7 & -9 \\ 0 & 0 & 0 & 12 \end{pmatrix}, \begin{pmatrix} 1 & 6 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 3 & 5 \\ 0 & 10 & 8 \\ 0 & 0 & -1 \end{pmatrix}$$

are upper triangular matrix

2.2 Matrix Addition and Scalar Multiplication

Let $A = (a_{ij})$ and $B = (b_{ij})$ be two matrices with the same size, say $m \times n$. The sum of A and B, written $A + B$, is the matrix obtained by adding corresponding elements from A and B. That is, if

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \vdots \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & & & \vdots \\ \vdots & & & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{pmatrix}$$

then,

$$A + B = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & & & \vdots \\ \vdots & & & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{pmatrix}$$

The product of the matrix A by a scalar k , written kA , is the matrix obtained by multiplying each element of A by k . That is,

$$kA = \begin{pmatrix} ka_{11} & ka_{12} & \dots & ka_{1n} \\ ka_{21} & ka_{22} & \dots & ka_{2n} \\ \vdots & & & \vdots \\ \vdots & & & \vdots \\ ka_{m1} & ka_{m2} & \dots & ka_{mn} \end{pmatrix}$$

The matrix $-A$ is called the negative of the matrix A, and the matrix $A - B$ is called the difference of A and B.

Subtraction is performed on matrices of the same size by subtracting corresponding elements. Thus, $A - B = A + (-B)$

 The sum of matrices with different sizes is not defined.

■ **Example 2.4** Let $A = \begin{bmatrix} -1 & 3 \\ 7 & 11 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 4 & 1 \\ 10 & -6 \end{bmatrix}$. Find

(a) $A + B$

(b) $2A$

(c) $3A - 2B$

Solution:

$$(a) A + B = \begin{bmatrix} -1 & 3 \\ 7 & 11 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 5 \\ 4 & 1 \\ 10 & -6 \end{bmatrix} = \begin{bmatrix} -1+3 & 3+5 \\ 7+4 & 11+1 \\ 0+10 & 1+(-6) \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ 11 & 12 \\ 10 & -5 \end{bmatrix}$$

$$(b) 2A = 2 \begin{bmatrix} -1 & 3 \\ 7 & 11 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 6 \\ 14 & 22 \\ 0 & 2 \end{bmatrix}$$

$$(c) 3A - 2B = 3 \begin{bmatrix} -1 & 3 \\ 7 & 11 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 3 & 5 \\ 4 & 1 \\ 10 & -6 \end{bmatrix} = \begin{bmatrix} -3 & 9 \\ 21 & 33 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 6 & 10 \\ 8 & 2 \\ 20 & -12 \end{bmatrix} = \begin{bmatrix} -9 & -1 \\ 13 & 31 \\ -20 & 15 \end{bmatrix}$$

Properties of Matrix Addition and Scalar Multiplication

If A, B , and C are any $m \times n$ matrices and if O is the zero $m \times n$ matrix, and k, l are any real number, then the following hold:

- (a) Associative law: $A + (B + C) = (A + B) + C$
- (b) Commutative law: $A + B = B + A$
- (c) Additive identity law: $A + O = O + A = A$
- (d) Additive inverse law: $A + (-A) = -A + A = O$
- (e) Distributive law: $k(A + B) = kA + kB$
Distributive law: $(l + k)A = lA + kA$
- (f) Scalar unit $1A = A$
- (g) Scalar zero $0A = O$

2.2.1 Transpose Matrix

Definition 2.2.1 The transpose of a matrix A , written A^t , is the matrix obtained by interchanging the columns and rows of A . That is, if $A = (a_{ij})$ is $m \times n$ matrix, then $A^t = (b_{ij})$ is $n \times m$ matrix where $(b_{ij}) = (a_{ji})$

■ **Example 2.5** Let $A = \begin{pmatrix} 4 & 2 & 3 \\ 2 & 5 & -9 \end{pmatrix}$. Then $A^t = \begin{pmatrix} 4 & 2 \\ 2 & 5 \\ 3 & -9 \end{pmatrix}$ ■

Properties

- (a) $(A + B)^t = A^t + B^t$
- (b) $(A^t)^t = A$
- (c) $(kA)^t = kA^t$, k scalar
- (d) $(AB)^t = B^t A^t$

- R** A square matrix A is symmetric if $A^t = A$.
 A square matrix A is skew symmetric if $A^t = -A$.

■ **Example 2.6** Let $A = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 9 & 12 \\ 4 & 12 & -7 \end{pmatrix}$ $B = \begin{pmatrix} 0 & -5 & 9 \\ 5 & 0 & 6 \\ -9 & -6 & 0 \end{pmatrix}$. Find A^t , B^t ■

Solution: $A^t = \begin{pmatrix} -2 & 5 & 4 \\ 5 & 9 & 12 \\ 4 & 12 & -7 \end{pmatrix} = A \implies$, Matrix A is symmetric

$B^t = \begin{pmatrix} 0 & 5 & 9 \\ -5 & 0 & -6 \\ 9 & 6 & 0 \end{pmatrix} = -B \implies$, Matrix B is skew symmetric

Theorem 2.2.1 Let A be a square matrix. Then

- (a) $A + A^t$ is symmetric.
 (b) $A - A^t$ is skew symmetric.

■ **Example 2.7** Let $A = \begin{pmatrix} 4 & 3 & 6 \\ -1 & 0 & 7 \\ 3 & -2 & 1 \end{pmatrix}$ ■

Solution: $A^t = \begin{pmatrix} 4 & -1 & 3 \\ 3 & 0 & -2 \\ 6 & 7 & 1 \end{pmatrix}$, then

$B = A + A^t = \begin{pmatrix} 4 & 3 & 6 \\ -1 & 0 & 7 \\ 3 & -2 & 1 \end{pmatrix} + \begin{pmatrix} 4 & -1 & 3 \\ 3 & 0 & -2 \\ 6 & 7 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 2 & 9 \\ 2 & 0 & 5 \\ 9 & 5 & 2 \end{pmatrix} = B^t$, which is symmetric

$C = A - A^t = \begin{pmatrix} 4 & 3 & 6 \\ -1 & 0 & 7 \\ 3 & -2 & 1 \end{pmatrix} - \begin{pmatrix} 4 & -1 & 3 \\ 3 & 0 & -2 \\ 6 & 7 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 4 & 3 \\ -4 & 0 & 9 \\ -3 & -9 & 0 \end{pmatrix} = -C^t$, which is skew symmetric

2.3 Matrix Multiplication

Definition 2.3.1 Let A be an $m \times n$ matrix and let B be an $n \times k$ matrix. The product matrix AB is the $m \times k$ matrix whose entry in the i th row and j th column is the product of the i th row of A and the j th column of B .

R The number of columns of A must equal the number of rows of B in order to get the product matrix AB .

■ **Example 2.8** Let $A = \begin{pmatrix} 2 & 3 & 4 \\ -1 & 2 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 6 & -2 \\ -4 & 2 \\ 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 10 & 2 \\ 4 & 3 \end{pmatrix}$. Find

(a) AB (b) BA (c) BC (d) CA (e) AC

Solution:

(a) Here, matrix A is 2×3 and matrix B is 3×2 , so matrix AB can be found and will be 2×2 .

$$AB = \begin{pmatrix} 2 & 3 & 4 \\ -1 & 2 & 2 \end{pmatrix} \begin{pmatrix} 6 & -2 \\ -4 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 \cdot 6 + 3 \cdot (-4) + 4 \cdot 0 & 2 \cdot (-2) + 3 \cdot 2 + 4 \cdot 1 \\ -1 \cdot 6 + 2 \cdot (-4) + 2 \cdot 0 & -1 \cdot (-2) + 2 \cdot 2 + 2 \cdot 1 \end{pmatrix}$$

$$= \begin{pmatrix} 12 + (-12) + 0 & -4 + 6 + 4 \\ -6 + (-8) + 2 & 2 + 4 + 2 \end{pmatrix} = \begin{pmatrix} 10 & 6 \\ -12 & 8 \end{pmatrix}$$

(b) Here, matrix B is 3×2 and matrix A is 2×3 , so matrix BA can be found and will be 3×3 .

$$BA = \begin{pmatrix} 6 & -2 \\ -4 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 4 \\ -1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 6 \times 2 + (-2) \times (-1) & 6 \times 3 + (-2) \times 2 & 6 \times 4 + (-2) \times 2 \\ -4 \times 2 + 2 \times (-1) & -4 \times 3 + 2 \times 2 & -4 \times 4 + 2 \times 2 \\ 0 \times 2 + 1 \times (-1) & 0 \times 3 + 1 \times 2 & 0 \times 4 + 1 \times 2 \end{pmatrix}$$

$$= \begin{pmatrix} 14 & 14 & 20 \\ -10 & -8 & -12 \\ -1 & 2 & 2 \end{pmatrix}$$

(c) Here, matrix B is 3×2 and matrix C is 2×2 , so matrix BC can be found and will be 3×2 .

$$BC = \begin{pmatrix} 6 & -2 \\ -4 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 10 & 2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 6 \times 10 + (-2) \times 4 & 6 \times 2 + (-2) \times 3 \\ -4 \times 10 + 2 \times 4 & -4 \times 2 + 2 \times 3 \\ 0 \times 10 + 1 \times 4 & 0 \times 2 + 1 \times 3 \end{pmatrix}$$

$$= \begin{pmatrix} 52 & 6 \\ -32 & -2 \\ 4 & 3 \end{pmatrix}$$

(d) Here, matrix C is 2×2 and matrix A is 2×3 , so matrix CA can be found and will be 2×3 .

$$CA = \begin{pmatrix} 10 & 2 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 2 & 3 & 4 \\ -1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 10 \times 2 + 2 \times (-1) & 10 \times 3 + 2 \times 2 & 10 \times 4 + 2 \times 2 \\ 4 \times 2 + 3 \times (-1) & 4 \times 3 + 3 \times 2 & 4 \times 4 + 3 \times 2 \end{pmatrix}$$

$$= \begin{pmatrix} 18 & 34 & 44 \\ 5 & 18 & 22 \end{pmatrix}$$

(e) Since the number of column in matrix A (2×3) is not equal to the number of rows in matrix C (2×2), the product AC is not defined.

Properties of Matrix Multiplication

Let A be a matrix of dimension $m \times k$, let B be a matrix of dimension $k \times r$, and let C be a matrix of dimension $r \times n$

- (a) Matrix multiplication is not commutative. That is, in general, $AB \neq BA$
- (b) Matrix multiplication is associative. That is, $A(BC) = (AB)C$
- (c) Distributive property: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
- (d) $k(AB) = (kA)B = A(kB)$, where k is a scalar.
- (e) If A is a square matrix of dimension $n \times n$, then $AI_n = I_nA = A$, where I_n is $n \times n$ identity matrix
- (f) If A is $m \times n$, then $AI_n = A = I_mA$

R If A and B are matrices such that the products AB and BA exist, then AB may not equal BA . The cancellation law do not hold for matrix multiplication. That is, if $AB = AC$ then it is not true in general that $B = C$.
If a product AB is the zero matrix, we cannot conclude in general that either $A = 0$ or $B = 0$

■ **Example 2.9** Let $A = \begin{bmatrix} 5 & -2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 2 \\ 4 & -1 & 6 \end{bmatrix}$, $C = \begin{bmatrix} 4 & -3 & -2 \\ 0 & 2 & 0 \\ 7 & 8 & 1 \end{bmatrix}$. Show that $A(BC) = (AB)C, AI_2 = I_2A = A$ ■

Solution: Since matrix A is 2×2 , matrix B is 2×3 and matrix C is 3×3 , the product $A(BC) = (AB)C$ can be found and will be 2×3

$$AB = \begin{bmatrix} 5 & -2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 4 & -1 & 6 \end{bmatrix} = \begin{bmatrix} -3 & 17 & -2 \\ 19 & 5 & 30 \end{bmatrix}$$

$$BC = \begin{bmatrix} 1 & 3 & 2 \\ 4 & -1 & 6 \end{bmatrix} \begin{bmatrix} 4 & -3 & -2 \\ 0 & 2 & 0 \\ 7 & 8 & 1 \end{bmatrix} = \begin{bmatrix} 18 & 19 & 0 \\ 58 & 34 & -2 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 5 & -2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 18 & 19 & 0 \\ 58 & 34 & -2 \end{bmatrix} = \begin{bmatrix} -26 & 27 & 4 \\ 286 & 193 & -8 \end{bmatrix} = \begin{bmatrix} -3 & 17 & -2 \\ 19 & 5 & 30 \end{bmatrix} \begin{bmatrix} 4 & -3 & -2 \\ 0 & 2 & 0 \\ 7 & 8 & 1 \end{bmatrix} = (AB)C$$

$$I_2A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = AI_2$$

■ **Example 2.10** Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$. Find AB, BA ■

Solution: $AB = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \times 2 + 2 \times -1 & 1 \times -4 + 2 \times 2 \\ 3 \times 2 + 6 \times -1 & 3 \times -4 + 6 \times 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$BA = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 2 \times 1 + -4 \times 3 & 2 \times 2 + -4 \times 6 \\ -1 \times 1 + 2 \times 3 & -1 \times 2 + 2 \times 6 \end{bmatrix} = \begin{bmatrix} -10 & -20 \\ 5 & 10 \end{bmatrix}$$

2.4 Elementary Row Operations and Echelon Form

Let A be an $m \times n$ matrix. The elementary row operation on A is

- (a) Interchanging two rows of a matrix. If we interchange the i^{th} row with the j^{th} row, then we usually denote the operation as $R_i \longleftrightarrow R_j$
- (b) Multiply the elements of a row by a non-zero constant. If the i^{th} row is multiplied by α then we usually denote this operation as $R_i \longrightarrow \alpha R_i$
- (c) Add a multiple of the elements of one row to the corresponding elements of another row i.e., $R_i \longrightarrow R_i + \alpha R_j$

Definition 2.4.1 Two matrices are equivalent written as $A \sim B$ if one can be obtained from the other by a sequence of elementary row operations.

■ **Example 2.11** Let $A = \begin{pmatrix} 2 & -3 & 6 \\ 1 & 1 & 4 \\ 5 & -4 & 1 \end{pmatrix}$ ■

Solution:

$$A \sim \begin{pmatrix} 2 & -3 & 6 \\ 4 & 4 & 16 \\ 5 & -4 & 1 \end{pmatrix} (R_2 \longrightarrow 4R_2) \sim \begin{pmatrix} 2 & -3 & 6 \\ 5 & -4 & 1 \\ 4 & 4 & 16 \end{pmatrix} (R_2 \longrightarrow R_3) \sim \begin{pmatrix} 2 & -3 & 6 \\ 5 & -4 & 1 \\ 0 & 10 & 4 \end{pmatrix} (R_3 \longrightarrow R_3 - 2R_1)$$

Definition 2.4.2 A matrix is in a row echelon form if it satisfies the following conditions

- (a) The first non-zero entry in each row is a 1 (called a Leading 1).
- (b) If a column contains a leading 1, then every entry of the column below the leading 1 is a zero.
- (c) As we move downwards through the rows of the matrix, the leading 1's move from left to right.
- (d) Any row (if any) consisting of entirely of zeros appears at the bottom of the matrix.

■ **Example 2.12** The following matrices are in row echelon form

$$A = \begin{pmatrix} 1 & -3 & 4 \\ 0 & 5 & 5 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 14 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & -13 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

■ **Example 2.13** The following matrices are not in row echelon form

$$E = \begin{pmatrix} 1 & 2 & 14 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 3 & 3 \\ 0 & 0 & 10 \\ 0 & 0 & 19 \end{pmatrix}, \quad G = \begin{pmatrix} 4 & 5 & 6 \\ 0 & 6 & 4 \\ 0 & 3 & 3 \end{pmatrix}$$

■ **Example 2.14** Let A be the matrix

$$A = \begin{pmatrix} 1 & -1 & -1 & 2 & 0 \\ 2 & 1 & -1 & 2 & 8 \\ 1 & -3 & 2 & 7 & 2 \end{pmatrix}$$

Reduce A to row-echelon form. ■

Solution:

1. Get a 1 as the first entry of Row 1. Done.
2. Use this first leading 1 to clear out column 1 as follows :

$$A \sim \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \begin{pmatrix} 1 & -1 & -1 & 2 & 0 \\ 0 & 3 & 1 & -2 & 8 \\ 0 & -2 & 3 & 5 & 2 \end{pmatrix}$$

3. Get a leading 1 as the second entry of Row 2, for example as follows:

$$A \sim \begin{matrix} R_2 \rightarrow R_2 + R_3 \end{matrix} \begin{pmatrix} 1 & -1 & -1 & 2 & 0 \\ 0 & 1 & 4 & 3 & 10 \\ 0 & -2 & 3 & 5 & 2 \end{pmatrix}$$

4. Use this leading 1 to clear out whatever appears below it in Column 2

$$A \sim \begin{matrix} R_3 \rightarrow R_3 + 2R_2 \end{matrix} \begin{pmatrix} 1 & -1 & -1 & 2 & 0 \\ 0 & 1 & 4 & 3 & 10 \\ 0 & 0 & 11 & 11 & 22 \end{pmatrix}$$

5. Get a leading 1 in Row 3:

$$\begin{matrix} R_3 \rightarrow \frac{1}{11} R_3 \end{matrix} \begin{pmatrix} 1 & -1 & -1 & 2 & 0 \\ 0 & 1 & 4 & 3 & 10 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix}$$

This matrix is now in row-echelon form.

■ **Example 2.15** Let H be the matrix

$$H = \begin{pmatrix} 1 & -1 & 1 & 4 \\ 4 & 2 & -2 & 4 \\ 1 & -3 & 5 & 6 \end{pmatrix}$$

Reduce H to row-echelon form. ■

Solution:

1. Get a 1 as the first entry of Row 1. Done.
2. Use this first leading 1 to clear out column 1 as follows :

$$H \sim \begin{matrix} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 6 & -6 & -12 \\ 0 & -2 & 4 & 4 \end{pmatrix}$$

3. Get a leading 1 as the second entry of Row 2, for example as follows:

$$H \sim \begin{matrix} R_2 \rightarrow \frac{1}{6} R_2 \end{matrix} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & -2 & 4 & 4 \end{pmatrix}$$

4. Use this leading 1 to clear out whatever appears below it in Column 2

$$H \sim \begin{matrix} R_3 \rightarrow R_3 + 2R_2 \end{matrix} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$

5. Get a leading 1 in Row 3:

$$\begin{matrix} R_3 \rightarrow \frac{1}{2} R_3 \end{matrix} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

This matrix is now in row-echelon form.

Definition 2.4.3 A matrix in a row echelon form is said to be in reduced row echelon form if all entries in any column containing the leading 1 is zero.

■ **Example 2.16** The following matrices are in reduced row echelon form

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, C = \begin{pmatrix} 1 & -13 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \blacksquare$$

■ **Example 2.17** Let A be the matrix

$$A = \begin{pmatrix} 1 & -1 & 1 & 4 \\ 4 & 2 & -2 & 4 \\ 1 & -3 & 5 & 6 \end{pmatrix}$$

Reduce A to reduced row-echelon form. ■

Solution:

1. Get a 1 as the first entry of Row 1. Done.
2. Use this first leading 1 to clear out column 1 as follows :

$$A \sim \begin{matrix} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \rightarrow \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 6 & -6 & -12 \\ 0 & -2 & 4 & 4 \end{pmatrix}$$

3. Get a leading 1 as the second entry of Row 2, for example as follows:

$$A \sim \xrightarrow{R_2 \rightarrow \frac{1}{6}R_2} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & -2 & 4 & 4 \end{pmatrix}$$

4. Use this leading 1 to clear out whatever appears below it in Column 2

$$A \sim \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$

5. Get a leading 1 in Row 3:

$$\xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \begin{pmatrix} 1 & -1 & 1 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

6. Make zero above the leading 1 in column 2

$$\xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

7. Make zero above the leading 1 in column 3

$$\xrightarrow{R_2 \rightarrow R_2 + R_3} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

This matrix is now in reduced row-echelon form.

2.4.1 Rank of a matrix

Definition 2.4.4 Let A be an $m \times n$ matrix. Let A_R be the row echelon form of A . The rank $\rho(A)$ is the number of non-zero rows of the row echelon form of A (A_R).

$$\rho(A) \leq \min(m, n)$$

■ **Example 2.18** Find the rank of the following matrices.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 & 0 \\ 2 & 9 & 1 \\ 1 & 5 & 1 \end{pmatrix}$$

Solution: (a) $\text{rank}(A) = 3$

(b) Write the augmented matrix and apply elementary row operations

$$B|I = \begin{pmatrix} 1 & 4 & 0 & 1 \\ 2 & 9 & 1 & 0 \\ 1 & 5 & 1 & 0 \end{pmatrix} \xrightarrow[R_2 \rightarrow R_2 - 2R_1]{R_3 \leftrightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 4 & 0 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 1 & 1 & -3 \end{pmatrix} \sim \begin{pmatrix} 1 & 4 & 0 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Hence, $\text{Rank}(B) = 2$

2.5 Determinant of a Matrix and its Properties

The determinant of a square matrix is a single number that results from performing a specific operation on the array. The determinant of a matrix A is denoted as $\det(A)$ or $|A|$.

Determinant of order two

Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ be a square matrix of order two. Then

$$\det(A) = a_{11}a_{22} - a_{21}a_{12}$$

Determinant of order three

Let $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ be a 3×3 matrix. Then

$$\begin{aligned} \det(A) &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{31} - a_{21}a_{33}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) \end{aligned}$$

■ **Example 2.19** Find the determinant of the matrix

$$A = \begin{pmatrix} 1 & 2 & 4 \\ -1 & 3 & 2 \\ 0 & -4 & 1 \end{pmatrix}$$

Solution: $\det(A) = 1(3(1) - (-4)2) - 2(-1(1) - 0(2)) + 4((-1)(-4) - 0(3)) = 27$

Properties of determinants

1. If matrix B results from matrix A by interchanging two rows (columns) of A , then $\det(B) = -\det(A)$
2. If two rows (columns) of A are equal, then $\det(A) = 0$
3. The determinant of the transpose of A is equal to the determinant of the given matrix A
 $\det(A^t) = \det(A)$
4. If a row (column) of A consists entirely of zeros, then $\det(A) = 0$.

5. If B is obtained from A by multiplying a row (column) of A by a real number c , then $\det(B) = c \det(A)$
6. If to any row (or column) is added k times the corresponding elements of another row (or column), the determinant remains unchanged.
7. The determinant of a product of two matrices is the product of their determinants; that is $\det(AB) = \det(A) \det(B)$.
8. If c is a real number and A is $n \times n$ matrix, then $\det(cA) = c^n \det(A)$.
9. The determinant of a diagonal matrix is the product of the diagonal elements.
10. The determinant of identity matrix is 1.

2.6 Inverse of a Matrix

Definition 2.6.1 Let A be a matrix of dimension $n \times n$. A matrix B of dimension $n \times n$ is called the inverse of A if

$$AB = BA = I_n$$

where, I_n is $n \times n$ identity matrix. We denote the inverse of a matrix A , if it exists, by A^{-1} .

■ **Example 2.20** Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{pmatrix}$.

Show that matrix A and B are inverse of each other. ■

Solution: Since $AB = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = BA$, then $A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{pmatrix}$



- Only square matrices possibly have an inverse.
- A non-square matrix has no inverse.
- The inverse of a square matrix, if it exists, is unique.
- A square matrix that does not have an inverse is called singular.
- If A is invertible, then $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$

An example of a singular matrix is given by

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

If B had an inverse given by

$$B^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

where a , b , c , and d are some appropriate numbers, then by the definition of an inverse we would have $BB^{-1} = I$; that is,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \implies \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

which implies that $0 = 1$ —an impossibility! This contradiction shows that B does not have an inverse.

Properties of Invertible matrices

1. $(A^t)^{-1} = (A^{-1})^t$
2. $(A^{-1})^{-1} = A$
3. $(AB)^{-1} = B^{-1}A^{-1}$

2.6.1 Gauss–Jordan elimination method

To find the inverse of a matrix A by using Gauss–Jordan elimination method the operations are:

Step 1 Augment the matrix A with identity matrix. i.e., Write in the form $A|I$.

Step 2 Reduce the augmented matrix in to reduced row-echelon form.

■ **Example 2.21** Find the inverse of $A = \begin{pmatrix} 7 & 3 \\ 2 & 1 \end{pmatrix}$ ■

Solution: Write the augmented matrix

$$A|I = \left(\begin{array}{cc|cc} 7 & 3 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array} \right)$$

Apply elementary row operations

$$\begin{aligned} A|I &\sim \xrightarrow{R_2 \rightarrow 7R_2 - 2R_1} \left(\begin{array}{cc|cc} 7 & 3 & 1 & 0 \\ 0 & 1 & -2 & 7 \end{array} \right) \sim \xrightarrow{R_1 \rightarrow R_1 - 3R_2} \left(\begin{array}{cc|cc} 7 & 0 & 7 & -21 \\ 0 & 1 & -2 & 7 \end{array} \right) \\ &\sim \xrightarrow{R_1 \rightarrow \frac{1}{7}R_1} \left(\begin{array}{cc|cc} 1 & 0 & 1 & -3 \\ 0 & 1 & -2 & 7 \end{array} \right) \\ \text{Thus, } A^{-1} &= \begin{pmatrix} 1 & -3 \\ -2 & 7 \end{pmatrix} \end{aligned}$$

■ **Example 2.22** Find the inverse of $A = \begin{pmatrix} 3 & 1 & -2 \\ 1 & 1 & -1 \\ -5 & -2 & 4 \end{pmatrix}$ ■

Solution: Write the augmented matrix

$$A|I = \left(\begin{array}{ccc|ccc} 3 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 1 & 0 \\ -5 & -2 & 4 & 0 & 0 & 1 \end{array} \right)$$

Apply elementary row operations

$$\begin{aligned} A|I &\sim \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 3 & 1 & -2 & 1 & 0 & 0 \\ -5 & -2 & 4 & 0 & 0 & 1 \end{array} \right) \sim \xrightarrow[R_3 \rightarrow R_3 + 5R_1]{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -2 & 1 & 1 & -3 & 0 \\ 0 & 3 & -1 & 0 & 5 & 1 \end{array} \right) \\ &\sim \xrightarrow{R_2 \rightarrow R_2 + R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 3 & -1 & 0 & 5 & 1 \end{array} \right) \sim \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \end{array} \right) \\ &\sim \xrightarrow{R_1 \rightarrow R_1 + R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 3 & 2 & 2 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \end{array} \right) \sim \xrightarrow{R_1 \rightarrow R_1 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \end{array} \right) \\ \text{Hence, } A^{-1} &= \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{pmatrix} \end{aligned}$$

■ **Example 2.23** Find the inverse of $B = \begin{pmatrix} 2 & 4 & -2 \\ -4 & -6 & 1 \\ 3 & 5 & -1 \end{pmatrix}$ ■

Solution: Write the augmented matrix and apply elementary row operations

$$\begin{aligned}
 B|I &= \left(\begin{array}{ccc|ccc} 2 & 4 & -2 & 1 & 0 & 0 \\ -4 & -6 & 1 & 0 & 1 & 0 \\ 3 & 5 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \rightarrow 2R_3 - 3R_1]{R_2 \leftrightarrow R_2 + 2R_1} \left(\begin{array}{ccc|ccc} 2 & 4 & -2 & 1 & 0 & 0 \\ 0 & 2 & -3 & 2 & 1 & 0 \\ 0 & -2 & 4 & -3 & 0 & 2 \end{array} \right) \\
 &\sim \xrightarrow{R_3 \rightarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 2 & 4 & -2 & 1 & 0 & 0 \\ 0 & 2 & -3 & 2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array} \right) \xrightarrow[R_2 \rightarrow R_2 + 3R_3]{R_1 \rightarrow R_1 + 2R_3} \left(\begin{array}{ccc|ccc} 2 & 4 & 0 & -1 & 2 & 4 \\ 0 & 2 & 0 & -1 & 4 & 6 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array} \right) \\
 &\sim \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & -6 & -8 \\ 0 & 2 & 0 & -1 & 4 & 6 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array} \right) \xrightarrow[R_2 \rightarrow \frac{1}{2}R_2]{R_1 \rightarrow \frac{1}{2}R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -3 & -4 \\ 0 & 1 & 0 & -\frac{1}{2} & 2 & 3 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array} \right) \\
 \text{Thus, } B^{-1} &= \begin{pmatrix} \frac{1}{2} & -3 & -4 \\ -\frac{1}{2} & 2 & 3 \\ -1 & 1 & 2 \end{pmatrix}
 \end{aligned}$$

R If there is a row to the left of the vertical line in the augmented matrix containing all zeros, then the matrix does not have an inverse.

■ **Example 2.24** Find the inverse of $A = \begin{bmatrix} 1 & 3 & -4 \\ -2 & -6 & 8 \\ 5 & -2 & 1 \end{bmatrix}$ if exist. ■

Solution: Write the augmented matrix

$$A|I = \left(\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ -2 & -6 & 8 & 0 & 1 & 0 \\ 5 & -2 & 1 & 0 & 0 & 1 \end{array} \right)$$

Apply elementary row operations

$$A|I \sim \xrightarrow[R_3 \rightarrow R_3 - 5R_1]{R_2 \rightarrow R_2 + 2R_1} \left(\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & -17 & 21 & -5 & 0 & 1 \end{array} \right) \sim \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & 3 & -4 & 1 & 0 & 0 \\ 0 & -17 & 21 & -5 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & 0 \end{array} \right)$$

Since there is a row to the left of the vertical line in the augmented matrix containing all zeros, then the matrix A does not have an inverse.

2.6.2 Cofactor matrix and adjoint

Definition 2.6.2 Let $A = (a_{ij})$ be an $n \times n$ matrix. Let M_{ij} be the $(n-1) \times (n-1)$ sub matrix of A obtained by deleting the i^{th} row and j^{th} column of A . The determinant $\det(M_{ij})$ (is called the minor of a_{ij} . The cofactor A_{ij} of a_{ij} is defined as

$$A_{ij} = (-1)^{i+j} \det(M_{ij}).$$

The cofactor of an $n \times n$ matrix A is the matrix

$$B = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{pmatrix}$$

Definition 2.6.3 The adjoint of a matrix A , written $\text{adj}A$, is the transpose of the cofactor matrix B .

■ **Example 2.25** Find the adjoint of

$$A = \begin{pmatrix} 3 & -3 & -2 \\ 1 & -1 & -1 \\ -3 & 4 & 2 \end{pmatrix}$$

Solution:

$$\begin{aligned} A_{11} &= |M_{11}| = \begin{vmatrix} -1 & -1 \\ 4 & 2 \end{vmatrix} = 2, A_{12} = -|M_{12}| = -\begin{vmatrix} 1 & -1 \\ -3 & 2 \end{vmatrix} = 1, A_{13} = |M_{13}| = \begin{vmatrix} 1 & -1 \\ -3 & 4 \end{vmatrix} = 2 \\ A_{21} &= -|M_{21}| = -\begin{vmatrix} -3 & -2 \\ 4 & 2 \end{vmatrix} = -2, A_{22} = |M_{22}| = \begin{vmatrix} 3 & -2 \\ -3 & 2 \end{vmatrix} = 0, A_{23} = -|M_{23}| = -\begin{vmatrix} 3 & -3 \\ -3 & 4 \end{vmatrix} = -3 \\ A_{31} &= |M_{31}| = \begin{vmatrix} -3 & -2 \\ -1 & -1 \end{vmatrix} = 1, A_{32} = -|M_{32}| = -\begin{vmatrix} 3 & -2 \\ 1 & -1 \end{vmatrix} = 1, A_{33} = |M_{33}| = \begin{vmatrix} 3 & -3 \\ 1 & -1 \end{vmatrix} = 0 \end{aligned}$$

The cofactor matrix is

$$B = \begin{pmatrix} 2 & 1 & 2 \\ -2 & 0 & -3 \\ 1 & 1 & 0 \end{pmatrix}$$

Hence,

$$\text{Adj } A = B^t = \begin{pmatrix} 2 & -2 & 1 \\ 1 & 0 & 1 \\ 2 & -3 & 0 \end{pmatrix}$$

Definition 2.6.4 If A is invertible, then $\det(A) \neq 0$ and

$$A^{-1} = \frac{1}{\det(A)} \text{Adj } A$$

■ **Example 2.26** Find the inverse of the matrix

$$A = \begin{pmatrix} 3 & -3 & -2 \\ 1 & -1 & -1 \\ -3 & 4 & 2 \end{pmatrix}$$

if exist. ■

Solution: $\det(A) = 3(-2+4) - (-3)(2-3) + (-2)(4-3) = 1$ and $\text{Adj } A = \begin{pmatrix} 2 & -2 & 1 \\ 1 & 0 & 1 \\ 2 & -3 & 0 \end{pmatrix}$. Then

2.7.1 Solving System of Equation using Cramer's Rule

Consider a system of linear equation consisting of n equations with n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots & \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned} \quad (2.2)$$

- (a) The system (2.2) has a unique solution if the determinant of the coefficient matrix ($D = \det(A)$) is nonzero. Hence, the solution is given by

$$x_1 = \frac{D_1}{D}, x_2 = \frac{D_2}{D}, \dots, x_n = \frac{D_n}{D}$$

where D_k is the determinant obtained from D by replacing the k^{th} column in D by the column with the entries $b_1, b_2, \dots, b_n$.

- (b) If the system is homogeneous and $D \neq 0$, it has only the trivial solutions $x_1 = x_2 = \cdots = x_n = 0$. If $D = 0$, the homogeneous system has non-trivial solution.

■ **Example 2.28** Solve the system

$$\begin{aligned} x - y &= 1 \\ -2x + 5y + z &= 3 \\ -x + 3y + z &= 2 \end{aligned}$$

Solution: The coefficient matrix is $A = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 5 & 1 \\ -1 & 3 & 1 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$

$$D = \det(A) = \begin{vmatrix} 1 & -1 & 0 \\ -2 & 5 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 1, \quad D_1 = \begin{vmatrix} 1 & -1 & 0 \\ 3 & 5 & 1 \\ 2 & 3 & 1 \end{vmatrix} = 3$$

$$D_2 = \begin{vmatrix} 1 & 1 & 0 \\ -2 & 3 & 1 \\ -1 & 2 & 1 \end{vmatrix} = 2, \quad D_3 = \begin{vmatrix} 1 & -1 & 1 \\ 3 & 5 & 3 \\ 2 & 3 & 2 \end{vmatrix} = -1$$

$$\text{Hence, } x = \frac{D_1}{D} = \frac{3}{1} = 3, \quad y = \frac{D_2}{D} = \frac{2}{1} = 2, \quad z = \frac{D_3}{D} = -1$$

2.7.2 Solving System of Equation using Gaussian Elimination Method

The method is based on the idea of reducing the given system of equations $Ax = b$, to an upper triangular system of equations $Ux = z$, using elementary row operations.

Consider the system of equation

$$AX = b$$

To solve this system:

Step 1: Augment the coefficient matrix with the constant (RHS) value i.e., $A|b$

Step 2: Perform elementary row operation to change the augmented matrix into upper triangular matrix.

- If $\text{Rank}(A) = \text{Rank}(A|b) = n$, then the system has a unique solution.
- If $\text{Rank}(A) = \text{Rank}(A|b) < n$, then the system has infinitely many solution.
- If $\text{Rank}(A) < \text{Rank}(A|b)$, then the system has no solution.

Step 3: Use back substitution to find the unknown values.

■ **Example 2.29** Solve the system of linear equations

$$3x - 2y + z = -1$$

$$5x - 4y - z = 3$$

$$-2x + y - z = 2$$

Solution: Since , the solution to the given system is $x = -2, y = -3$, and $z = -1$.

■ **Example 2.30** Solve the system of linear equation by using Gaussian elimination methods.

$$x - 3y + z = 2$$

$$2x - y - z = 9$$

$$-3x + 14y - 6z = -1$$

Solution: The coefficient matrix is $A = \begin{pmatrix} 1 & -3 & 1 \\ 2 & -1 & -1 \\ -3 & 14 & -6 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $b = \begin{pmatrix} 2 \\ 9 \\ -1 \end{pmatrix}$ Write the augmented matrix and apply row operations

$$\begin{aligned} A|b &\sim \left(\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 2 & -1 & -1 & 9 \\ -3 & 14 & -6 & -1 \end{array} \right) \xrightarrow[R_3 \rightarrow R_3 + 3R_1]{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 0 & 5 & -3 & 5 \\ 0 & 5 & -3 & 5 \end{array} \right) \\ &\sim \xrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 0 & 5 & -3 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

The given system reduced to

$$x - 3y + z = 2$$

$$5y - 3z = 5, \quad 0 = 0$$

The system has infinitely many solution.

Let $z = t$

$$\Rightarrow 5y = 5 + 3z \Rightarrow y = 1 + \frac{3}{5}t \quad \text{and} \quad x = 2 + 3y - z = 2 + 3\left(1 + \frac{3}{5}t\right) - t = 5 + \frac{9}{5}t - t = 5 + \frac{4}{5}t$$

where t is any real number. Hence,

$$x = 5 + \frac{4}{5}t, \quad y = 1 + \frac{3}{5}t, \quad \text{and} \quad z = t \quad \forall t \in \mathbb{R}$$

For example, if $t = 0$ then $x = 5, y = 1, z = 0$, if $t = -5$ then $x = 1, y = -2, z = -5$

■ **Example 2.31** Solve the system of linear equation by using Gaussian elimination methods.

$$x - 3y + z = 2$$

$$2x - y - z = 5$$

$$-3x + 14y - 6z = 1$$

Solution: The coefficient matrix is $A = \begin{pmatrix} 1 & -3 & 1 \\ 2 & -1 & -1 \\ -3 & 14 & -6 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $b = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$. Write the augmented matrix and apply row operations

$$\begin{aligned} A|b &\sim \left(\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 2 & -1 & -1 & 5 \\ -3 & 14 & -6 & 1 \end{array} \right) \xrightarrow[R_3 \rightarrow R_3 + 3R_1]{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 0 & 5 & -3 & 1 \\ 0 & 5 & -3 & 7 \end{array} \right) \\ &\sim \xrightarrow{R_3 \rightarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & -3 & 1 & 2 \\ 0 & 5 & -3 & 5 \\ 0 & 0 & 0 & 6 \end{array} \right) \end{aligned}$$

The given system reduced to

$$\begin{aligned} x - 3y + z &= 2 \\ 5y - 3z &= 5 \\ 0 &= 6 \text{ (impossible)} \end{aligned}$$

Therefore, the given system has no solutions.

Solving a System of linear Equations Using Inverse Method

Consider the system of linear equation

$$Ax = b \quad (2.3)$$

where A is invertible square matrix. Multiply equation (2.3) by A^{-1} we get

$$x = A^{-1}b$$

which is the unique solution of the given system.

■ Example 2.32 Solve

$$\begin{aligned} 2x + 4y - 2z &= 6 \\ -4x - 6y + z &= 1 \\ 3x + 5y - z &= -1 \end{aligned}$$

Solution: The coefficient matrix is $A = \begin{pmatrix} 2 & 4 & -2 \\ -4 & -6 & 1 \\ 3 & 5 & -1 \end{pmatrix}$.

From example (2.23) the inverse of A is

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & -3 & -4 \\ -\frac{1}{2} & 2 & 3 \\ -1 & 1 & 2 \end{pmatrix}$$

Using this result, we find that the solution of the given system is

$$X = A^{-1}b = \begin{pmatrix} \frac{1}{2} & -3 & -4 \\ -\frac{1}{2} & 2 & 3 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 6 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 - 3 + 4 \\ -3 + 2 - 3 \\ -6 + 1 - 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ -7 \end{pmatrix}$$

OR $x = 4$, $y = -4$, $z = -7$

Definition of Limit
 Basic Limit Theorems
 One sided Limits
 Infinite Limits, Limit at infinity and Asymptotes
 Infinite limits and Vertical Asymptotes
 Limit at infinity and Horizontal asymptotes
 Continuity of a function
 One sided continuity
 Continuity on Intervals
 Intermediate Value Theorem (IVT)

3 — LIMIT AND CONTINUITY

3.1 Definition of Limit

Definition 3.1.1 Formal Definition of Limit Let f be a function defined on an open interval containing a , with the possible exception of a itself. Then the limit of $f(x)$ as x approaches a is the number L , written as

$$\lim_{x \rightarrow a} f(x) = L \quad \text{iff}$$

for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that if

$$0 < |x - a| < \delta, \quad \text{then} \quad |f(x) - L| < \varepsilon.$$

A Geometric Interpretation

Let $\varepsilon > 0$ be given. Draw the lines $y = L + \varepsilon$ and $y = L - \varepsilon$. Since $|f(x) - L| < \varepsilon$ is equivalent to $L - \varepsilon < f(x) < L + \varepsilon$, $\lim_{x \rightarrow a} f(x) = L$ exists provided that we can find a number δ such that if we restrict x to lie in the interval $(a - \delta, a + \delta)$ with $x \neq a$, then the graph of $y = f(x)$ lies inside the band of width 2ε determined by the lines $y = L + \varepsilon$ and $y = L - \varepsilon$. (See Figure 3.1) You can see from Figure 3.1 that once a number $\delta > 0$ has been found, then any number smaller than δ will also satisfy the requirement.

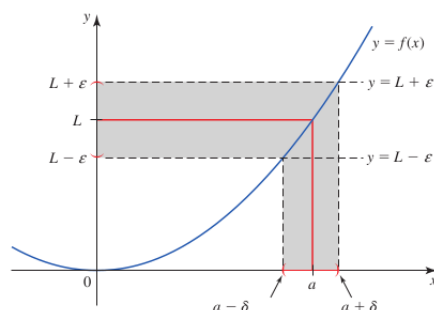


Figure 3.1:

1. δ is a function of ε
2. δ is not unique
3. For all $\delta' < \delta$ it is true

■ **Example 3.1** Show that

$$\lim_{x \rightarrow 4} (3x - 4) = 8$$

Proof: Let $\varepsilon > 0$ be given. We must find $\delta > 0$ such that

$$0 < |x - 4| < \delta \Rightarrow |f(x) - 8| < \varepsilon$$

Consider

$$\begin{aligned} |f(x) - 8| < \varepsilon &\Rightarrow |(3x - 4) - 8| < \varepsilon \\ &\Rightarrow |3x - 12| < \varepsilon \\ &\Rightarrow |x - 4| < \frac{\varepsilon}{3} \end{aligned}$$

Now choose $\delta = \frac{\varepsilon}{3}$. Thus

$$0 < |x - 4| < \delta \Rightarrow |(3x - 4) - 8| = |3x - 12| = 3|x - 4| < 3 \frac{\varepsilon}{3} = \varepsilon$$

■ **Example 3.2** Show that $\lim_{x \rightarrow 2} (x^2 - 1) = 3$

Proof: Let $\varepsilon > 0$ be given. We must find $\delta > 0$ such that

$$0 < |x - 2| < \delta \Rightarrow |f(x) - 3| < \varepsilon$$

Consider

$$\begin{aligned} |f(x) - 3| < \varepsilon &\Rightarrow |(x^2 - 1) - 3| < \varepsilon \\ &\Rightarrow |x^2 - 4| < \varepsilon \\ &\Rightarrow |(x - 2)(x + 2)| < \varepsilon \end{aligned}$$

Let $\delta = 1$, thus,

$$\begin{aligned} |x - 2| < 1 &\Rightarrow -1 < x - 2 < 1 \\ &\Rightarrow 3 < x + 2 < 5 \\ \text{Hence, } &|x + 2| < 5 \end{aligned}$$

Thus we have

$$|x^2 - 4| = |x + 2||x - 2| < 5|x - 2| < \varepsilon \Rightarrow |x - 2| < \frac{\varepsilon}{5}$$

Now choose $\delta = \min(\frac{\varepsilon}{5}, 1)$

Thus

$$0 < |x - 2| < \delta \Rightarrow |(x^2 - 1) - 3| < 5|x - 2| < 5 \frac{\varepsilon}{5} = \varepsilon$$

■ **Example 3.3** Using $\varepsilon - \delta$ definition of limit prove that ■

1. $\lim_{x \rightarrow 3} (x^2 + x) = 12$
2. $\lim_{x \rightarrow -3} (x^2 + x) = 6$
3. $\lim_{x \rightarrow 3} \frac{2}{x+3} = \frac{1}{3}$
4. $\lim_{x \rightarrow 9} (2 + \sqrt{x}) = 5$

Proof. 1. Let $\varepsilon > 0$ be given. We must find $\delta > 0$ such that

$$0 < |x - 3| < \delta \Rightarrow |(x^2 + x) - 12| < \varepsilon$$

Consider

$$\begin{aligned} |f(x) - 12| < \varepsilon &\Rightarrow |(x^2 + x) - 12| < \varepsilon \\ &\Rightarrow |(x+4)(x-3)| < \varepsilon \\ &\Rightarrow |x+4||x-3| < \varepsilon \end{aligned}$$

Take $\delta = 1$, then

$$\begin{aligned} |x-3| < \delta &\Rightarrow |x-3| < 1 \\ &\Rightarrow -1 < x-3 < 1 \\ &\Rightarrow 2 < x < 4 \\ &\Rightarrow 6 < |x+4| < 8 \end{aligned}$$

Thus we have

$$|f(x) - 12| = |x+4||x-3| < 8|x-3| < \varepsilon \quad \text{iff} \quad |x-3| < \frac{\varepsilon}{8}$$

Now choose $\delta = \min(1, \frac{\varepsilon}{8})$

Thus,

$$0 < |x-3| < \delta \Rightarrow |x^2 + x - 12| = |x+4||x-3| < 8 \frac{\varepsilon}{8} = \varepsilon$$

2. Let $\varepsilon > 0$ be given. We must find $\delta > 0$ such that

$$0 < |x - (-3)| < \delta \Rightarrow |(x^2 + x) - 6| < \varepsilon$$

Consider

$$\begin{aligned} |f(x) - 6| < \varepsilon &\Rightarrow |(x^2 + x) - 6| < \varepsilon \\ &\Rightarrow |(x-2)(x+3)| < \varepsilon \\ &\Rightarrow |x+3||x-2| < \varepsilon \end{aligned}$$

Take $\delta = 1$, then

$$\begin{aligned} |x+3| < \delta &\Rightarrow |x+3| < 1 \\ &\Rightarrow -1 < x+3 < 1 \\ &\Rightarrow -4 < x < -2 \\ &\Rightarrow -6 < x-2 < -4 \\ &\Rightarrow 4 < 2-x < 6 \\ &\Rightarrow |2-x| < 6 \end{aligned}$$

Thus we have

$$|f(x) - 6| = |x+3||x-2| < 6|x+3| < \varepsilon \quad \text{iff} \quad |x+3| < \frac{\varepsilon}{6}$$

Now choose $\delta = \min(1, \frac{\varepsilon}{6})$

Thus,

$$0 < |x-3| < \delta \Rightarrow |x^2 + x - 6| = |x+3||x-2| < 6 \frac{\varepsilon}{6} = \varepsilon$$

$$\therefore \lim_{x \rightarrow -3} (x^2 + x) = 6$$

3. Let $\varepsilon > 0$ be given. We must find δ such that

$$0 < |x-3| < \delta \Rightarrow \left| f(x) - \frac{1}{3} \right| < \varepsilon$$

Consider

$$\begin{aligned} \left| f(x) - \frac{1}{3} \right| < \varepsilon &\Rightarrow \left| \frac{2}{x+3} - \frac{1}{3} \right| < \varepsilon \\ &\Rightarrow \left| \frac{6 - (x+3)}{3(x+3)} \right| < \varepsilon \\ &\Rightarrow \frac{1}{3} \left| \frac{x-3}{x+3} \right| < \varepsilon \end{aligned}$$

Take $\delta = 1$, then

$$\begin{aligned} |x-3| < \delta &\Rightarrow |x-3| < 1 \\ &\Rightarrow -1 < x-3 < 1 \\ &\Rightarrow 2 < x < 4 \\ &\Rightarrow 5 < |x+3| < 7 \\ &\Rightarrow \frac{1}{7} < \frac{1}{|x+3|} < \frac{1}{5} \end{aligned}$$

Thus we have

$$\frac{1}{3} |x-3| \frac{1}{|x+3|} < \frac{1}{15} |x-3| < \varepsilon \quad \text{iff} \quad |x-3| < 15\varepsilon$$

Now choose $\delta = \min(1, 15\varepsilon)$

Thus,

$$0 < |x-3| < \delta \Rightarrow \left| \frac{2}{x+3} - \frac{1}{3} \right| = \frac{1}{3} \left| \frac{x-3}{x+3} \right| < \frac{1}{15} 15\varepsilon = \varepsilon$$

4. Let $\varepsilon > 0$ be given. We must find δ such that

$$0 < |x-9| < \delta \Rightarrow |f(x) - 5| < \varepsilon$$

Consider

$$\begin{aligned} |f(x) - 5| < \varepsilon &\Rightarrow |(2 + \sqrt{x}) - 5| < \varepsilon \\ &\Rightarrow |\sqrt{x} - 3| < \varepsilon \\ &\Rightarrow \left| \frac{x-9}{\sqrt{x}+3} \right| < \varepsilon \end{aligned}$$

Take $\delta = 1$, then

$$\begin{aligned} |x-9| < \delta &\Rightarrow |x-9| < 1 \\ &\Rightarrow -1 < x-9 < 1 \\ &\Rightarrow 8 < x < 10 \\ &\Rightarrow \sqrt{8} + 3 < \sqrt{x} + 3 < \sqrt{10} + 3 \\ &\Rightarrow \frac{1}{\sqrt{10} + 3} < \frac{1}{|\sqrt{x} + 3|} < \frac{1}{\sqrt{8} + 3} \end{aligned}$$

Thus we have

$$|x-9| \frac{1}{|\sqrt{x} + 3|} < \frac{1}{\sqrt{8} + 3} |x-9| < \varepsilon \quad \text{iff} \quad |x-9| < (\sqrt{8} + 3)\varepsilon$$

Now choose $\delta = \min(1, (\sqrt{8} + 3)\varepsilon)$

Thus,

$$\begin{aligned} 0 < |x-9| < \delta &\Rightarrow |(2 + \sqrt{x}) - 5| = \left| \frac{x-9}{\sqrt{x}+3} \right| \\ &< \frac{1}{\sqrt{8} + 3} (\sqrt{8} + 3)\varepsilon = \varepsilon \end{aligned}$$

■

3.2 Basic Limit Theorems

Theorem 3.2.1 If m and b are any constants, then $\lim_{x \rightarrow a} (mx + b) = ma + b$

Theorem 3.2.2 If c is a constant, then for any number a , $\lim_{x \rightarrow a} c = c$

Theorem 3.2.3 If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ then;

1. $\lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L \pm M$
2. $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L \cdot M$
3. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}$ provided that $M \neq 0$

Theorem 3.2.4 If $\lim_{x \rightarrow a} f(x) = L$ and n is any positive integer, then

$$\lim_{x \rightarrow a} [f(x)]^n = L^n$$

Theorem 3.2.5 If $\lim_{x \rightarrow a} f(x) = L$, then

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L}$$

if $L > 0$ and n is any positive integer, or if $L \leq 0$ and n is a positive odd integer

Theorem 3.2.6 If f and g are functions such that $\lim_{x \rightarrow a} g(x) = L$ and $\lim_{x \rightarrow a} f(x) = f(L)$, then

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(L)$$

■ **Example 3.4** Evaluate the following limit

- | | | |
|--|---|---|
| (a) $\lim_{x \rightarrow -1} (3x^2 - 2x + 1)^3$ | (d) $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$ | (g) $\lim_{x \rightarrow 5} \sqrt[3]{3x^2 - 4x + 9}$ |
| (b) $\lim_{x \rightarrow 1} \frac{3x^3 - 5x^2 - 1}{2x - 3}$ | (e) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - \sqrt{1-x}}{x}$ | (h) $\lim_{x \rightarrow 9} \frac{x^3 - 1}{x - 1}$ |
| (c) $\lim_{x \rightarrow 2} \frac{2x^2 - 5x + 2}{5x^2 - 7x - 6}$ | (f) $\lim_{x \rightarrow 8} \frac{x^{2/3} + 3\sqrt{x}}{4 - 16/x}$ | (i) $\lim_{x \rightarrow 1} \frac{3x - 3}{\sqrt[3]{x+7} - 2}$ |

Solution: (i) Let $t = \lim_{x \rightarrow 1} \sqrt[3]{x+7} \Rightarrow t^3 = x+7 \Rightarrow x = t^3 - 7$

Observe that as x approaches 1, t approaches 2. Therefore

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{3x - 3}{\sqrt[3]{x+7} - 2} &= \lim_{t \rightarrow 2} \frac{3(t^3 - 7) - 3}{t - 2} = \lim_{t \rightarrow 2} \frac{3t^3 - 24}{t - 2} \\ &= \lim_{t \rightarrow 2} \frac{3(t-2)(t^2 + 2t + 4)}{t - 2} = \lim_{t \rightarrow 2} 3(t^2 + 2t + 4) = 36 \end{aligned}$$

3.3 One sided Limits

Definition 3.3.1 1. Let f be a function which is defined at every number in some open interval (a, c) . Then the limit of $f(x)$, as x approaches a from the right, is L , written

$$\lim_{x \rightarrow a^+} f(x) = L$$

if for any $\varepsilon > 0$, however small, there exists a $\delta > 0$ such that,

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < x - a < \delta$$

2. Let f be a function which is defined at every number in some open interval (d, a) . Then the limit of $f(x)$, as x approaches a from the left, is L , written

$$\lim_{x \rightarrow a^-} f(x) = L$$

if for any $\varepsilon > 0$, however small, there exists a $\delta > 0$ such that,

$$|f(x) - L| < \varepsilon \text{ whenever } -\delta < x - a < 0$$

Theorem 3.3.1 If f is defined throughout an open interval containing a , except possibly at a itself, then $\lim_{x \rightarrow a} f(x) = L$ if and only if both $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$.

■ **Example 3.5** Find $\lim_{x \rightarrow 2^+} (1 + \sqrt{x-2})$ ■

■ **Example 3.6** Suppose $f(x) = \frac{|x|}{x}$ if $x \neq 0$ and $f(0) = 1$. Find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$. What is $\lim_{x \rightarrow 0} f(x)$? ■

■ **Example 3.7** Suppose $f(x) = \begin{cases} x+3 & \text{if } x \neq 1 \\ 2, & \text{if } x = 1 \end{cases}$ ■

■ **Example 3.8** Suppose $f(x) = \begin{cases} -x+3 & \text{if } x \leq 2 \\ \sqrt{x-2}, & \text{if } x \geq 2 \end{cases}$ ■

The Squeeze Theorem

Theorem 3.3.2 The Squeeze Theorem

Suppose that $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing a , except possibly at a , and

$$\lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} h(x)$$

Then

$$\lim_{x \rightarrow a} g(x) = L$$

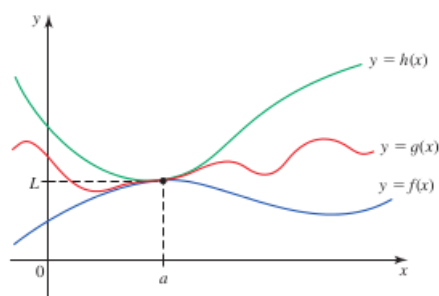


Figure1: An illustration of the Squeeze Theorem

■ **Example 3.9** Find $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ ■

Solution: Since $-1 \leq \sin t \leq 1$ for every real number t , we have

$$-1 \leq \sin \frac{1}{x} \leq 1 \quad \text{for every } x \neq 0$$

Therefore

$$-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2, \quad x \neq 0$$

Let $f(x) = -x^2$, $g(x) = x^2 \sin \frac{1}{x}$, and $h(x) = x^2$. Then

$$f(x) \leq g(x) \leq h(x).$$

Since

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (-x^2) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} x^2 = 0$$

The squeeze theorem implies that

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

Theorem 3.3.3

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

Proof: First, suppose that $0 < x < \frac{\pi}{2}$. The Figure below shows a sector of a circle of radius 1.

The area A of a sector with radius r and a central angle of θ radians can be derived by:

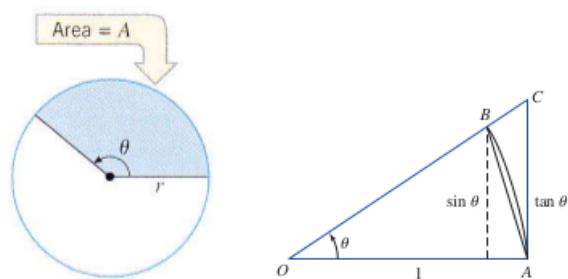
$$\frac{A}{\pi r^2} = \frac{\theta}{2\pi} \left[\frac{\text{area of the sector}}{\text{area of the circle}} = \frac{\text{central angle of the sector}}{\text{central angle of the circle}} \right]$$

From the figure we see that

$$\text{Area of } \triangle OAB = \frac{1}{2} \text{base} \cdot \text{height} = \frac{1}{2}(1) \sin \theta = \frac{1}{2} \sin \theta$$

$$\text{Area of sector } OAB = \frac{1}{2} r^2 \theta = \frac{1}{2} \theta$$

$$\text{Area of } \triangle OAC = \frac{1}{2} \text{base} \cdot \text{height} = \frac{1}{2}(1) \tan \theta = \frac{1}{2} \tan \theta$$



Since $0 < \text{Area of } \triangle OAB < \text{Area of sector } OAB < \text{Area of } \triangle OAC$, we have,

$$0 < \frac{1}{2} \sin \theta < \frac{1}{2} \theta < \frac{1}{2} \tan \theta$$

Multiplying through by $(2/\sin \theta)$ and keeping in mind that $\sin \theta > 0$ and $\cos \theta > 0$ for $0 < \theta < \pi/2$ we obtain

$$1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta}$$

or, upon taking reciprocals,

$$\cos \theta < \frac{\sin \theta}{\theta} < 1$$

Let $f(\theta) = \cos \theta$, $g(\theta) = \frac{\sin \theta}{\theta}$, $h(\theta) = 1$ Observe that,

$$\lim_{\theta \rightarrow 0} f(\theta) = \lim_{\theta \rightarrow 0} \cos \theta = 1 = \lim_{\theta \rightarrow 0} h(\theta)$$

Then the squeeze Theorem implies that

$$\lim_{\theta \rightarrow 0} g(\theta) = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

Theorem 3.3.4

$$\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$$

■ Example 3.10 Find

$$\begin{aligned} \text{(a)} \quad & \lim_{x \rightarrow 0} \frac{\sin 2x}{3x} \\ \text{(b)} \quad & \lim_{x \rightarrow 0} \frac{\tan x}{x} \end{aligned}$$

$$\text{(c)} \quad \lim_{x \rightarrow 0} \frac{2x + 1 - \cos x}{4x}$$

Solution:

$$\begin{aligned} \text{(a)} \quad & \lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \lim_{x \rightarrow 0} \frac{2}{3} \frac{\sin 2x}{2x} = \frac{2}{3} (1) = \frac{2}{3} \\ \text{(b)} \quad & \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{1}{\cos x} \frac{\sin x}{x} = \frac{1}{\cos 0} (1) = 1 \\ \text{(c)} \quad & \lim_{x \rightarrow 0} \frac{2x + 1 - \cos x}{4x} = \lim_{x \rightarrow 0} \frac{2x}{4x} - \frac{\cos x - 1}{4x} = \frac{2}{4} - (1)(0) = \frac{1}{2} \end{aligned}$$

3.4 Infinite Limits, Limit at infinity and Asymptotes

3.4.1 Infinite limits and Vertical Asymptotes

Definition 3.4.1 Let f be defined on an interval that contains a except possibly at a itself. Then

1. $\lim_{x \rightarrow a} f(x) = \infty$ iff for every $M > 0$, there is a $\delta > 0$, such that

$$\text{if } 0 < |x - a| < \delta, \text{ then } f(x) > M$$

2. $\lim_{x \rightarrow a} f(x) = -\infty$ iff for every $N > 0$, there is a $\delta > 0$,

$$\text{if } 0 < |x - a| < \delta, \text{ then } f(x) < -N$$

Definition 3.4.2 The line $x = a$ is called a vertical asymptote of the graph of $y = f(x)$ if any one of the following limits holds true:

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \lim_{x \rightarrow a^+} f(x) = \pm\infty, \lim_{x \rightarrow a} f(x) = \pm\infty$$

■ **Example 3.11** Show that $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$ ■

Solution: Let $M > 0$ be given. We want to show that there exists a $\delta > 0$ such that

$$\frac{1}{x^2} > M \text{ whenever } 0 < |x - 0| < \delta.$$

To find δ , consider

$$\begin{aligned} \frac{1}{x^2} &> M \\ x^2 &< \frac{1}{M} \implies |x| < \frac{1}{\sqrt{M}} \end{aligned}$$

Choose $\delta = \frac{1}{\sqrt{M}}$. Thus,

$$|x| < \delta \implies x^2 < \delta^2$$

so
$$\frac{1}{x^2} > \frac{1}{\delta^2} \geq M$$

■ **Example 3.12**

(a) $\lim_{x \rightarrow 3^+} \frac{2x}{x-3} = \infty$ (c) $\lim_{x \rightarrow 3^+} \frac{x^2 + x + 2}{x^2 - 2x - 3} = \infty$

(b) $\lim_{x \rightarrow 3^-} \frac{2x}{x-3} = -\infty$

■

Therefore, the line $x = 3$ is the vertical asymptote of $f(x) = \frac{2x}{x-3}$

3.4.2 Limit at infinity and Horizontal asymptotes

Definition 3.4.3 1. Let f be defined on an interval (a, ∞) , then

$$\lim_{x \rightarrow \infty} f(x) = L \text{ iff}$$

for every $\varepsilon > 0$ there is a number $M > 0$ such that if $x > M$, then $|f(x) - L| < \varepsilon$.

2. Let f be defined on an interval $(-\infty, a)$, then

$$\lim_{x \rightarrow -\infty} f(x) = L \text{ iff}$$

for every $\varepsilon > 0$ there is a number $M < 0$ such that if $x < M$, then $|f(x) - L| < \varepsilon$.

Definition 3.4.4 The line $y = L$ is called horizontal asymptote of the graph of $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L, \text{ or } \lim_{x \rightarrow -\infty} f(x) = L$$

■ **Example 3.13** Find

$$(a) \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$(b) \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

$$(c) \lim_{x \rightarrow \infty} \frac{2x^2 - x + 1}{3x^2 + 2x - 1}$$

$$(d) \lim_{x \rightarrow -\infty} \frac{x^2 + 1}{x - 2}$$

$$(e) \lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{x^2 + 1}}$$

Therefore, the line $y = 0$ is the horizontal asymptote of $f(x) = \frac{1}{x}$

■ **Example 3.14** Find the vertical and horizontal asymptotes of the graph of

$$f(x) = \frac{\sqrt{9x^2 + 1}}{3x - 5}$$

Solution:

1. To find vertical asymptote

$$\lim_{x \rightarrow \frac{5}{3}^-} \frac{\sqrt{9x^2 + 1}}{3x - 5} = -\infty$$

Thus, the line $x = \frac{5}{3}$ is the vertical asymptote of $f(x)$

2. To find the horizontal Asymptote

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 1}}{3x - 5} &= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2(9 + \frac{1}{x^2})}}{3x - 5} \\ &= \lim_{x \rightarrow \infty} \frac{x\sqrt{9 + \frac{1}{x^2}}}{3x - 5}, \text{ since } x > 0, \sqrt{x^2} = x \\ &= \lim_{x \rightarrow \infty} \frac{x\sqrt{9 + \frac{1}{x^2}}}{x(3 - \frac{5}{x})} = \lim_{x \rightarrow \infty} \frac{\sqrt{9 + \frac{1}{x^2}}}{(3 - \frac{5}{x})} = \frac{\sqrt{9}}{3} = 1 \end{aligned}$$

Again,

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + 1}}{3x - 5} &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2(9 + \frac{1}{x^2})}}{3x - 5} = \lim_{x \rightarrow -\infty} \frac{(-x)\sqrt{9 + \frac{1}{x^2}}}{3x - 5}, \text{ since } x < 0, \sqrt{x^2} = -x \\ &= \lim_{x \rightarrow -\infty} \frac{(-x)\sqrt{9 + \frac{1}{x^2}}}{x(3 - \frac{5}{x})} = \lim_{x \rightarrow -\infty} \frac{(-1)\sqrt{9 + \frac{1}{x^2}}}{(3 - \frac{5}{x})} = \frac{-\sqrt{9}}{3} = -1 \end{aligned}$$

Thus, the lines $y = 1$ and $y = -1$ are the horizontal asymptote of $f(x)$

Exercise 3.1 Find the asymptote of the following functions

1. $f(x) = \frac{2x}{1-x^2}$
2. $g(x) = \frac{2x-5}{x^3-2x^2+x}$

■

3.5 Continuity of a function

Definition 3.5.1 A function f is said to be continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$

Note: If f is continuous at a , then

1. $f(a)$ is defined
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

If f is not continuous at a , then we say that f is discontinuous at a .

■ **Example 3.15** Find the points of discontinuity of

$$1. f(x) = \frac{x^2 - x - 2}{x - 2}$$

$$2. f(x) = \begin{cases} \frac{1}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

$$3. f(x) = \frac{2x}{1-x^2}$$

$$4. f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2}, & \text{if } x \neq 2 \\ 1, & \text{if } x = 2 \end{cases}$$

■

3.5.1 One sided continuity

Definition 3.5.2 1. f is said to be continuous from the right at a if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

2. f is said to be continuous from the left at a if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

■ **Example 3.16** Let $f(x) = \begin{cases} 2x+1 & \text{if } x < 1 \\ x^2-2 & \text{if } x \geq 1 \end{cases}$

Determine whether f is continuous from the right at 1 and/or from the left at 1. ■

Solution: Because

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2x+1) = 3 \neq f(1) = -1.$$

Hence, f is not continuous from the left at 1

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2x+1) = -1 = f(1) = -1.$$

Hence, f is continuous from the right at 1.

(R) A function f is continuous at a if and only if f is simultaneously continuous from the right and from the left at a .

3.5.2 Continuity on Intervals

Definition 3.5.3 A function f is said to be **continuous on an open interval** (a, b) if it is continuous at each point of (a, b) . A function f is **continuous on a closed interval** $[a, b]$ if it is continuous on (a, b) and is also continuous from the right at a and from the left at b . A function f is **continuous on a half-open interval** $[a, b)$ or $(a, b]$ if f is continuous on (a, b) and f is continuous from the right at a or f is continuous from the left at b , respectively.

■ **Example 3.17** Show that the function f defined by $f(x) = \sqrt{9 - x^2}$ is continuous on the closed interval $[-3, 3]$. ■

Solution: We first show that f is continuous on $(-3, 3)$. Let a be any number in $(-3, 3)$. Then, using the laws of limits, we have

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \sqrt{9 - x^2} = \sqrt{9 - a^2} = f(a)$$

and this proves the assertion.

Next, let us show that f is continuous from the right at -3 and from the left at 3 .

Again, by invoking the limit properties, we see that

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \sqrt{9 - x^2} = \sqrt{9 - 3^2} = 0 = f(-3)$$

and

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sqrt{9 - x^2} = \sqrt{9 - 3^2} = 0 = f(3) \quad \text{and this proves the}$$

assertion. Therefore, f is continuous on $[-3, 3]$.

Theorem 3.5.1 If f and g are continuous at a , and c is a constant then the following are also continuous at a .

- (a) $f \pm g$, (b) cf , (c) fg , (d) $\frac{f}{g}$, if $g(a) \neq 0$

Theorem 3.5.2 The following functions:

Polynomial, Rationals, Root functions, Trigonometric functions, Inverse trigonometric functions, exponential functions and Logarithmic functions are continuous on their domain.

Theorem 3.5.3 If $\lim_{x \rightarrow a} g(x) = b$ and f is continuous at b , then

$$\lim_{x \rightarrow a} f(g(x)) = f(b) \text{ i.e., } \lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$$

Theorem 3.5.4 If g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a i.e.,

$$\lim_{x \rightarrow a} f(g(x)) = f(g(a))$$

■ **Example 3.18** Find the constant a and b such that

$$f(x) = \begin{cases} ax + 3, & \text{if } x > 1 \\ 4 & \text{if } x = 1 \\ x^2 + b & \text{if } x < 1 \end{cases}$$

is continuous at $x = 1$ ■

Solution: f is continuous at a iff $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) = f(1) &\Rightarrow \lim_{x \rightarrow 1^+} (ax + 3) = f(1) \\ &\Rightarrow a + 3 = 4 \Rightarrow a = 4 - 3 = 1 \end{aligned}$$

and

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) = f(1) &\Rightarrow \lim_{x \rightarrow 1^-} (x^2 + b) = f(1) \\ &\Rightarrow 1 + b = 4 \Rightarrow b = 4 - 1 = 3 \end{aligned}$$

Exercise 3.2 Determine whether f is continuous or discontinuous at a .

1. $f(x) = \sqrt{x-2}$; $a = 2$
2. $f(x) = \begin{cases} \frac{|x-4|}{x-4} & \text{for } x \neq 4 \\ e^{4-x} & \text{for } x = 4 \end{cases} \quad a = 4$
3. $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 2 & \text{if } x = 0 \end{cases} \quad a = 0$

3.5.3 Intermediate Value Theorem (IVT)

Theorem 3.5.5 Intermediate Value Theorem (IVT)

If $f \in C[a, b]$ and k is any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$ then there exist a number c in (a, b) for which $f(c) = k$

Theorem 3.5.6 IVT for Locating root

If $f(x)$ is continuous on $[a, b]$, $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one number x_0 in (a, b) such that $f(x_0) = 0$.

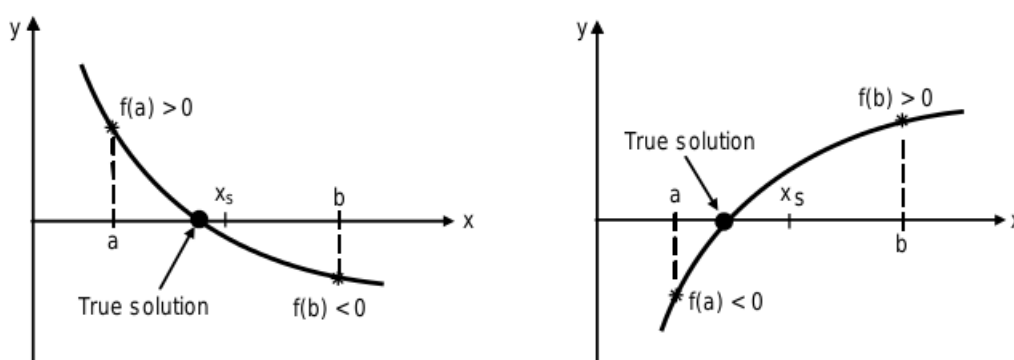


Figure 3.2: Solution of $f(x) = 0$ between $x = a$ and $x = b$

■ **Example 3.19** Show that the equation

$$x^5 - 2x^4 - 2x^3 + 8x^2 - 3x - 3$$

has a solution between 1 and 2

Solution: Let

$$f(x) = x^5 - 2x^4 - 2x^3 + 8x^2 - 3x - 3$$

Since f is a polynomial function, then f is continuous on $\mathbb{R}$.

Note that $f(1) = -1$ and $f(2) = 7$

Thus, $f(1) < 0 < f(2)$

i.e., $k = 0$ is between $f(1)$ and $f(2)$

So, by the IVT there is a number c between 1 and 2 such that $f(c) = 0$.

Exercise 3.3 Show that the equation given below has at least one real root.

1. $2x^3 + x^2 - x + 1 = 5$ in $[1, 2]$
2. $x + \tan x = 1$ in $[0, \frac{\pi}{4}]$
3. $x^5 - x^2 + 2x + 3 = 0$ in $[-1, 0]$

- Tangent Line
- Derivatives
- Basic Rules of Differentiation
- Derivatives of Trigonometric Functions
- Chain Rule
- Higher Derivatives
- Implicit Differentiation
- Related Rate
- Derivatives of Inverse Trigonometric Functions
- Hyperbolic Functions
- Derivatives of Hyperbolic Functions
- Derivatives of Inverse Hyperbolic Functions
- Applications of Derivatives
 - Extrema of a Function f
 - The Mean Value Theorem
 - Increasing and Decreasing Functions
 - and The First Derivative Test
 - Concavity and Inflection Point
 - Curve Sketching
- Indeterminate Forms and l'Hospital's Rule
 - The Indeterminate Form $0/0$ and ∞/∞
 - The Indeterminate Form $\infty \cdot 0$ and $0 \cdot \infty$
 - The Indeterminate Form $\infty - \infty$ and $0 - 0$

4 — Derivatives and Application of Derivatives

4.1 Tangent Line

Let P and Q be two distinct points on a curve, and consider the secant line passing through P and Q . (See Figure (4.1)). If we let Q move along the curve toward P , then the secant line rotates about P and approaches the fixed line T . We define T to be the tangent line at P on the curve. Let's make

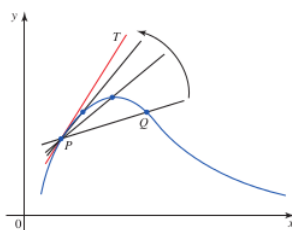


Figure 4.1: :As Q approaches P along the curve, the secant lines approach the tangent line T .

this notion more precise: Suppose that the curve is the graph of a function f defined by $y = f(x)$. (See Figure (4.2)). Let $P(a, f(a))$ be a point on the graph of f , and let Q be a point on the graph of f distinct from P . Then the x -coordinate of Q has the form $x = a + h$, where h is some appropriate nonzero number. If $h > 0$, then Q lies to the right of P ; and if $h < 0$, then Q lies to the left of P . The corresponding y -coordinate of Q is $y = f(a + h)$. In other words, we can specify Q in the usual manner by writing $Q(a + h, f(a + h))$. Observe that we can make Q approach P along the graph of f by letting h approach 0. This situation is illustrated in Figure (4.2)b. Next, using the formula

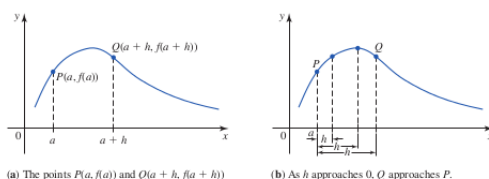


Figure 4.2:

for the slope of a line, we can write the slope of the secant line passing through $P(a, f(a))$ and $Q(a+h, f(a+h))$ as:

$$M = \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{f(a+h) - f(a)}{h} \quad (4.1)$$

The expression on the right-hand side of Equation 4.1 is called a difference quotient. As we observed earlier, if we let h approach 0, then Q approaches P and the secant line passing through P and Q approaches the tangent line T . This suggests that if the tangent line does exist at P , then its slope m should be the limit of M obtained by letting h approach zero. This leads to the following definition.

Definition 4.1.1 The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad (4.2)$$

provided that this limit exists.

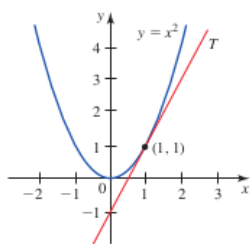
We can find an equation of the tangent line at P by using the point-slope form of an equation of a line. Thus,

$$y = m(x - a) + f(a)$$

Definition 4.1.2 The normal line to a curve at a given point is the line perpendicular to the tangent line at that Point.

■ **Example 4.1** Find an equation of the tangent line to the parabola $y = x^2$ at the point $(1, 1)$ ■

Solution: To find the slope of the tangent line at the point $P(1, 1)$, we use Equation 4.1 with $a = 1$, obtaining



$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^2 - (1)^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1+2h+h^2) - (1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} = \lim_{h \rightarrow 0} (h + 2) = 2 \end{aligned}$$

To find an equation of the tangent line, we use the point-slope form of an equation of a line to obtain

$$\begin{aligned} y - f(a) &= m(x - a) \Rightarrow y - 1 = 2(x - 1) \\ &\Rightarrow y = 2x - 1 \end{aligned}$$

■ **Example 4.2** Find an equation of the normal line to the curve $y = \sqrt{x-3}$ which is parallel to the line $6x + 3y - 4 = 0$. ■

Solution: Let l be the given line. To find the slope of l , we write its equation in the slope-intercept form, which is

$$y = -2x + \frac{4}{3}$$

Therefore, the slope of l is -2 and the slope of the desired normal line is also -2 because the two lines are parallel.

Now we need to find the slope of the given curve at any point $(a, f(a))$, we have

$$\begin{aligned} m &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{\sqrt{x-3} - \sqrt{a-3}}{x - a} \\ &= \lim_{x \rightarrow a} \left(\frac{\sqrt{x-3} - \sqrt{a-3}}{x - a} \right) \left(\frac{\sqrt{x-3} + \sqrt{a-3}}{\sqrt{x-3} + \sqrt{a-3}} \right) \\ &= \lim_{x \rightarrow a} \frac{(x-3) - (a-3)}{(x-a)(\sqrt{x-3} + \sqrt{a-3})} \\ &= \lim_{x \rightarrow a} \frac{1}{\sqrt{x-3} + \sqrt{a-3}} = \frac{1}{2\sqrt{a-3}} \end{aligned}$$

Because the normal line at a point is perpendicular to the tangent line at that point, the product of their slopes is -1 . Hence, the slope of the normal line at $(a, f(a))$ is given by $-2\sqrt{a-3}$. Since the slope of the desired line is -2 , so we solve the equation

$$-2\sqrt{a-3} = -2 \Rightarrow a = 4$$

Therefore, the desired line is the line through point $(4, 1)$ on the curve and has a slope of -2 . Using the point-slope form of an equation of a line we obtain,

$$y = -2(x-4) + 1 \Rightarrow y = -2x + 9$$

4.2 Derivatives

Definition 4.2.1 The derivative of a function f with respect to x is the function f' defined by the rule

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (4.3)$$

The domain of f' consists of all values of x for which the limit exists.

R If we let $x = a + h$ then $h = x - a$ hence $h \rightarrow 0$ iff $x \rightarrow a$.
Therefore,

$$f'(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Two interpretations of the derivative follow.

1. **Geometric Interpretation of the Derivative:** The derivative f' of a function f is a measure of the slope of the tangent line to the graph of f at any point $(x, f(x))$, provided that the derivative exists.
2. **Physical Interpretation of the Derivative:** The derivative f' of a function f measures the instantaneous rate of change of f at x . (See Figure 4.3.)

Note:

1. The value of the derivative of f at a is denoted by $f'(a)$
2. The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$.

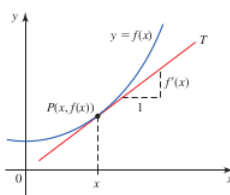


Figure 4.3: $f'(x)$ is the slope of T at P; $f(x)$ is changing at the rate of $f'(x)$ units per unit change in x at x .

■ **Example 4.3** Let $f(x) = \sqrt{5-x^2}$. Find

1. $f'(x)$ and determine the domain of the derivative function
2. an equation of the tangent line to the graph of $f(x)$ at $x = -1$

Solution:

1. From the above definition we have,

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{5-(x+h)^2} - \sqrt{5-x^2}}{h} \\
 &= \lim_{h \rightarrow 0} \left(\frac{\sqrt{5-(x^2+2xh+h^2)} - \sqrt{5-x^2}}{h} \right) \left(\frac{\sqrt{5-(x^2+2xh+h^2)} + \sqrt{5-x^2}}{\sqrt{5-(x^2+2xh+h^2)} + \sqrt{5-x^2}} \right) \\
 &= \lim_{h \rightarrow 0} \frac{(5-x^2-2xh-h^2) - (5-x^2)}{h(\sqrt{5-(x^2+2xh+h^2)} + \sqrt{5-x^2})} \\
 &= \lim_{h \rightarrow 0} \frac{-2xh-h^2}{h(\sqrt{5-(x^2+2xh+h^2)} + \sqrt{5-x^2})} \\
 &= \lim_{h \rightarrow 0} \frac{-2x-h}{\sqrt{5-(x^2+2xh+h^2)} + \sqrt{5-x^2}} = \frac{-2x}{\sqrt{5-x^2} + \sqrt{5-x^2}} \\
 &= \frac{-x}{\sqrt{5-x^2}}
 \end{aligned}$$

Since $f'(x)$ exist if $-\sqrt{5} < x < \sqrt{5}$, then the domain of $f'(x)$ is $(-\sqrt{5}, \sqrt{5})$

2. The slope m of the tangent line to the graph of $f(x)$ at $x = -1$ is

$$f'(-1) = \frac{-(-1)}{\sqrt{5-(-1)^2}} = \frac{1}{2}$$

■ **Example 4.4** Let $y = \frac{x-2}{-x+1}$. Find $\frac{dy}{dx}$

Solution:

$$\begin{aligned}
 \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{(x+h)-2}{-(x+h)+1} - \frac{x-2}{-x+1}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x+h-2)(-x+1) - (x-2)(-x-h+1)}{h(-x-h+1)(-x+1)} \\
 &= \lim_{h \rightarrow 0} \frac{-x^2+x+h-xh-2+2x-[-x^2+2x-xh+2h+x-2]}{h(-x-h+1)(-x+1)}
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{-h}{h(-x-h+1)(-x+1)} \\
 &= \lim_{h \rightarrow 0} \frac{-1}{(-x-h+1)(-x+1)} = \frac{-1}{(x-1)^2}
 \end{aligned}$$

■ **Example 4.5** Calculate the instantaneous velocity at time $t = 5$ of an automobile whose position at time t seconds is given by $g(t) = t^3 + 4t^2 + 10$ feet. ■

Solution: We know that the required instantaneous velocity is $g'(5)$. We calculate

$$\begin{aligned}
 g'(5) &= \lim_{h \rightarrow 0} \frac{g(5+h) - g(5)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(5+h)^3 + 4(5+h)^2 + 10 - [5^3 + 4 \cdot 5^2 + 10]}{h} \\
 &= \lim_{h \rightarrow 0} \frac{115h + 19h^2 + h^3}{h} \\
 &= \lim_{h \rightarrow 0} 115 + 19h + h^2 = 115
 \end{aligned}$$

We conclude that the instantaneous velocity of the moving body at time $t = 5$ is $g'(5) = 115$ ft/sec.

■ **Example 4.6** A rubber balloon is losing air steadily. At time t minutes the balloon contains $75 - 10t^2 + t$ cubic inches of air. What is the rate of loss of air in the balloon at time $t = 1$? ■

Solution: Let $f(t) = 75 - 10t^2 + t$. We calculate

$$\begin{aligned}
 f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-19h - 10h^2}{h} \\
 &= \lim_{h \rightarrow 0} (-19h - 10h^2) = -19
 \end{aligned}$$

In conclusion, the rate of air loss in the balloon at time $t = 1$ is $f'(1) = -19$ ft³/sec. Observe that the negative sign in this answer indicates that the change is negative, i.e., that the quantity is decreasing.

Other Notation:

If we denote the dependent variable by y so that $y = f(x)$, then some common alternative notations for the derivative are as follows:

$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$ The value of the derivative at a is denoted by $\left. \frac{dy}{dx} \right|_{x=a}$

Definition 4.2.2 A function f is differentiable at a if $f'(a)$ exists.

Definition 4.2.3 A function is said to be differentiable if it is differentiable at every number in its domain.

■ **Example 4.7** Show that the function $f(x) = |x|$ is differentiable everywhere except at 0. ■

Solution: To prove that f is not differentiable at 0, we will show that $f'(0)$ does not exist by demonstrating that the one-sided limits of the quotient

$$\frac{f(0+h) - f(0)}{h} = \frac{|h| - 0}{h} = \frac{|h|}{h}$$

as h approaches 0 are not equal.

First, suppose $h > 0$. Then $|h| = h$, so $\lim_{h \rightarrow 0} \frac{|h|}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1$

Next, if $h < 0$, then $|h| = -h$, and therefore,

$$\lim_{h \rightarrow 0} \frac{|h|}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = \lim_{h \rightarrow 0} -1 = -1$$

Therefore

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

does not exist, and f is not differentiable at 0.

To show that f is differentiable at all other numbers, we rewrite $f(x)$ in the form

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

and then differentiate $f(x)$ to obtain

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Theorem 4.2.1 If a function f is differentiable at a then f is continuous at a

Proof. If x is in the domain of f and $x \neq 0$ then we can write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} (x - a) \quad \text{We have}$$

$$\lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} (x - a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) = f'(x) \cdot 0 = 0$$

So,

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(a) + (f(x) - f(a))] = \lim_{x \rightarrow a} f(a) + \lim_{x \rightarrow a} [f(x) - f(a)] = f(a) + 0 = f(a)$$

and this shows that f is continuous at a , as asserted. ■

4.3 Basic Rules of Differentiation

Theorem 4.3.1 — Derivative of a Constant Function.

If c is a constant, then

$$\frac{d}{dx}(c) = 0$$

Theorem 4.3.2 — The Power Rule.

If n is any real number and $f(x) = x^n$, then

$$f'(x) = \frac{d}{dx}(x^n) = nx^{n-1}$$

■ **Example 4.8** If

1. $f(x) = x^{12}$, then $f'(x) = \frac{d}{dx}(x^{12}) = 12x^{11}$
2. $f(x) = \frac{1}{x^5}$ then $f'(x) = \frac{d}{dx}\left(\frac{1}{x^5}\right) = -5x^{-5-1} = \frac{-5}{x^6}$

Theorem 4.3.3 — The Constant Multiple Rule.

If f is a differentiable function and c is a constant, then

$$\frac{d}{dx}[cf(x)] = cf'(x)$$

■ **Example 4.9** If $f(x) = -6x^2$, then $f'(x) = -12x$ ■

Theorem 4.3.4 — The Sum Rule.

If f and g are differentiable functions, then

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x) = f'(x) \pm g'(x)$$

■ **Example 4.10** Find the derivative of $f(x) = 6x^8 + 3x^4 - 4x^3 + x^2 - 5x + 9$ ■

Solution:

$$\begin{aligned} f'(x) &= \frac{d}{dx}[6x^8 + 3x^4 - 4x^3 + x^2 - 5x + 9] \\ &= 6\frac{d}{dx}(x^8) + 3\frac{d}{dx}(x^4) - 4\frac{d}{dx}(x^3) + \frac{d}{dx}(x^2) - 5\frac{d}{dx}(x) + \frac{d}{dx}(9) \\ &= 48x^7 + 12x^3 - 12x^2 + 2x - 5 \end{aligned}$$

Theorem 4.3.5 — The Product Rule.

If f and g are differentiable functions, then

$$\frac{d}{dx}[f(x)g(x)] = g(x)\frac{d}{dx}f(x) + f(x)\frac{d}{dx}g(x) = f'(x)g(x) + g'(x)f(x)$$

■ **Example 4.11** Find the derivative of $f(x) = (x^2 - 1)(\sqrt{x} - 2x)$ ■

Solution:

$$\begin{aligned} f'(x) &= (\sqrt{x} - 2x)\frac{d}{dx}(x^2 - 1) + (x^2 - 1)\frac{d}{dx}(\sqrt{x} - 2x) \\ &= (\sqrt{x} - 2x)(2x) + (x^2 - 1)\left[\frac{d}{dx}(\sqrt{x}) - \frac{d}{dx}(2x)\right] \\ &= 2x\sqrt{x} - 4x^2 + (x^2 - 1)\left[\frac{1}{2\sqrt{x}} - 2\right] \\ &= 2x\sqrt{x} - 4x^2 + \frac{x^2}{2\sqrt{x}} - 2x^2 - \frac{1}{2\sqrt{x}} + 2 \\ &= 2x\sqrt{x} - 6x^2 + \frac{x^2 - 1}{2\sqrt{x}} + 2 \end{aligned}$$

■ **Example 4.12** Find the derivative of $f(x) = (xe^x + x)(x^3 - \ln x)$ ■

Theorem 4.3.6 — The Quotient Rule.

If f and g are differentiable functions and $g(x) \neq 0$, then

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$$

■ **Example 4.13** Find the derivative of $f(x) = \frac{x^2 - x}{x^3 + 1}$ ■

Solution:

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{x^2 - x}{x^3 + 1} \right) = \frac{(x^3 + 1) \frac{d}{dx}(x^2 - x) - (x^2 - x) \frac{d}{dx}(x^3 + 1)}{(x^3 + 1)^2} \\ &= \frac{(x^3 + 1)(2x - 1) - (x^2 - x)(3x^2)}{(x^3 + 1)^2} = \frac{2x^4 - x^3 + 2x - 1 - (3x^4 - 3x^3)}{(x^3 + 1)^2} \\ &= \frac{5x^4 + 2x^3 + 2x - 1}{(x^3 + 1)^2} \end{aligned}$$

4.4 Derivatives of Trigonometric Functions

Derivatives of Sines and Cosines

Theorem 4.4.1 Derivatives of $\sin x$

$$\frac{d}{dx}(\sin x) = \cos x$$

Let $f(x) = \sin x$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cosh + \cos x \sinh - \sin x}{h} = \lim_{h \rightarrow 0} \frac{\sin x (\cosh - 1) + \cos x \sinh}{h} \\ &= \sin x \lim_{h \rightarrow 0} \frac{(\cosh - 1)}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sinh}{h} \\ &= \sin x \cdot 0 + \cos x \cdot 1 \\ &= \cos x \end{aligned}$$

Theorem 4.4.2 — Derivative of $\cos x$.

$$\frac{d}{dx} \cos x = -\sin x$$

Theorem 4.4.3 — Rules for Differentiating Trigonometric Functions.

$$\begin{array}{lll} a. \frac{d}{dx} \sin x = \cos x & b. \frac{d}{dx} \cos x = -\sin x & c. \frac{d}{dx} (\sec x) = \sec x \tan x \\ d. \frac{d}{dx} (\csc x) = -\csc x \cot x & e. \frac{d}{dx} (\tan x) = \sec^2 x & f. \frac{d}{dx} (\cot x) = -\csc^2 x \end{array}$$

■ **Example 4.14** Differentiate $y = (\sec x)(x - \tan x)$.

■ **Example 4.15** Find the derivative of

(a) $f(x) = (\sin x + x)(x^3 - \ln x)$

(b) $h(x) = \frac{\sin x}{1 - 2 \cos x}$

(c) $g(x) = \left(\frac{e^x + x \sin x}{\tan x} \right)$

Exercise 4.1 1. Differentiate $\frac{\sec x}{1 + \tan x}$. For what values of x does the graph of f have a horizontal tangent?

2. $y = (1 + \cos 3x^2)^4$. Find $\frac{dy}{dx}$

4.5 Chain Rule

Theorem 4.5.1 — The chain Rule.

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

Also, if we write $u = g(x)$ and $y = f(u) = f(g(x))$, then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

■ **Example 4.16** Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$.

■ **Example 4.17** Find the derivative of

(a) $f(x) = \sin(x^3 - x^2)$

(b) $\ln\left(\frac{x^2}{x-2}\right)$

(c) $g(x) = \cos\left(\frac{2x}{1+x}\right)$

■ **Example 4.18** Find $\frac{dy}{dx}$ if $y = u^3 - u^2 + u + 1$ and $u = x^3 + 1$

Solution:

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} = (3u^2 - 2u + 1)(3x^2) \\ &= 3x^2(3x^6 + 4x^3 + 2) \end{aligned}$$

4.6 Higher Derivatives

The derivative f' of a function f is itself a function. As such, we may consider differentiating the function f' . The derivative of f' , if it exists, is denoted by f'' and is called the second derivative of f . Continuing in this fashion, we are led to the third, fourth, fifth, and higher-order derivatives of f , whenever they exist. Notations for the first, second, third, and in general, the n^{th} derivative of f are

$$f', f'', f''', \dots, f^{(n)}$$

or

$$\frac{d}{dx}[f(x)], \frac{d^2}{dx^2}[f(x)], \dots, \frac{d^n}{dx^n}[f(x)]$$

respectively.

■ **Example 4.19** Find the third derivative of $y = \frac{1}{x} \implies y''' = -\frac{6}{x^4}$ ■

■ **Example 4.20** The position of a particle moving along a straight line is given by

$$s = f(t) = 2t^3 - 15t^2 + 24t, \quad t \geq 0$$

where t is measured in seconds and s in feet.

- Find an expression giving the velocity of the particle at any time t . What are the velocity and speed of the particle when $t = 2$?
- Determine the position of the particle when it is stationary.
- Find the acceleration function of the particle. What is the acceleration of the particle when $t = 2$?
- When is the acceleration zero? Positive? Negative?

Solution:

- (a) The required velocity of the particle is given by

$$v(t) = \frac{ds}{dt} = f'(t) = 6t^2 - 30t + 24 = 6(t-1)(t-4)$$

The velocity of the particle when $t = 2$ is

$$v(2) = 6(2-1)(2-4) = -12$$

The speed of the particle when $t = 2$ is $|v(2)| = 12$ ft/sec. In short, the particle is moving in the negative direction at a speed of 12 ft/sec.

- (b) The particle is stationary when its velocity is equal to zero. Setting $v(t) = 0$ gives

$$v(t) = 6(t-1)(t-4) = 0$$

and we see that the particle is stationary at $t = 1$ and $t = 4$.

Its position at $t = 1$ is given by

$$f(1) = 2(1)^3 - 15(1)^2 + 24(1) = 11$$

Its position at $t = 4$ is given by

$$f(4) = 2(4)^3 - 15(4)^2 + 24(4) = -16$$

- (c) $a(t) = \frac{dv}{dt} = f''(t) = 12t - 30$

In particular, the acceleration of the particle when $t = 2$ is

$$a(2) = 12(2) - 30 = -6$$

The particle is decelerating at 6 ft/sec² when $t = 2$.

- (d) The acceleration of the particle is zero when $a(t) = 0$, or

$$12t - 30 = 0 \implies t = \frac{5}{2}.$$

Since $2t - 5 < 0$ when $t < \frac{5}{2}$ and $2t - 5 > 0$ when $t > \frac{5}{2}$, we also conclude that the acceleration is negative for $0 < t < \frac{5}{2}$ and positive for $t > \frac{5}{2}$.

4.7 Implicit Differentiation

Suppose that a function $y = f(x)$ is defined implicitly via an equation in x and y . To compute $\frac{dy}{dx}$:

1. Differentiate both sides of the equation with respect to x . Make sure that the derivative of any term involving y includes the factor $\frac{dy}{dx}$.
2. Solve the resulting equation for $\frac{dy}{dx}$ in terms of x and y .

■ **Example 4.21** Find $\frac{dy}{dx}$ if $2x^2y^2 - 3x^3 + 5y^3 + 6xy^2 = 5$ ■

Solution: $\frac{dy}{dx} = \frac{9x^2 - 6y^2 - 4xy^2}{4xy^2 + 15y^2 + 12xy}$

■ **Example 4.22** Find $\frac{dy}{dx}$ at $(\frac{\pi}{2}, \pi)$ if $x \sin y - y \cos 2x = 2x$ ■

Solution: $\frac{dy}{dx} = \frac{4}{2 - \pi}$

■ **Example 4.23** Find an equation of the tangent line to the bifolium $4x^4 + 8x^2y^2 - 25x^2y + 4y^4 = 0$ at the point $(2, 1)$. ■

Solution: The slope of the tangent line to the bifolium at any point (x, y) is given by $\frac{dy}{dx}$. To compute $\frac{dy}{dx}$, we differentiate both sides of the equation with respect to x to obtain

$$\begin{aligned} \frac{d}{dx}(4x^4 + 8x^2y^2 - 25x^2y + 4y^4) &= \frac{d}{dx}(0) \\ \implies 16x^3 + 16x^2y\frac{dy}{dx} + 16xy^2 - 25x^2\frac{dy}{dx} - 50xy + 16y^3\frac{dy}{dx} &= 0 \end{aligned}$$

By substituting $x = 2$ and $y = 1$ into the last equation, we obtain

$$\frac{dy}{dx} = 3$$

Using the slope-intercept form for an equation of a line, we see that an equation of the tangent line is

$$y = 3x - 5$$

■ **Example 4.24** Find $y'(x)$ implicitly for $y^4 + x^4 = 16$. Then find the value of y'' at the point $(-2, 0)$. ■

Solution: $4y^3y' + 4x^3 = 0 \implies y' = -\frac{x^3}{y^3}$

To find y'' we differentiate this expression for y' using the Quotient Rule and remembering that y is a function of x :

$$\begin{aligned} y'' &= \frac{d}{dx} \left(-\frac{x^3}{y^3} \right) = -\frac{3x^2y^3 - 3x^3y^2y'}{y^6} = -\frac{3x^2y^3 - 3x^3y^2 \left(-\frac{x^3}{y^3} \right)}{y^6} \\ &= -\frac{3x^2(y^4 + x^4)}{y^7} = -\frac{3x^2(16)}{y^7} = -\frac{48x^2}{y^7} \end{aligned}$$

4.8 Related Rate

In a related rates problem the idea is to compute the rate of change of one quantity in terms of the rate of change of another quantity (which may be more easily measured). The procedure is to find an equation that relates the two quantities and then use the Chain Rule to differentiate both sides with respect to time.

Guidelines for Solving a Related Rates Problem

- Draw a diagram, and label the variable quantities.
- Write down the given values of the variables and their rates of change with respect to t .
- Find an equation that relates the variables.
- Differentiate both sides of this equation implicitly with respect to t .
- Replace the variables and derivative in the resulting equation by the values found in Step 2, and solve this equation for the required rate of change.

■ **Example 4.25** At a distance of 12,000 feet from the launch site, a spectator is observing a rocket being launched vertically. What is the speed of the rocket at the instant when the distance of the rocket from the spectator is 13,000 ft and is increasing at the rate of 480 ft/sec? ■

Solution:

Step 1 Let y = the altitude of the rocket and z = the distance of the rocket from the spectator at any time t . (See Figure 4.4.)

Step 2 We are given that at a certain instant of time

$$z = 13,000 \text{ and } \frac{dz}{dt} = 480, \frac{dy}{dt} = ?$$

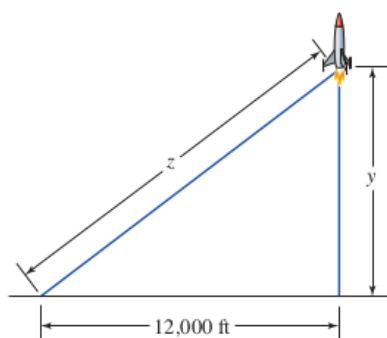


Figure 4.4:

Step 3 Applying the Pythagorean Theorem to the right triangle in Figure 4.4, we find that

$$z^2 = y^2 + (12,000)^2 \quad (4.4)$$

Step 3 Differentiating Equation (4.4) implicitly with respect to t , we obtain

$$2z \frac{dz}{dt} = 2y \frac{dy}{dt} \quad (4.5)$$

Step 3 Using Equation (4.4) we see that if $z = 13,000$, then

$$y = \sqrt{13000^2 - 12000^2} = 5000$$

Finally, substituting $z = 13,000$, $y = 5000$, and $\frac{dz}{dt} = 480$ in Equation (4.5), we find

$$(13000)(480) = 5000 \frac{dy}{dt} \implies \frac{dy}{dt} = 1248 \quad (4.6)$$

Therefore, the rocket is rising at the rate of 1248 ft/sec.

■ **Example 4.26** A 10-foot ladder leans against the side of a building. If the top of the ladder begins to slide down the wall at the rate of 2 ft/sec, how fast is the bottom of the ladder sliding away from the wall when the top of the ladder is 8 feet off the ground? ■

Solution: Let y = the height of the top of the ladder and x = the distance from the wall to the bottom of the ladder. Since the ladder is sliding down the wall at the rate of 2 ft/sec, we have that

$$\frac{dy}{dt} = -2$$

By the Pythagorean Theorem, we have

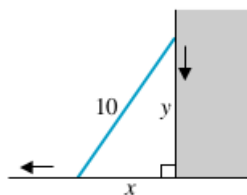


Figure 4.5:

$$x^2 + y^2 = 100$$

Differentiating both sides of this equation with respect to time gives us

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \implies \frac{dx}{dt} = -\frac{y}{x} \frac{dy}{dt}$$

Since the height above ground of the top of the ladder at the point in question is 8 feet, we have that $y = 8$ and from the Pythagorean Theorem, we get

$$100 = x^2 + 8^2,$$

so that $x = 6$. We now have that at the point in question,

$$\frac{dx}{dt} = -\frac{y}{x} \frac{dy}{dt} = -\frac{8}{6}(-2) = \frac{8}{3}$$

So, the bottom of the ladder is sliding away from the building at the rate of $\frac{8}{3}$ ft/sec.

■ **Example 4.27** A water tank has the shape of an inverted circular cone with base radius $2m$ and height $4m$. If water is being pumped into the tank at a rate of $2m^3/min$, find the rate at which the water level is rising when the water is $3m$ deep. ■

Solution: Let V , r , and h be the volume of the water, the radius of the surface, and the height of the water at time t , where t is measured in minutes.

We are given that $\frac{dV}{dt} = 2m^3/min$ and we are asked to find $\frac{dh}{dt}$ when h is $3m$. The quantities V and h

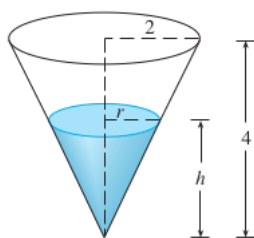


Figure 4.6:

are related by the equation

$$V = \frac{1}{3}\pi r^2 h$$

To do this, we use similar triangles and deduce that

$$\frac{r}{h} = \frac{2}{4} \implies r = \frac{1}{2}h$$

Substituting this value of r into the expression for V , we obtain

$$V = \frac{1}{3}\pi \left(\frac{1}{2}h\right)^2 h = \frac{1}{12}\pi h^3$$

Now we can differentiate each side with respect to t :

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt} \implies \frac{dh}{dt} = \frac{4}{\pi h^2} \frac{dV}{dt}$$

Substituting $h = 3m$ and $\frac{dV}{dt} = 2m^3/min$, we have

$$\frac{dh}{dt} = \frac{4}{\pi(3)^2} \cdot 2 = \frac{8}{9\pi}$$

■ **Example 4.28** A boat is being pulled toward a dock by means of a rope attached to the front tip of the bow. Initially there are 30 feet of rope out and the rope is taught and being reeled in by a circular device the top of which is 10 feet higher than the point where the rope is attached to the boat. This circular device has a radius of 1 foot and turns at the rate of one revolution every pi seconds. How fast is the boat moving along the water when there are 15 feet of rope out? ■

Let

y = amount of rope out

x = distance the boat is from the base of the dock

The circumference of the circular device is 2π ft. so 2π feet of rope are pulled in every π seconds.

Thus y is decreasing by $\frac{2\pi}{\pi} = 2$ ft/sec. i.e.,

$$\frac{dy}{dt} = -2$$

We want to find $\frac{dx}{dt}$ when $y = 15$ (and so $x = \sqrt{(15^2 - 10^2)} = 5\sqrt{5}$ by the Pythagorean Theorem).

$$\begin{aligned} x^2 + 10^2 &= y^2 \implies 2x \frac{dx}{dt} = 2y \frac{dy}{dt} \\ \implies \frac{dx}{dt} &= \frac{y}{x} \frac{dy}{dt} = -2 \frac{15}{5\sqrt{5}} = \frac{-6}{\sqrt{5}} \text{ ft/sec} \end{aligned}$$

when $y = 15$ ft (negative because x is decreasing).

What happens when the boat gets very close to the base of the dock?

■ **Example 4.29** At 1:00 P.M., ship A is 25 mile due south of ship B. If ship A is sailing west at a rate of 16 mi/hr and ship B is sailing south at a rate of 20 mi/hr, find the rate at which the distance between the ships is changing at 1:30 P.M. ■

Let x and y denote the miles covered by ships A and B, respectively, in t hours after 1:00 P.M. We then have the situation sketched in Figure 4.7, where P and Q are their respective positions at 1:00 P.M. and z is the distance between the ships at time t .

By the Pythagorean Theorem,

$$z^2 = x^2 + (25 - y)^2$$

Differentiating with respect to t , we obtain

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} - 2(25 - y) \frac{dy}{dt} \implies z \frac{dz}{dt} = x \frac{dx}{dt} - (25 - y) \frac{dy}{dt}$$

It is given that

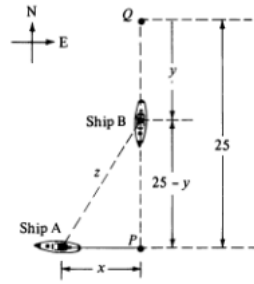


Figure 4.7:

$$\frac{dx}{dt} = 16 \text{ mi/hr} \quad \text{and} \quad \frac{dy}{dt} = 20 \text{ mi/hr}$$

Our objective is to find $\frac{dz}{dt}$.

At 1:30 P.M. the ships have traveled for half an hour and we have $x = 8$, $y = 10$, $25 - y = 15$, and, therefore, $z^2 = 64 + 225 = 289$, or $z = 17$.

Substituting in the equation involving $\frac{dz}{dt}$ gives us

$$17 \frac{dz}{dt} = 8(16) - 20(25 - 10) \implies \frac{dz}{dt} = -\frac{172}{17} \approx -10.12 \text{ mi/hr}$$

The negative sign indicates that the distance between the ships is decreasing at 1:30 P.M.

■ **Example 4.30** Water is poured into a conical funnel at the constant rate of $1 \text{ in.}^3/\text{sec}$ and flows out at the rate of $12 \text{ in.}^3/\text{sec}$ (Figure 4.8a). The funnel is a right circular cone with a height of 4 in. and a radius of 2 in. at the base (Figure 4.8b). How fast is the water level changing when the water is 2 in. high? ■

Solution: Let V = the volume of the water in the funnel

h = the height of the water in the funnel

and r = the radius of the surface of the water in the funnel We are given that

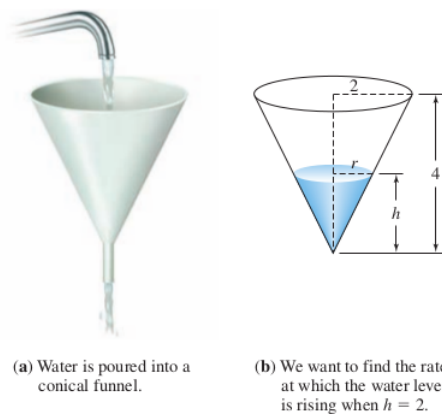


Figure 4.8:

$$\frac{dV}{dt} = 1 - \frac{1}{2} = \frac{1}{2} \quad (\text{Rate of flow in minus rate of flow out})$$

and are asked to find $\frac{dh}{dt}$ when $h = 2$.

The volume of water in the funnel is equal to the volume of the shaded cone in Figure 4.8b. Thus,

$$V = \frac{1}{3}\pi r^2 h$$

To do this, we use similar triangles and deduce that

$$\frac{r}{h} = \frac{2}{4} \implies r = \frac{1}{2}h$$

Substituting this value of r into the expression for V , we obtain

$$V = \frac{1}{3}\pi \left(\frac{1}{2}h\right)^2 h = \frac{1}{12}\pi h^3$$

Now we can differentiate each side with respect to t :

$$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt} \implies \frac{dh}{dt} = \frac{4}{\pi h^2} \frac{dV}{dt}$$

Substituting $h = 2m$ and $\frac{dV}{dt} = \frac{1}{2}m^3/min$, we have

$$\frac{dh}{dt} = \frac{4}{\pi(2)^2} \cdot \frac{1}{2} = \frac{1}{2\pi} \approx 0.159$$

and we see that the water level is rising at the rate of $0.159in./sec$.

- Exercise 4.2**
1. A passenger ship and an oil tanker left port sometime in the morning; the former headed north, and the latter headed east. At noon the passenger ship was 40 mi from port and moving at 30 mph, while the oil tanker was 30 mi from port and moving at 20 mph. How fast was the distance between the two ships changing at that time?
 2. A man who is 6 ft tall walks away from a streetlight that is 15 ft from the ground at a speed of $4ft/sec$. How fast is the tip of his shadow moving along the ground when he is $30ft$ from the base of the light pole?(See figure 4.9)
 3. A balloon is rising at a constant speed $4m/sec$. A boy is cycling along a straight road at a speed of $8m/sec$. When he passes under the balloon, it is 36 metres above him. How fast is the distance between the boy and balloon increasing 3 seconds later.
 4. Suppose you have a cylindrical tank that has radius $10m$ and height $25m$. You pump water into the tank at a constant rate of $20m^3/min$. There is a hole in the bottom of the tank so that you lose water at a rate proportional to the height of water in the tank, say if h is the height of water, the tank loses $\frac{h}{4}m^3/min$. How fast the height of water is changing when there are $15m$ of water in the tank. Does this tank ever overflow?
 5. A coffee pot that has the shape of a circular cylinder of radius $4in.$ is being filled with water flowing at a constant rate. At what rate is the water flowing into the coffee pot when the water level is rising at the rate of $0.4in./sec$? (See figure 4.10)
 6. A car is traveling at 50 mph due south at a point 12 mile north of an intersection. A police car is traveling at 40 mph due west at a point 14 mile east of the same intersection. At that instant, the radar in the police car measures the rate at which the distance between the two cars is changing. What does the radar gun register?

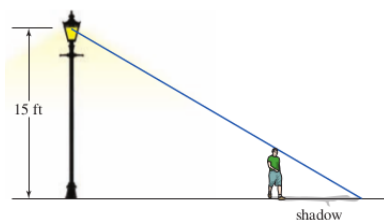


Figure 4.9:



Figure 4.10:

4.9 Derivatives of Inverse Trigonometric Functions

Definition 4.9.1 — Inverse Trigonometric Functions.

	Domain
$y = \sin^{-1} x$ iff $x = \sin y$	$[-1, 1]$
$y = \cos^{-1} x$ iff $x = \cos y$	$[-1, 1]$
$y = \tan^{-1} x$ iff $x = \tan y$	$(-\infty, \infty)$
$y = \csc^{-1} x$ iff $x = \csc y$	$(-\infty, -1] \cup [1, \infty)$
$y = \sec^{-1} x$ iff $x = \sec y$	$(-\infty, -1] \cup [1, \infty)$
$y = \cot^{-1} x$ iff $x = \cot y$	$(-\infty, \infty)$

The graphs of the six inverse trigonometric functions are shown in Figures 4.11a -4.11f

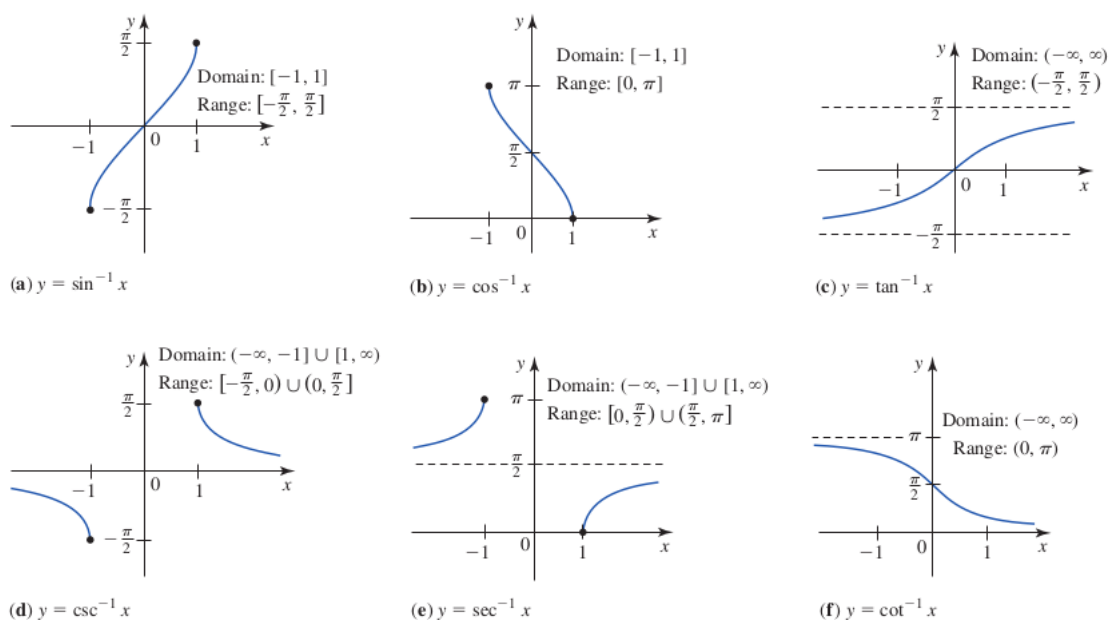


Figure 4.11:

Inverse Properties of Trigonometric Functions

$$\begin{aligned}
 \sin(\sin^{-1} x) &= x & \text{for } -1 \leq x \leq 1 \\
 \sin^{-1}(\sin x) &= x & \text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\
 \cos(\cos^{-1} x) &= x & \text{for } -1 \leq x \leq 1 \\
 \cos^{-1}(\cos x) &= x & \text{for } 0 \leq x \leq \pi \\
 \tan(\tan^{-1} x) &= x & \text{for } -\infty \leq x \leq \infty \\
 \tan^{-1}(\tan x) &= x & \text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}
 \end{aligned}$$

Derivatives of Inverse Trigonometric Functions

$$\begin{aligned}\frac{d}{dx}(\sin^{-1} u) &= \frac{1}{\sqrt{1-u^2}} \frac{du}{dx} \\ \frac{d}{dx}(\cos^{-1} u) &= -\frac{1}{\sqrt{1-u^2}} \frac{du}{dx} \\ \frac{d}{dx}(\tan^{-1} u) &= \frac{1}{1+u^2} \frac{du}{dx}\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}(\csc^{-1} u) &= -\frac{1}{|u|\sqrt{u^2-1}} \frac{du}{dx} \\ \frac{d}{dx}(\sec^{-1} u) &= \frac{1}{|u|\sqrt{u^2-1}} \frac{du}{dx} \\ \frac{d}{dx}(\cot^{-1} u) &= -\frac{1}{1+u^2} \frac{du}{dx}\end{aligned}$$

Proof. Let $y = \sin^{-1} x \implies \sin y = x$. Differentiating implicitly with respect to x , we obtain

$$\cos y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{\cos y}$$

Now, $\cos y \geq 0$, since $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, so we can write,

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

Therefore,

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}, \text{ for } -1 < x < 1$$

Finally, if u is a differentiable function of x , then the Chain Rule gives

$$\frac{d}{dx}(\sin^{-1} u) = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$$

■ **Example 4.31** Find the derivative of

$$(a) f(x) = \cos^{-1} 5x \quad (b) g(x) = \tan^{-1} \sqrt{1+2x} \quad (c) y = \sec^{-1} e^{-2x}$$

4.10 Hyperbolic Functions

Definition 4.10.1 — The Hyperbolic Functions.

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}, x \neq 0$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\operatorname{coth} x = \frac{\cosh x}{\sinh x}, x \neq 0$$

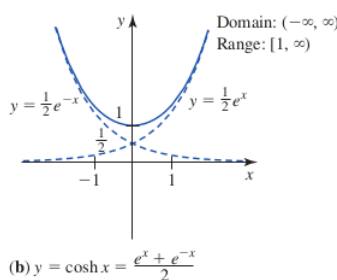
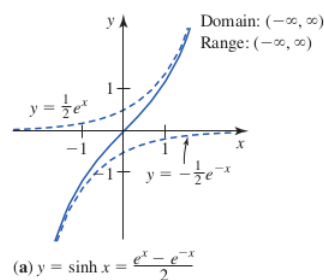


Figure 4.12: The graphs of the hyperbolic sine and cosine functions

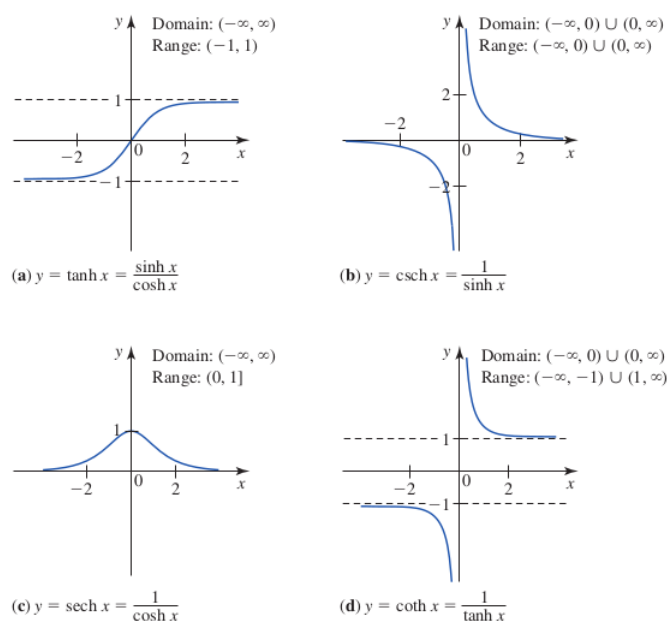


Figure 4.13: The graphs of the hyperbolic tangent, cosecant, secant, and cotangent functions

Hyperbolic Identities

$$\sinh(-x) = -\sinh x$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\cosh^2 x = \frac{1}{2}(1 + \cosh 2x)$$

$$\cosh(-x) = \cosh x$$

$$\operatorname{sech}^2 x = 1 - \tanh^2 x$$

$$\sinh 2x = 2 \sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\sinh^2 x = \frac{1}{2}(-1 + \cosh 2x)$$

4.11 Derivatives of Hyperbolic Functions**Derivatives of Hyperbolic Functions**

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$

$$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

■ **Example 4.32** Find the derivative of

(a) $\sinh(x^2 + 2)$

(b) $\cosh^2(\ln 2x)$

4.12 Derivatives of Inverse Hyperbolic Functions

Definition 4.12.1 — Inverse Trigonometric Functions.

	Domain
$y = \sinh^{-1} x$ iff $x = \sinh y$	$(-\infty, \infty)$
$y = \cosh^{-1} x$ iff $x = \cosh y$	$[1, \infty)$
$y = \tanh^{-1} x$ iff $x = \tanh y$	$(-1, 1)$
$y = \operatorname{csch}^{-1} x$ iff $x = \operatorname{csch} y$	$(-\infty, 0) \cup (0, \infty)$
$y = \operatorname{sech}^{-1} x$ iff $x = \operatorname{sech} y$	$(0, 1]$
$y = \operatorname{coth}^{-1} x$ iff $x = \operatorname{coth} y$	$(-\infty, -1) \cup (1, \infty)$

■ **Example 4.33** Show that $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$ ■

Solution: Let $y = \sinh^{-1} x$. Then

$$x = \sinh y = \frac{e^y - e^{-y}}{2} \quad (4.7)$$

$$\implies e^y - 2x - e^{-y} = 0 \implies e^{2y} - 2xe^y - 1 = 0 \quad (4.8)$$

which is a quadratic in e^y . Using the quadratic formula, we have

$$e^y = x \pm \sqrt{x^2 + 1}$$

Only the root $x + \sqrt{x^2 + 1}$ is admissible. To see why, observe that $e^y > 0$, but $x - \sqrt{x^2 + 1} < 0$ since $x < \sqrt{x^2 + 1}$. Therefore, we have

$$e^y = x + \sqrt{x^2 + 1}$$

so

$$y = \ln(x + \sqrt{x^2 + 1})$$

That is

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

Representations of Inverse Hyperbolic Functions in Terms of Logarithmic Functions

$$\begin{aligned} \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) & (\infty, \infty) \\ \cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) & [1, \infty) \\ \tanh^{-1} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) & (-1, 1) \end{aligned}$$

The derivatives of the inverse hyperbolic functions can be found by differentiating the function in question directly. For example,

$$\begin{aligned} \frac{d}{dx}(\cosh^{-1} x) &= \frac{d}{dx}(\ln(x + \sqrt{x^2 - 1})) \\ &= \frac{1}{x + \sqrt{x^2 - 1}} \left(1 + \frac{x}{\sqrt{x^2 - 1}}\right) \\ &= \frac{1}{x + \sqrt{x^2 - 1}} \left(\frac{\sqrt{x^2 - 1} + x}{\sqrt{x^2 - 1}}\right) \\ &= \frac{1}{\sqrt{x^2 - 1}} \end{aligned}$$

Derivatives of Inverse Hyperbolic Functions

$$\frac{d}{dx}(\sinh^{-1} u) = \frac{1}{\sqrt{u^2 + 1}} \frac{du}{dx}$$

$$\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}$$

$$\frac{d}{dx}(\tanh^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx}$$

$$\frac{d}{dx}(\operatorname{csch}^{-1} u) = -\frac{1}{|u|\sqrt{u^2 + 1}} \frac{du}{dx}$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} u) = -\frac{1}{u\sqrt{1 - u^2}} \frac{du}{dx}$$

$$\frac{d}{dx}(\operatorname{coth}^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx}$$

■ **Example 4.34** Find the derivative of $y = x^2 \operatorname{sech}^{-1} 3x$ ■

4.13 Applications of Derivatives**4.13.1 Extrema of a Function f** **Definition 4.13.1 — Extrema of a Function f .**

A function f has an absolute maximum at c if $f(x) \leq f(c)$ for all x in the domain D of f . The number $f(c)$ is called the maximum value of f on D . Similarly, f has an absolute minimum at c if $f(x) \geq f(c)$ for all x in D . The number $f(c)$ is called the minimum value of f on D . The absolute maximum and absolute minimum values of f on D are called the extreme values, or extrema, of f on D .

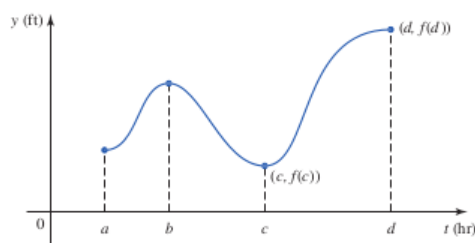


Figure 4.14:

Definition 4.13.2 — Relative Extrema of a Function.

A function f has a relative (or local) maximum at c if $f(c) \geq f(x)$ for all values of x in some open interval containing c . Similarly, f has a relative (or local) minimum at c if $f(c) \leq f(x)$ for all values of x in some open interval containing c .

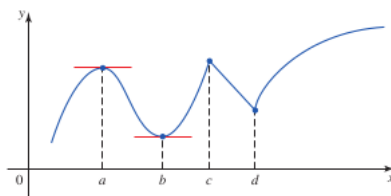


Figure 4.15: The function f has relative extrema at a, b, c , and d . The tangent lines at a and b are horizontal. There are no tangent lines at c and d .

Theorem 4.13.1 — Fermat's Theorem.

If f has a relative extremum at c , then either $f'(c) = 0$ or $f'(c)$ does not exist.

Definition 4.13.3 — Critical Number of f .

A critical number of a function f is any number c in the domain of f at which $f'(c) = 0$ or $f'(c)$ does not exist.

■ **Example 4.35** Find the critical numbers of $f(x) = x - 3\sqrt[3]{x}$ ■

Solution: $f'(x) = \frac{x^{2/3} - 1}{x^{2/3}}$. Then the critical numbers are $-1, 0$, and 1 .

Theorem 4.13.2 — The Extreme Value Theorem.

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ for some number c in $[a, b]$ and an absolute minimum value $f(d)$ for some number d in $[a, b]$.

Guidelines for Finding the Extrema of a Continuous Function f on $[a, b]$

1. Find the critical numbers of f that lie in (a, b) .
2. Compute the value of f at each of these critical numbers, and also compute $f(a)$ and $f(b)$.
3. The absolute maximum value of f and the absolute minimum value of f are precisely the largest and the smallest numbers found in Step 2.

■ **Example 4.36** Find the extreme values of the function ■

(a) $f(x) = 3x^4 - 4x^3 - 8$ on $[-1, 2]$

(b) $f(x) = 2\cos x - x$ on $[0, 2\pi]$

(c) $f(x) = \frac{1}{x}$ on $[1, 3]$ ■

Solution:

- (a) Since f is a polynomial function, it is continuous everywhere; in particular, it is continuous on the closed interval $[-1, 2]$. Therefore, we can use the Extreme Value Theorem.

First, we find the critical numbers of f in $(-1, 2)$:

$$f'(x) = 12x^3 - 12x^2 = 12x^2(x - 1)$$

Observe that $f'(x)$ is continuous on $(-1, 2)$. Next, setting $f'(x) = 0$ gives $x = 0$ or $x = 1$. Therefore, 0 and 1 are the only critical numbers of f in $(-1, 2)$.

Next, we compute $f(x)$ at these critical numbers as well as at the endpoints -1 and 2 .

These values are $f(0) = -8$, $f(1) = -9$, $f(-1) = -1$ and $f(2) = 8$

Thus, f attains the absolute maximum value of 8 at 2 and the absolute minimum value of -9 at 1 .

- (b) The function f is continuous everywhere; in particular, it is continuous on the closed interval $[0, 2\pi]$. Therefore, the Extreme Value Theorem is applicable. First, we find the critical numbers of f in $(0, 2\pi)$. We have $f'(x) = -2\sin x - 1 = 0$

Observe that $f'(x)$ is continuous on $(0, 2\pi)$. Next, setting $f'(x) = 0$ gives

$$-2\sin x - 1 = 0 \Rightarrow \sin x = -\frac{1}{2}$$

Thus, $x = \frac{7\pi}{6}$ or $\frac{11\pi}{6}$. (Remember x lies in $(0, 2\pi)$.) So $\frac{7\pi}{6}$ and $\frac{11\pi}{6}$ are the only critical numbers of f in $(0, 2\pi)$.

Next, we compute the values of f at these critical numbers as well as at the endpoints 0 and 2π . These values are

$$f(0) = 2, f(2\pi) = 2 - 2\pi = -4.28, f\left(\frac{7\pi}{6}\right) = -\sqrt{3} - \frac{7\pi}{6} = -5.40,$$

$$f\left(\frac{11\pi}{6}\right) = \sqrt{3} - \frac{7\pi}{6} = -4.03$$

Thus, f attains the absolute maximum value of 2 at 0 and the absolute minimum value of approximately -5.4 at $\frac{7\pi}{6}$.

- (c) Notice that on the interval $[1, 3]$, f is continuous. Consequently, the Extreme Value Theorem guarantees that f has both an absolute maximum and an absolute minimum on $[1, 3]$.

Hence, $f(x)$ reaches its maximum value of 1 at $x = 1$ and its minimum value of $\frac{1}{3}$ at $x = 3$.

4.13.2 The Mean Value Theorem

Theorem 4.13.3 Rolle's Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b)$, then there exists at least one number c in (a, b) such that $f'(c) = 0$.

■ **Example 4.37** Let $f(x) = x^3 - x$

- Show that f satisfies the hypotheses of Rolle's Theorem on $[-1, 1]$.
- Find the number(s) c in $(-1, 1)$ such that $f'(c) = 0$ as guaranteed by Rolle's Theorem.

Solution:

- (a) The polynomial function f is continuous and differentiable on $(-\infty, \infty)$. In particular, it is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. Furthermore,

$$f(-1) = (-1)^3 - (-1) = 0 \text{ and } f(1) = 1^3 - 1 = 0$$

and the hypotheses of Rolle's theorem are satisfied.

- (b) Rolle's Theorem guarantees that there exists at least one number c in $(-1, 1)$ such that $f'(c) = 0$. But $f'(x) = 3x^2 - 1$, so to find c , we solve

$$3c^2 - 1 = 0 \implies c = \pm \frac{\sqrt{3}}{3}$$

Theorem 4.13.4 The Mean Value Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists at least one number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (4.9)$$

■ **Example 4.38** Let $f(x) = x^3 - 5x^2 - 3x$

- Show that f satisfies the hypothesis of the Mean Value Theorem on $[1, 3]$.
- Find the number(s) c in $(1, 3)$ such that $f'(c) = \frac{f(3) - f(1)}{3 - 1}$ as guaranteed by the Mean Value Theorem.

Solution:

- (a) f is a polynomial function, so it is continuous and differentiable on $(-\infty, \infty)$. In particular, it is continuous on $[1, 3]$ and differentiable on $(1, 3)$. So the hypotheses of the Mean Value Theorem are satisfied.

(b) $f'(x) = 3x^2 - 10x - 3$, so $f'(c) = 3c^2 - 10c - 3$. With $a = 1$ and $b = 3$, Equation (4.9) gives

$$\begin{aligned} f'(c) &= \frac{f(3) - f(1)}{3 - 1} \\ 3c^2 - 10c - 3 &= \frac{-27 - (-7)}{3 - 1} \\ 3c^2 - 10c - 3 &= -10 \implies (3c - 7)(c - 1) \\ c = 1 \text{ or } c &= \frac{7}{3} \end{aligned}$$

So there are two numbers, $c_1 = 1$ and $c_2 = \frac{7}{3}$, in $(1, 3)$ that satisfy Equation (4.9).

4.13.3 Increasing and Decreasing Functions and The First Derivative Test

Definition 4.13.4 Increasing and Decreasing Functions

A function f is increasing on an interval I , if for every pair of numbers x_1 and x_2 in I ,

$$x_1 < x_2 \text{ implies that } f(x_1) < f(x_2)$$

f is decreasing on I if, for every pair of numbers x_1 and x_2 in I ,

$$x_1 < x_2 \text{ implies that } f(x_1) > f(x_2)$$

f is monotonic on I if it is either increasing or decreasing on I .

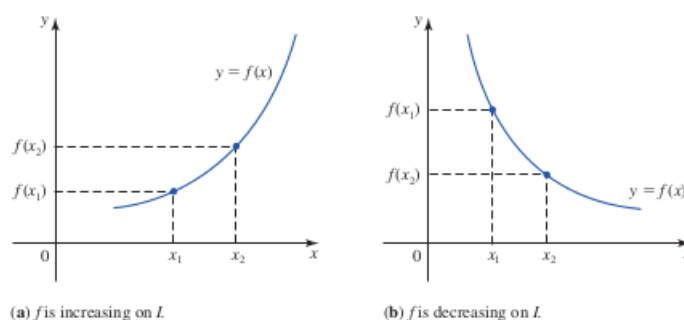


Figure 4.16:

Theorem 4.13.5 Suppose f is differentiable on an open interval (a, b) .

1. If $f'(x) > 0$ for all x in (a, b) , then f is increasing on (a, b) .
2. If $f'(x) < 0$ for all x in (a, b) , then f is decreasing on (a, b) .
3. If $f'(x) = 0$ for all x in (a, b) , then f is constant on (a, b) .

Determining the Intervals Where a Function Is Increasing or Decreasing

1. Find all the values of x for which $f'(x) = 0$ or $f'(x)$ does not exist. Use these values of x to partition the domain of f into open intervals.
2. Select a test number c in each interval found in Step 1, and determine the sign of $f'(c)$ in that interval.
 - (a) If $f'(c) > 0$, then f is increasing on that interval.
 - (b) If $f'(c) < 0$, then f is decreasing on that interval.
 - (c) If $f'(c) = 0$, then f is constant on that interval.

■ **Example 4.39** Determine the intervals where the function $f(x) = x^3 + x^2 - 5x - 5$ is increasing and where it is decreasing. ■

Solution: $f'(x) = 3x^2 + 2x - 5 = (x - 1)(3x + 5)$

from which we see that f' is continuous everywhere and has zeros at $-\frac{5}{3}$ and 1. These zeros of f' partition the domain of f into the intervals $(-\infty, -\frac{5}{3})$, $(-\frac{5}{3}, 1)$, and $(1, \infty)$.

To determine the sign of $f'(x)$ on each of these intervals, we evaluate $f'(x)$ at a convenient test number in each interval. These results are summarized in the following table.

Interval	k	Test value $f'(k)$	sign of $f'(x)$	variation of f
$(-\infty, -\frac{5}{3})$	-2	3	+	increasing on $(-\infty, -\frac{5}{3})$
$(-\frac{5}{3}, 1)$	0	-5	-	decreasing on $(-\frac{5}{3}, 1)$
$(1, \infty)$	2	11	+	increasing on $(1, \infty)$

We conclude that f is increasing on $(-\infty, -\frac{5}{3})$ and $(1, \infty)$ and decreasing on $(-\frac{5}{3}, 1)$.

■ **Example 4.40** Determine the intervals where the function $f(x) = x + \frac{1}{x}$ is increasing and where it is decreasing. ■

Solution: The derivative of f is

$$f'(x) = \frac{(x+1)(x-1)}{x^2}$$

from which we see that $f'(x)$ is continuous everywhere except at $x = 0$ and has zeros at $x = -1$ and $x = 1$. These values of x partition the domain of f into the intervals $(-\infty, -1)$, $(-1, 0)$, $(0, 1)$, and $(1, \infty)$.

By evaluating $f'(x)$ at each of the test numbers $x = -2, -\frac{1}{2}, \frac{1}{2}$, and 2, we find

$f'(-2) = \frac{3}{4}$, $f'(-0.5) = -3$, $f'(0.5) = -3$, and $f'(2) = \frac{3}{4}$. We conclude that f is increasing on $(-\infty, -1)$ and $(1, \infty)$ and decreasing on $(-1, 0)$ and $(0, 1)$. Note that $f'(x)$ does not change sign as we move across the point of discontinuity.

Theorem 4.13.6 The First Derivative Test

Let c be a critical number of a continuous function f in the interval (a, b) and suppose that f is differentiable at every number in (a, b) with the possible exception of c itself.

If $f'(x) > 0$ on (a, c) and $f'(x) < 0$ on (c, b) , then f has a relative maximum at c

2. If $f'(x) < 0$ on (a, c) and $f'(x) > 0$ on (c, b) , then f has a relative minimum at c

3. If $f'(x)$ has the same sign on (a, c) and (c, b) , then f does not have a relative extremum at c

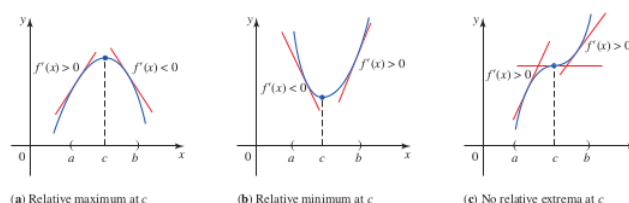


Figure 4.17:

■ **Example 4.41** Find the relative extrema of $f(x) = x^4 - 4x^3 + 12$ ■

Solution: The derivative of f ,

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$$

is continuous everywhere. Therefore, the zeros of f' , which are 0 and 3, are the only critical numbers of f . The sign diagram of f' is shown in Figure 4.18. Since f' has the same sign on $(-\infty, 0)$ and $(0, 3)$, the First Derivative Test tells us that f does not have a relative extremum at 0. Next, we note that f' changes sign from negative to positive as we move across 3, so 3 does give rise to a relative minimum of f . The relative minimum value of f is $f(3) = -15$. The graph of f is shown in Figure 4.19 and confirms these results.



Figure 4.18: The sign diagram of f'

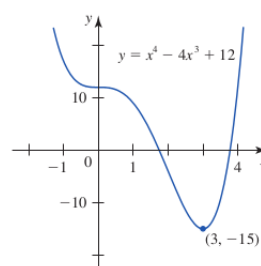


Figure 4.19: The graph of f

■ **Example 4.42** Find the local extrema of f if $f(x) = x^{2/3}(x^2 - 8)$. ■

Solution: The derivatives of f

$$f'(x) = \frac{8(x^2 - 2)}{3x^{1/3}}.$$

The critical numbers are $-\sqrt{2}$, 0, and $\sqrt{2}$. These suggest an examination of the sign of $f'(x)$ in the intervals $(-\infty, -\sqrt{2})$, $(-\sqrt{2}, 0)$, $(0, \sqrt{2})$ and $(\sqrt{2}, \infty)$.

Interval	k	Test value $f'(k)$	sign of $f'(x)$	variation of f
$(-\infty, -\sqrt{2})$	-8	$-\frac{248}{3}$	-	decreasing on $(-\infty, -\sqrt{2}]$
$(-\sqrt{2}, 0)$	-1	$\frac{8}{3}$	+	increasing on $[-\sqrt{2}, 0]$
$(0, \sqrt{2})$	1	$-\frac{8}{3}$	-	decreasing on $(0, \sqrt{2})$
$(\sqrt{2}, \infty)$	8	$\frac{248}{3}$	+	increasing on $(\sqrt{2}, \infty)$

By the First Derivative Test, f has local minima at $-\sqrt{2}$ and $\sqrt{2}$ and a local maxima at 0.

■ **Example 4.43** Find the relative extrema of $x^3 - 6x^2 + 9x + 1$ ■

4.13.4 Concavity and Inflection Point

Definition 4.13.5 Concavity of the Graph of a Function

Suppose f is differentiable on an open interval I . Then

1. the graph of f is concave upward on I if f' is increasing on I .
2. the graph of f is concave downward on I if f' is decreasing on I .

Theorem 4.13.7 Suppose f has a second derivative on an open interval I .

1. If $f''(x) > 0$ for all x in I , then the graph of f is concave upward on I .
2. If $f''(x) < 0$ for all x in I , then the graph of f is concave downward on I .

Definition 4.13.6 — Inflection Point. Let the function f be continuous on an open interval containing the point c , and suppose the graph of f has a tangent line at $P(c, f(c))$. If the graph of f changes from concave upward to concave downward (or vice versa) at P , then the point P is called an inflection point of the graph of f .

Theorem 4.13.8 The Second Derivative Test

Suppose that f has a continuous second derivative on an interval (a, b) containing a critical number c of f .

1. If $f''(c) > 0$, then f has a relative minimum at c .
2. If $f''(c) < 0$, then f has a relative maximum at c .
3. If $f''(c) = 0$, then the test is inconclusive.

■ **Example 4.44** Find the relative extrema of $f(x) = x^4 + \frac{4}{3}x^3 - x^2$ using the Second Derivative Test. ■

Solution: $f'(x) = 4x^3 + 4x^2 - 2x$.

Setting $f'(x) = 0$, we see that 0 , $\frac{-1 - \sqrt{3}}{2}$ and $\frac{-1 + \sqrt{3}}{2}$ are critical numbers of f . Next, we compute

$$f''(x) = 12x^2 + 8x - 2$$

Evaluating $f''(x)$ at the critical number 0 , we find

$$f''(0) = -2 < 0$$

and the Second Derivative Test implies that 0 gives rise to a relative maximum of f . Also

$$f''\left(\frac{-1 - \sqrt{3}}{2}\right) = > 0 \text{ and } f''\left(\frac{-1 + \sqrt{3}}{2}\right) = > 0$$

so $\frac{-1 \pm \sqrt{3}}{2}$ gives rise to a relative minimum of f .

4.13.5 Curve Sketching

Guidelines for Sketching a Curve

1. Find the domain of f .
2. Find the x - and y -intercepts of f .
3. Determine whether the graph of f is symmetric with respect to the y -axis or the origin. That is
 - (a) If $f(-x) = f(x)$ for all x in the domain D , then f is an even function and the curve is symmetric about the y -axis.
 - (b) If $f(-x) = -f(x)$ for all x in the domain D , then f is an odd function and the curve is symmetric about the origin.
4. Determine the behavior of f for large absolute values of x .
5. Find the asymptotes of the graph of f .
6. Find the intervals where f is increasing and where f is decreasing.
7. Find the relative extrema of f .
8. Determine the concavity of the graph of f .
9. Find the inflection points of f .
10. Sketch the graph of f .

■ **Example 4.45** Let $y = \frac{x^2}{x^2 - 1}$. Then sketch the graph of y . ■

Solution:

1. The denominator of f is $(-\infty, -1) \cup (1, 1) \cup (1, \infty)$.
2. Setting $x = 0$ gives 0 as the y -intercept. Next, setting $f(x) = 0$ gives $x^2 = 0$, or $x = 0$. So the x -intercept is 0.
3. $f(-x) = \frac{(-x)^2}{(-x)^2 - 1} = \frac{x^2}{x^2 - 1} = f(x)$ and this shows that the graph of f is symmetric with respect to the y -axis.

$$4. \lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 1} = 1$$

5. Because the denominator of $f(x)$ is equal to zero at -1 and 1 , the lines $x = -1$ and $x = 1$ are candidates for the vertical asymptotes of the graph of f . Since

$$\lim_{x \rightarrow -1^-} \frac{x^2}{x^2 - 1} = \infty \text{ and } \lim_{x \rightarrow 1^-} \frac{x^2}{x^2 - 1} = -\infty$$

we see that $x = -1$ and $x = 1$ are indeed vertical asymptotes. From part (4) we see that $y = 1$ is a horizontal asymptote of the graph of f .

$$6. f'(x) = -\frac{2x}{(x^2 - 1)^2}$$

Notice that f' is continuous everywhere except at ± 1 and that it has a zero when $x = 0$. The sign diagram of f' is shown in Figure 4.20.

From the diagram we see that f is increasing on $(-\infty, -1)$ and on $(-1, 0)$ and decreasing on

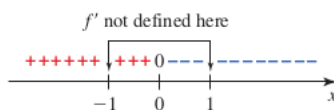


Figure 4.20:

$(0, 1)$ and on $(1, \infty)$.

7. From the results of part (6) we see that 0 is a critical number of f . The numbers -1 and 1 are not in the domain of f and, therefore, are not critical numbers of f .

Also, from Figure 4.20 we see that f has a relative maximum at $x = 0$. Its value is $f(0) = 0$.

$$8. f''(x) = \frac{2(3x^2 + 1)}{(x^2 - 1)^3}$$

Notice that f'' is continuous everywhere except at ± 1 and that f'' has no zero. The sign diagram of f'' is shown in Figure 4.21, we see that the graph of f is concave upward on $(-\infty, -1)$ and on $(1, \infty)$ and concave downward on $(-1, 1)$.

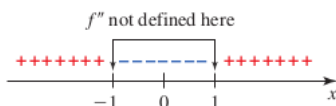


Figure 4.21:

9. f has no inflection points. Remember that -1 and 1 are not in the domain of f .

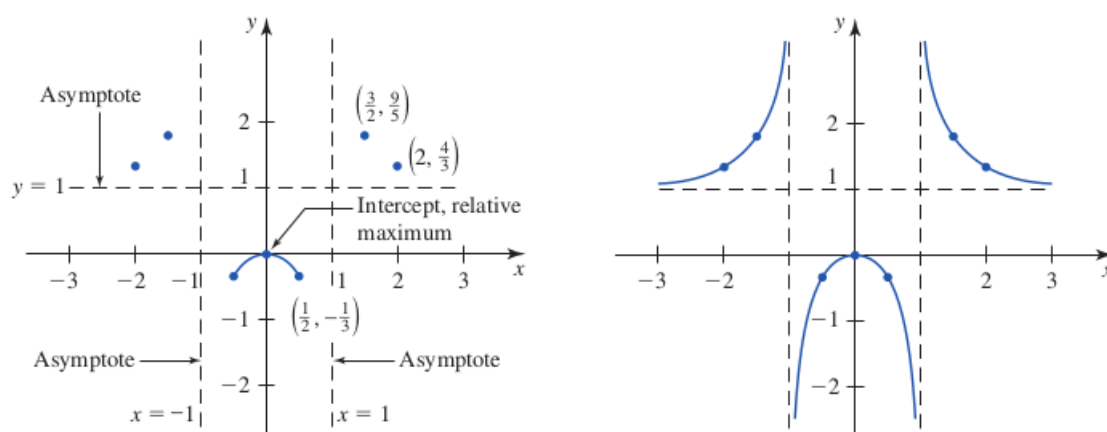


Figure 4.22:

Slant Asymptotes

The graph of a function f may have an asymptote that is neither vertical nor horizontal but slanted. We call the line with equation $y = mx + b$ a **slant** or **oblique** (right) asymptote of the graph of f if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = m \quad \text{and} \quad \lim_{x \rightarrow \infty} (f(x) - mx) = b$$

4.14 Indeterminate Forms and l'Hospital's Rule

4.14.1 The Indeterminate Form $0/0$ and ∞/∞

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, then the limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

is called an **indeterminate form of the type $0/0$** . As the name implies, the undefined expression $0/0$ does not provide us with a definitive answer concerning the existence of the limit or its value, if the limit exists.

If $\lim_{x \rightarrow a} f(x) = \pm\infty$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$, then the limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

is called an **indeterminate form of the type ∞/∞** .

To see why this limit is an indeterminate form, we simply write

$$\lim_{x \rightarrow a} \frac{\frac{1}{\frac{1}{g(x)}}}{\frac{1}{\frac{1}{f(x)}}}$$

which has the form $0/0$ and, therefore, is indeterminate.

Theorem 4.14.1 — l'Hôpital's Rule. Suppose that f and g are differentiable on an open interval I that contains a , with the possible exception of a itself, and $g'(x) \neq 0$ for all x in I . If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is indeterminate form of the type $0/0$ or ∞/∞ , then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided that the limit on the right-hand side exists or is infinite.

R l'Hôpital's Rule is also valid for one-sided limits as well as limits at infinity or negative infinity; that is, we can replace " $x \rightarrow a$ " by any of the symbols $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, or $x \rightarrow -\infty$.

■ **Example 4.46** Evaluate

$$(a) \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x}$$

$$(b) \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$$(c) \lim_{x \rightarrow 1^+} \frac{\sin \pi x}{\sqrt{x-1}}$$

$$(d) \lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$$

$$(e) \lim_{x \rightarrow \infty} \frac{x^3}{e^{2x}}$$

$$(f) \lim_{x \rightarrow 0} \frac{x^3}{x - \tan x}$$

4.14.2 The Indeterminate Form $\infty - \infty$ and $0 \cdot \infty$

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$, then the limit

$$\lim_{x \rightarrow a} (f(x) - g(x))$$

is said to be an **indeterminate form of the type $\infty - \infty$** . An indeterminate form of this type can often be expressed as one of the type $0/0$ or ∞/∞ by algebraic manipulation.

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \pm\infty$, then the limit

$$\lim_{x \rightarrow a} (f(x)g(x))$$

is said to be an **indeterminate form of the type $0 \cdot \infty$** . An indeterminate form of this type can often be expressed as one of the type $0/0$ or ∞/∞ by algebraic manipulation.

■ **Example 4.47** Evaluate

$$(a) \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$$

$$(b) \lim_{x \rightarrow 0} \left(\frac{1}{\ln(x+1)} - \frac{1}{x} \right)$$

$$(c) \lim_{x \rightarrow 0^+} x \ln x$$

$$(d) \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

4.14.3 The Indeterminate Form $\infty - \infty$ and $0 \cdot \infty$

The limit

$$\lim_{x \rightarrow a} [f(x)]^{g(x)}$$

is said to be an **indeterminate form of the type**

$$0^0 \text{ if } \lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

$$\infty^0 \text{ if } \lim_{x \rightarrow a} f(x) = \infty \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

$$1^0 \text{ if } \lim_{x \rightarrow a} f(x) = 1 \text{ and } \lim_{x \rightarrow a} g(x) = \infty$$

These indeterminate forms can usually be converted to indeterminate forms of the type $0 \cdot \infty$ by taking logarithms or by using the identity

$$[f(x)]^{g(x)} = e^{g(x) \ln f(x)}$$

■ **Example 4.48** Evaluate

$$(a) \lim_{x \rightarrow 0^+} x^x$$

$$(b) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$$

$$(c) \lim_{x \rightarrow 1^+} x^{\frac{1}{x-1}}$$